Smarter Regular Expression Matching

From Theory to Tools

Ilian Kohl

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1 Introduction

1.1 Motivation

Regular expressions are a fundamental formalism for describing regular languages and are among the most widely used abstractions in computer science. They underpin applications including lexical analysis, input validation, log processing, search tools, and network security. Across these applications, the problem is regular expression matching. Given a regular expression and an input string, determine whether the string belongs to the language denoted by the expression. Formally, this is the *membership problem*.

For example, the expression `(ab|a)*` matches `"aba"`, as it decomposes into `"ab"` followed by `"a"`, but does not match `"abb"`, since no sequence of subexpressions can produce it.

For many applications, a Boolean answer is enough. An input validator may only need to accept or reject a password or URL. Others require more. For instance, a lexer must determine how input decomposes into tokens. Capturing groups, schema validation, and information extraction require identifying which input parts correspond to parts of the expression. Recognition alone is not enough. The matcher must also recover the *structure*.

Efficiently solving these problems remains an active research area. The choice of algorithm influences performance, memory consumption, and security. Once matching becomes parsing, the way in which match structure is recovered also influences correctness and raises the question of how to handle ambiguity when multiple matches are possible.

Existing approaches include backtracking and finite-automata-based techniques, such as nondeterministic (NFA) and deterministic (DFA) automata. These approaches have different trade-offs in implementation complexity, memory consumption, and worst-case performance. Backtracking engines are flexible and widely used, but may exhibit exponential running time on constructed inputs, leading to regular expression denial-of-service (ReDoS) vulnerabilities [Cox, 2007]. Automata-based approaches provide stronger worst-case guarantees but may require preprocessing or suffer from state-space explosion.

Derivative-based algorithms provide an alternative by transforming the regular expression as input symbols are consumed. Each derivative represents the language remaining after an input symbol has been consumed. By maintaining a correspondence between an expression and its successive residuals, derivative-based methods provide an elegant foundation for regular expression matching and, with suitable extensions, structured parsing.

However, derivative-based techniques are less common in practical regular expression tooling. Their use for practical matching, parsing, and diagnostic tools motivates this thesis: "From Theory to Tools".

1.2 Problem Statement

Derivative-based approaches provide a principled foundation for regular expression matching and parsing. Brzozowski introduced regular expression derivatives as a way of transforming a regular expression after consuming an input symbol [Brzozowski, 1964]. Antimirov later introduced partial derivatives, which represent sets of residual expressions and are closely related to nondeterministic automata [Antimirov, 1996].

Building on Brzozowski's derivatives, Sulzmann and Lu developed a derivative-based algorithm for POSIX regular expression parsing [Sulzmann and Lu, 2014]. Their work also presents a bit-coded variant that represents parse information as a compact bit sequence, which can be decoded after a successful match rather than maintaining an explicit parse tree throughout the computation. A Haskell implementation of the POSIX parser is available in Sulzmann's *Programming Paradigms* teaching material [Sulzmann, 2026].

Sulzmann and Lu also studied partial derivatives in the context of regular expression sub-matching [Sulzmann and Lu, 2012]. While sub-matching is not the focus of this thesis, their work provides relevant background for the partial-derivative approach. The *Programming Paradigms* teaching material also presents a bit-coded partial-derivative-based parser in Haskell.

These constructions have primarily been presented in a theoretical and functional programming setting, leaving open practical questions about execution time, memory consumption, and data representation.

This thesis investigates these questions through implementations in Rust. As a systems programming language, Rust provides fine-grained control over memory allocation and data representation, allowing the derivative-based and partial-derivative-based parsers, including their bit-coded variants, to be evaluated in practice.

The implementation also provides traced versions of the parsing algorithms that record intermediate derivative computations during matching. These traces show the transformations performed by the algorithms and help explain successful and unsuccessful matches.

1.3 Contributions

The following contributions are being made:

- **Rust implementation of derivative-based parsers.** The thesis implements derivative-based and partial-derivative-based parsing constructions in Rust, including direct and bit-coded variants. The implementations are based on the constructions described in the literature and the corresponding Haskell material, resulting in four parser variants in total.
- **Empirical evaluation of the parser variants.** The four implementations are evaluated with respect to execution time and memory consumption using synthetic benchmarks and a corpus of real-world regular expressions from Snort/Suricata, SpamAssassin, and RegexLib.

1.4 Thesis Outline

The remainder of this thesis follows the parsing constructions from their formal concepts through their Rust implementation to their evaluation. Theory and implementation are presented together.

- **Chapter 2 - Regular Expressions.** This chapter introduces the syntax and semantics of regular expressions and the Rust representation used throughout the implementation.
- **Chapter 3 - Derivatives.** This chapter introduces Brzozowski's derivatives and Antimirov's partial derivatives, together with their basic matching algorithms and Rust implementations.
- **Chapter 4 - Matching vs. Parsing.** This chapter distinguishes regular expression matching from parsing and introduces the disambiguation problem.
- **Chapter 5 - Derivative Parser.** This chapter presents the derivative-based POSIX parsing approach of Sulzmann and Lu, along with a Rust implementation developed in this work, in both direct and bit-coded variants.
- **Chapter 6 - Partial-Derivative Parser.** This chapter presents the partial-derivative-based parsing approach of Sulzmann and Lu, along with a Rust implementation developed in this work, in both direct and bit-coded variants.
- **Chapter 7 - External versus Internal Syntax.** This chapter describes the parsing of regular expression strings into an external AST and their translation into the internal representation used by the parsers.
- **Chapter 8 - Evaluation.** This chapter evaluates the four parser variants with respect to execution time, memory consumption, and correctness. The evaluation uses synthetic benchmarks and a corpus of real-world regular expressions. It also examines optimization challenges and the insights gained from the evaluation.
- **Chapter 9 - Conclusion.** Finally, this chapter summarizes the findings of the thesis and discusses the practical implications of the results and directions for future work.

2 Regular Expressions

2.1 Syntax

Let Σ be an alphabet. Regular expressions over Σ are defined inductively by the grammar:

$$r, s ::= x \mid \varepsilon \mid \emptyset \mid r + s \mid r \cdot s \mid r^* \qquad (x \in \Sigma). \tag{2.1}$$

In the grammar, r and s range over regular expressions, while x ranges over the alphabet Σ and denotes an arbitrary symbol, such as a, b, or c. The constants ε and $\emptyset$ represent the *empty-word* and *empty-language* expressions. The operators $+$, $\cdot$, and * denote alternation, concatenation, and Kleene star, respectively.

Concatenation is written implicitly, so rs denotes $r \cdot s$. Given this notation, we adopt the precedence and associativity conventions. Unless parentheses are shown explicitly, Kleene star binds most tightly, followed by concatenation, followed by alternation. For example,

$$ab^*c = (a \cdot (b^*)) \cdot c,$$
$$a + bc^* = a + (b \cdot (c^*)).$$

For repeated operators we use left association, so

$$a + b + c = (a + b) + c,$$
$$abc = (ab)c.$$

The grammar above is the core syntax. Most regular expression libraries also offer one-or-more, optional, and bounded repetition (r^+, $r?$, $r^{\{m,n\}}$), but these add no expressive power since each can be defined by the core constructs: $r^+ = r \cdot r^*$, $r? = r + \varepsilon$, and

$$r^{\{m,n\}} = \underbrace{r \cdot \ldots \cdot r}_{m \text{ times}} \cdot \underbrace{(r + \varepsilon) \cdot \ldots \cdot (r + \varepsilon)}_{n-m \text{ times}}.$$

In the implementation, expressions using such operators are translated into the core constructs before they reach the parsing algorithms.

For example, consider the regular expression r_{01} over $\Sigma = \{0, 1\}$

$$r_{01} = (0 + 1)^* \cdot 0 \cdot 1. \tag{2.2}$$

By the conventions above, its syntax tree is $((0 + 1)^* \cdot 0) \cdot 1$. This tree represents the hierarchical structure of the expression: a Kleene star applied to the alternation $0 + 1$, followed by concatenation with the literals 0 and 1.

The syntactic structure must be given a semantic interpretation.

2.2 Semantics

A semantic interpretation assigns a language to each regular expression. A *word* over Σ is a finite sequence of symbols from Σ. The set of all words over Σ, including the empty word ε, is denoted by Σ^*. A *language* over Σ is a subset of Σ^*.

For a regular expression r, we write $L(r)$ for the language it denotes, where $L(r) \subseteq \Sigma^*$. The definition follows the structure of the syntax tree by *structural recursion*. Each case corresponds to one production of the grammar in (2.1):

$$\begin{aligned} L(x) &= \{x\}, \\ L(\varepsilon) &= \{\varepsilon\}, \\ L(\emptyset) &= \emptyset, \\ L(r + s) &= L(r) \cup L(s), \\ L(r \cdot s) &= \{vw \mid v \in L(r),\ w \in L(s)\}, \\ L(r^*) &= \{\varepsilon\} \cup \{w_1 \cdots w_n \mid n \geq 1,\ w_1, \ldots, w_n \in L(r)\}. \end{aligned} \tag{2.3}$$

Table 2.1 applies these definitions to small expressions. The literal a denotes the language containing only the word a. The expression ε denotes the language containing only the empty word, while $\emptyset$ denotes the language containing no words. The alternation $a + b$ combines the languages of a and b to give $\{a, b\}$, and concatenation ab combines words from the languages of a and b to give $\{ab\}$. The Kleene star a^* allows zero or more concatenations of words from the language of a, giving $\{\varepsilon, a, aa, aaa, \ldots\}$.

Expression r	Language $L(r)$
a	$\{a\}$
ε	$\{\varepsilon\}$
$\emptyset$	$\emptyset$
$a + b$	$\{a, b\}$
ab	$\{ab\}$
a^*	$\{\varepsilon, a, aa, aaa, \ldots\}$

Table 2.1: Languages denoted by basic expressions.

For example $r_{01} = (0 + 1)^* \cdot 0 \cdot 1$ from (2.2), we compute its language step by step:

$$\begin{aligned} L(0) &= \{0\}, \\ L(1) &= \{1\}, \\ L(0 + 1) &= L(0) \cup L(1) = \{0, 1\}, \\ L((0 + 1)^*) &= \{0, 1\}^*, \\ L(r_{01}) &= L((0 + 1)^*) \cdot L(0) \cdot L(1) = \{\, w \cdot 0 \cdot 1 \mid w \in \{0, 1\}^* \,\}. \end{aligned} \tag{2.4}$$

This is the set of binary words ending in 01. For example, the words 01, 001, and 1101 belong to $L(r_{01})$, whereas ε, 0, and 010 do not.

This is the question of whether a word *matches* a regular expression: w *matches* r if $w \in L(r)$. Determining this is the *membership problem*.

2.3 Rust Representation

The grammar from (2.1) is represented directly as a Rust enum with one variant for each syntactic construct, as shown in Figure 2.1:

```
#[derive(Debug, Clone, PartialEq, Eq, Hash)]
pub enum Regex {
    Phi, Eps, Lit(char),
    Seq(Box<Regex>, Box<Regex>),  // r . s
    Alt(Box<Regex>, Box<Regex>),  // r + s
    Star(Box<Regex>),              // r*
}

impl Regex {
    pub fn seq(r: Regex, s: Regex) -> Regex {
        Regex::Seq(Box::new(r), Box::new(s))
    }
    pub fn alt(r: Regex, s: Regex) -> Regex {
        Regex::Alt(Box::new(r), Box::new(s))
    }
    pub fn star(r: Regex) -> Regex { Regex::Star(Box::new(r)) }
    pub fn lit(c: char) -> Regex { Regex::Lit(c) }
}
```

Figure 2.1: The `Regex` enum and constructors. Derived traits support operations needed by parsing: `Clone` for duplication, `PartialEq/Eq` for comparison, `Hash` for collections.

The type carries no precedence or associativity information, as these are determined by the syntax tree. Rust requires explicit indirection because enum sizes must be known at compile time. Recursive variants therefore use `Box<Regex>`. A direct definition such as `Seq(Regex, Regex)` would have infinite size, whereas `Box` stores recursive values separately, providing finite size [The Rust Project Developers, 2026a,b]. This differs from Haskell's heap-allocated data types.

The example $r_{01} = (0 + 1)^* \cdot 0 \cdot 1$ from (2.2) is represented by the syntax tree:

$$\mathrm{Seq}\Big(\mathrm{Seq}(\mathrm{Star}(\mathrm{Alt}(0, 1)), 0), 1\Big). \tag{2.5}$$

In Rust, this is represented as shown in Figure 2.2, with the outermost `Seq` containing `Lit('1')` and `Seq(Star(Alt(0,1)), 0)` as its right and left children.

```
Regex::seq(
    Regex::seq(
        Regex::star(Regex::alt(Regex::lit('0'), Regex::lit('1'))),
        Regex::lit('0')),
    Regex::lit('1'))
```

Figure 2.2: Rust representation of r_{01}.

3 Derivatives and Partial Derivatives

Chapter 2 introduced the syntax and semantics of regular expressions, and defined the *membership problem*: Decide whether a word w belongs to $L(r)$. This chapter introduces three algorithms for solving it, each returning a Boolean answer: match or no match.

The first is a naive recursive matcher that follows the structure of the regular-expression semantics. The other two transform the regular expression as input symbols are consumed. Brzozowski derivatives transform it into a single residual expression, while Antimirov's partial derivatives represent the possible residual expressions as a set.

3.1 Naive Matching

A direct way to decide whether $w \in L(r)$ is to interpret the regular-expression semantics as recursive matching rules, by following the structure of the expression and searching for possible decompositions of the input. Let $w[i:j]$ denote the substring from position i to j. A recursive procedure $\mathrm{match}(w, i, j, r)$ determines whether $w[i:j] \in L(r)$:

$$\begin{aligned}
\mathrm{match}(w,i,j,\varepsilon) &\iff i = j,\\
\mathrm{match}(w,i,j,x) &\iff j = i+1 \wedge w[i] = x,\\
\mathrm{match}(w,i,j,r+s) &\iff \mathrm{match}(w,i,j,r) \vee \mathrm{match}(w,i,j,s),\\
\mathrm{match}(w,i,j,r\cdot s) &\iff \exists k \in [i,j].\ \mathrm{match}(w,i,k,r) \wedge \mathrm{match}(w,k,j,s),\\
\mathrm{match}(w,i,j,r^*) &\iff i = j \vee \exists k \in (i,j].\ \mathrm{match}(w,i,k,r) \wedge \mathrm{match}(w,k,j,r^*).
\end{aligned} \tag{3.1}$$

Each rule corresponds directly to the semantics. Concatenation does not specify where to split, so the matcher must consider all possible split points k.

For matching $(a \cdot a) \cdot a$ against `aaa`, the split points $k \in \{0,1,2,3\}$ are:

$$\begin{aligned}
&\underbrace{\mathrm{match}(w,0,0,a\cdot a)}_{a\cdot a \text{ against } w[0:0]=\epsilon} \wedge \underbrace{\mathrm{match}(w,0,3,a)}_{a \text{ against } w[0:3]=\texttt{aaa}}\\
\vee\ &\underbrace{\mathrm{match}(w,0,1,a\cdot a)}_{a\cdot a \text{ against } w[0:1]=\texttt{a}} \wedge \underbrace{\mathrm{match}(w,1,3,a)}_{a \text{ against } w[1:3]=\texttt{aa}}\\
\vee\ &\underbrace{\mathrm{match}(w,0,2,a\cdot a)}_{a\cdot a \text{ against } w[0:2]=\texttt{aa}} \wedge \underbrace{\mathrm{match}(w,2,3,a)}_{a \text{ against } w[2:3]=\texttt{a}}\\
\vee\ &\underbrace{\mathrm{match}(w,0,3,a\cdot a)}_{a\cdot a \text{ against } w[0:3]=\texttt{aaa}} \wedge \underbrace{\mathrm{match}(w,3,3,a)}_{a \text{ against } w[3:3]=\epsilon}.
\end{aligned} \tag{3.2}$$

At $k = 2$, both operands match, triggering a recursive search on $(a \cdot a)$ against `aa` with split points $k \in \{0, 1, 2\}$, where $k = 1$ succeeds:

$$\begin{aligned} & \underbrace{\text{match}(w, 0, 0, a)}_{a \text{ against } w[0:0]=\epsilon} \wedge \underbrace{\text{match}(w, 0, 2, a)}_{a \text{ against } w[0:2]=\texttt{aa}} \\ \vee\ & \underbrace{\text{match}(w, 0, 1, a)}_{a \text{ against } w[0:1]=\texttt{a}} \wedge \underbrace{\text{match}(w, 1, 2, a)}_{a \text{ against } w[1:2]=\texttt{a}} \\ \vee\ & \underbrace{\text{match}(w, 0, 2, a)}_{a \text{ against } w[0:2]=\texttt{aa}} \wedge \underbrace{\text{match}(w, 2, 2, a)}_{a \text{ against } w[2:2]=\epsilon}. \end{aligned} \tag{3.3}$$

This completes the original match through the nested split points $k = 2$ and $k = 1$.

In Rust. The recursive rules translate directly into the corresponding Rust match arms, shown in Figure 3.1. Alternation tries both branches on the same input, while concatenation and star search possible split points. The star arm clones the body to construct the recursive `Star` expression. This is necessary because the recursive call needs to own the newly constructed expression. The range-based search short-circuits at the first successful split. Nested concatenations repeatedly search smaller substrings, exploring multiple paths before succeeding. This leads to exponential backtracking in the worst case.

```
Regex::Alt(r, s) => {
    match_naive_range(word, i, j, r)
        || match_naive_range(word, i, j, s)
}
Regex::Seq(r, s) => {
    (i..=j).any(|k| {
        match_naive_range(word, i, k, r)
            && match_naive_range(word, k, j, s)
    })
}
Regex::Star(r) => {
    i == j || (i + 1..=j).any(|k| {
        match_naive_range(word, i, k, r)
            && match_naive_range(word, k, j,
                &Regex::star(*r.clone()))
    })
}
```

Figure 3.1: Alternation, concatenation, and star arms of the naive matcher.

Although inefficient, this matcher is useful as a reference implementation because it follows the regular-expression semantics directly without any intermediate transformation. The cost is visible in the worked examples: nested concatenations repeatedly search smaller substrings, each exploring all possible split points before finding a match.

The derivative approach takes a different approach. Rather than searching for a decomposition of w while keeping r fixed, it transforms r as each character of w is consumed. The resulting expression describes the language that can still match the remaining suffix.

3.2 Brzozowski Derivatives

Brzozowski's derivative of r with respect to a symbol x, written $r\backslash x$, is the expression describing exactly what remains of r's language after consuming the leading symbol x of a word $w \in L(r)$ [Brzozowski, 1964]:

$$L(r\backslash x) \;=\; \{\, w \mid xw \in L(r) \,\}. \tag{3.4}$$

For $w = x_1 x_2 \cdots x_n$, consume one symbol at a time by repeatedly taking derivatives:

$$r, \quad r\backslash x_1, \quad (r\backslash x_1)\backslash x_2, \quad \ldots$$

Nullability. After all symbols have been consumed, the remaining input is ε. Thus, the resulting expression must accept ε for the original word to belong to $L(r)$. This is the question of whether the expression is *nullable*. Let $n(r)$ be true exactly when $\varepsilon \in L(r)$:

$$\begin{array}{llll} n(\emptyset) = \text{false} & (\mathsf{N}_\emptyset) & n(r+s) = n(r) \vee n(s) & (\mathsf{N}_+) \\ n(\varepsilon) = \text{true} & (\mathsf{N}_\varepsilon) & n(r \cdot s) = n(r) \wedge n(s) & (\mathsf{N}_{\text{seq}}) \\ n(x) = \text{false} & (\mathsf{N}_{\text{lit}}) & n(r^*) = \text{true} & (\mathsf{N}_*) \end{array} \tag{3.5}$$

The empty language $\emptyset$ is not nullable because it contains no words, including ε, while a literal cannot match the empty word. ε and r^* always match ε. An alternation is nullable when either branch is nullable, and a concatenation only when both components are.

In Rust. The definition is transcribed one `Regex` variant at a time, using the enum of Figure 2.1. The case split of (3.5) becomes the arms of a `match`, shown in Figure 3.2.

```
pub fn nullable(r: &Regex) -> bool {
    match r {
        Regex::Phi => false,
        Regex::Eps => true,
        Regex::Lit(_) => false,
        Regex::Alt(r, s) => nullable(r) || nullable(s),
        Regex::Seq(r, s) => nullable(r) && nullable(s),
        Regex::Star(_) => true,
    }
}
```

Figure 3.2: `nullable` implementation and its six cases: $\mathsf{N}_\emptyset$, N_ε, N_{lit}, N_+, N_{seq} and N_*.

The expression is passed by reference because `nullable` only inspects r and does not modify or consume it. The underscores in N_{lit} and N_* indicate that the corresponding operands are irrelevant. A literal is never nullable, regardless of its symbol, while a Kleene-star expression is always nullable, regardless of its body. The short-circuiting operators `||` and `&&` avoid evaluating the second operand when the result is already determined, while preserving the corresponding logical semantics.

Derivative Rules. The rules specify the expression that remains after consuming x:

$$\begin{aligned}
&\emptyset\backslash x = \emptyset && (\mathsf{D}_\emptyset) && (r+s)\backslash x = (r\backslash x) + (s\backslash x) && (\mathsf{D}_+)\\
&\varepsilon\backslash x = \emptyset && (\mathsf{D}_\varepsilon) && r^*\backslash x = (r\backslash x)\cdot r^* && (\mathsf{D}_*)\\
&y\backslash x = \begin{cases} \varepsilon & x = y \\ \emptyset & x \neq y \end{cases} && (\mathsf{D}_{\text{lit}}) && (r\cdot s)\backslash x = \begin{cases} (r\backslash x)\cdot s + s\backslash x & n(r) = \text{true} \\ (r\backslash x)\cdot s & n(r) = \text{false} \end{cases} && \begin{matrix}(\mathsf{D}^n_{\text{seq}})\\(\mathsf{D}_{\text{seq}})\end{matrix}
\end{aligned} \tag{3.6}$$

The empty language and ε cannot consume x. A literal y succeeds exactly when $x = y$, leaving ε. Alternation retains both derivatives and for r^*, one occurrence of r consumes x while further occurrences remain available.

Concatenation is the only rule that depends on nullability. If r is not nullable, r must consume the leading x, leaving $r\backslash x$ as the remainder of r while s remains untouched. If r is nullable, it may either consume x giving $(r\backslash x)\cdot s$, or r matches ε and s consumes x leaving $s\backslash x$. Hence the nullable case produces the alternation $(r\backslash x)\cdot s + s\backslash x$.

In Rust. Figure 3.3 implements (3.6) similarly, one variant at a time.

```
pub fn deriv(r: &Regex, x: char) -> Regex {
    match r {
        Regex::Phi => Regex::Phi,
        Regex::Eps => Regex::Phi,
        Regex::Lit(c) => if *c == x { Regex::Eps } else { Regex::Phi },
        Regex::Alt(r1, r2) =>
            Regex::alt(deriv(r1, x), deriv(r2, x)),
        Regex::Seq(r1, r2) => {
            let dr1 = Regex::seq(deriv(r1, x), *r2.clone());
            if nullable(r1) {
                Regex::alt(dr1, deriv(r2, x))
            } else { dr1 }
        }
        Regex::Star(r1) =>
            Regex::seq(deriv(r1, x), Regex::star(*r1.clone())),
    }
}
```

Figure 3.3: `deriv` implementation with six cases.

`deriv` differs from `nullable` in that it borrows its input but returns an owned `Regex`, since a derivative is a new expression. The `if nullable(r1)` implements the case split of (3.6), with branches in order $\mathsf{D}^n_{\text{seq}}$ then D_{seq}. This is the only place where a derivative rule consults nullability and where the output can be larger than the input, since the nullable branch produces an alternation while the other produces a concatenation.

The cloning in `*r2.clone()` is necessary because s appears untouched in both branches of the concatenation case, and the star arm must preserve r^*'s body both derived and underived. This copying mechanism drives expression growth, which justifies simplification after every derivative step.

Example: Growth without simplification. Consider $(x+y)^*$ against the word $\mathtt{xx}$: The expression is a star, so D_* applies with body $r = x + y$:

$$\begin{aligned} (x+y)^*\backslash\mathtt{x} &= ((x+y)\backslash\mathtt{x}) \cdot (x+y)^* && \mathsf{D}_*,\ r = x+y \\ &= (\varepsilon + \emptyset) \cdot (x+y)^* && \mathsf{D}_+,\ \mathsf{D}_{\text{lit}} \end{aligned} \tag{3.7}$$

The expression is now the concatenation $(\varepsilon + \emptyset) \cdot (x+y)^*$, and deriving it consumes the second $\mathtt{x}$. The result is an alternation because $\mathsf{D}^n_{\text{seq}}$ applies with $r = \varepsilon + \emptyset$ and $s = (x+y)^*$, its nullable case being selected since $n(r) = n(\varepsilon + \emptyset) = \text{true}$ by N_+ and N_ε:

$$\begin{aligned} &((\varepsilon + \emptyset) \cdot (x+y)^*)\backslash\mathtt{x} \\ &= ((\varepsilon + \emptyset)\backslash\mathtt{x}) \cdot (x+y)^* + (x+y)^*\backslash\mathtt{x} && \mathsf{D}^n_{\text{seq}} \\ &= (\emptyset + \emptyset) \cdot (x+y)^* + (x+y)^*\backslash\mathtt{x} && \mathsf{D}_\varepsilon \\ &= (\emptyset + \emptyset) \cdot (x+y)^* + (\varepsilon + \emptyset) \cdot (x+y)^* && \mathsf{D}_*,\ \mathsf{D}_+,\ \mathsf{D}_{\text{lit}} \end{aligned} \tag{3.8}$$

After both symbols are consumed, the result is checked for nullability:

$$\begin{aligned} &n((\emptyset + \emptyset) \cdot (x+y)^* + (\varepsilon + \emptyset) \cdot (x+y)^*) \\ &= n((\emptyset + \emptyset) \cdot (x+y)^*) \vee n((\varepsilon + \emptyset) \cdot (x+y)^*) && \mathsf{N}_+ \\ &= (n(\emptyset + \emptyset) \wedge n((x+y)^*)) \vee (n(\varepsilon + \emptyset) \wedge n((x+y)^*)) && \mathsf{N}_{\text{seq}} \\ &= (\text{false} \wedge \text{true}) \vee (\text{true} \wedge \text{true}) && \mathsf{N}_+,\ \mathsf{N}_\emptyset,\ \mathsf{N}_\varepsilon,\ \mathsf{N}_* \\ &= \text{true}. \end{aligned} \tag{3.9}$$

Thus $(x+y)^*$ matches $\mathtt{xx}$.

Simplification. Derivatives can grow unboundedly. Repeatedly deriving $(x+y)^*$ by the same symbol produces expressions of increasing size. This growth results from the nullable branch of $\mathsf{D}^n_{\text{seq}}$, which introduces an alternation as a redundant structure that is carried into subsequent derivatives. ε and $\emptyset$ accumulate without bound:

$$\begin{aligned} & (x+y)^* && 4 \\ \xrightarrow{\mathtt{x}}\ & (\varepsilon + \emptyset) \cdot (x+y)^* && 8 \\ \xrightarrow{\mathtt{x}}\ & (\emptyset + \emptyset) \cdot (x+y)^* \ +\ (\varepsilon + \emptyset) \cdot (x+y)^* && 17 \\ \xrightarrow{\mathtt{x}}\ & (\emptyset + \emptyset) \cdot (x+y)^* \ +\ \left((\emptyset + \emptyset) \cdot (x+y)^* \ +\ (\varepsilon + \emptyset) \cdot (x+y)^*\right) && 26 \end{aligned} \tag{3.10}$$

To counter the growth, expressions are simplified after each derivative step, using identities that preserve the denoted language:

$$\begin{aligned} \emptyset \cdot r &= \emptyset && (\mathsf{S}_1) \\ \varepsilon \cdot r &= r && (\mathsf{S}_2) \\ \emptyset + r &= r && (\mathsf{S}_3) \end{aligned} \tag{3.11}$$

S_1 eliminates dead concatenations with $\emptyset$, S_2 eliminates identity concatenations with ε, and S_3 eliminates $\emptyset$ from alternations. Each holds equally with the unit on the right.

Applying them to (3.8), after each step, returns the expression back to $(x+y)^*$, keeping the size bounded regardless of input length:

$$\begin{aligned}
&(\emptyset + \emptyset) \cdot (x+y)^* + (\varepsilon + \emptyset) \cdot (x+y)^* \\
&= \emptyset \cdot (x+y)^* + \varepsilon \cdot (x+y)^* && \mathsf{S}_3,\ \text{twice} \\
&= \emptyset + (x+y)^* && \mathsf{S}_1,\ \mathsf{S}_2 \\
&= (x+y)^* && \mathsf{S}_3
\end{aligned} \tag{3.12}$$

In Rust. The identities are implemented recursively, applying the rules bottom-up over the expression tree. Each arm of the inner match implements an identity from (3.11).

`simplify` takes ownership of its input and returns simplified subexpressions directly, unlike `deriv`, which borrows and constructs. This model is visible where the two compose:

```
current = simplify(deriv(&current, c));
```

`deriv` borrows `current`, constructs a new expression, and `simplify` takes ownership of that result to return it without additional copying. The recursive calls ensure that all dead code is eliminated before an expression is returned, keeping size bounded at each step. Together, derive-then-simplify forms the core loop of the matcher, transforming and cleaning the expression at every input symbol.

```
pub fn simplify(r: Regex) -> Regex {
    match r {
        Regex::Seq(r, s) => {
            let r = simplify(*r);
            let s = simplify(*s);
            match (&r, &s) {
                (Regex::Phi, _) | (_, Regex::Phi) => Regex::Phi,
                (Regex::Eps, _) => s,
                (_, Regex::Eps) => r,
                _ => Regex::seq(r, s),
            }
        }
        Regex::Alt(r, s) => {
            let r = simplify(*r);
            let s = simplify(*s);
            match (&r, &s) {
                (Regex::Phi, _) => s,
                (_, Regex::Phi) => r,
                _ if r == s => r,
                _ => Regex::alt(r, s),
            }
        }
        Regex::Star(r) => match simplify(*r) {
            Regex::Eps | Regex::Phi => Regex::Eps,
            r => Regex::star(r),
        },
        other => other,
    }
}
```

Figure 3.4: Simplification rules: S_1 and S_2 in concatenation, S_3 in both branches.

The matcher. The implementation can be summarized by the sequence of residual expressions produced while consuming the input. For $w = x_1 x_2 \cdots x_n$, the initial residual is $r_0 = r$, and each subsequent residual is obtained by deriving with respect to the next symbol and simplifying the result:

$$r_0 = r, \qquad r_i = \text{simplify}(r_{i-1} \backslash x_i) \quad (1 \le i \le n). \tag{3.13}$$

The residual r_i denotes exactly the set of suffixes that complete the consumed prefix $x_1 \cdots x_i$ to a word in $L(r)$. Since simplification preserves the represented language, each r_i is equivalent to the unsimplified derivative after consuming that prefix. After the final symbol has been consumed, the matcher accepts when the remaining residual is nullable:

$$w \in L(r) \iff n(r_n) = \text{true}.$$

Each step depends only on the current residual and the next symbol, so the matcher never revisits a prefix and never backtracks into an earlier residual. The cost of one step is the cost of `deriv` on the current residual, which is proportional to its size.

So $r_0 \xrightarrow{x_1} r_1 \xrightarrow{x_2} \cdots \xrightarrow{x_n} r_n$ can be viewed as a path through a lazily constructed DFA. Each residual expression represents a state, and $\xrightarrow{x}$ denotes the transition obtained by taking the derivative with respect to x. Unlike an explicitly constructed DFA, only the states reached by the input are constructed. A residual is an accepting state exactly when it is nullable, so acceptance is determined by whether the final residual is nullable.

In Rust. Figure 3.5 implements (3.13) by composing `deriv`, `simplify`, and `nullable`. The implementation follows the mathematical definition directly. The current residual is transformed at each step, and acceptance is determined after the complete input has been consumed.

```
pub fn match_deriv(input: &str, r: &Regex) -> bool {
    let mut current = r.clone();
    for c in input.chars() {
        current = simplify(deriv(&current, c));
    }
    nullable(&current)
}
```

Figure 3.5: `match_deriv`: the core matcher loop. For each input character, derive the current residual expression and simplify it in place. Accept if the final residual is nullable.

The loop implements the residual sequence. At each iteration, `current` contains the current residual, which is then derived and simplified before processing the next input symbol. Only the current residual is retained, since the previous residual is dropped when `current` is reassigned. In contrast to parsing, where intermediate results may be required to reconstruct a parse tree, Boolean matching requires only the final residual.

The derivative-based matcher avoids the exponential backtracking of naive matching by transforming the expression rather than searching for decompositions. Simplification keeps the resulting expressions compact, preventing the growth shown in (3.10).

Worked Example: Applying the derivative-based matcher Apply `match_deriv` to $r_0 = (a + ab)(b + \varepsilon)$ and $w = \mathtt{ab}$ with the expression written as:

$$r_0 = \mathit{Seq}(s,\, t), \qquad s = a + ab, \qquad t = b + \varepsilon,$$

Derive by `a`. The first factor s is not nullable, so D_{seq} applies to the sequence r_0:

$$r_1 = r_0 \backslash \mathtt{a} = (s \backslash \mathtt{a}) \cdot t \qquad (\mathsf{D}_{\text{seq}}).$$

Since s cannot match the empty word, the second factor t is not touched by this derivative and appears unchanged. To derive s, the alternation rule D_{+} is applied to give:

$$s \backslash \mathtt{a} = (a \backslash \mathtt{a}) + (ab \backslash \mathtt{a}) \qquad (\mathsf{D}_{+}).$$

The two summands are derived separately, each by its own rule:

$$\begin{aligned} a \backslash \mathtt{a} &= \varepsilon && (\mathsf{D}_{\text{lit}}), \\ ab \backslash \mathtt{a} &= (a \backslash \mathtt{a}) \cdot b = \varepsilon \cdot b && (\mathsf{D}_{\text{seq}},\ \mathsf{D}_{\text{lit}}). \end{aligned}$$

Substituting both summands into the alternation and the result into the outer sequence:

$$\begin{aligned} s \backslash \mathtt{a} &= \varepsilon + \varepsilon \cdot b && (\mathsf{D}_{+}), \\ r_0 \backslash \mathtt{a} &= (\varepsilon + \varepsilon \cdot b) \cdot t && (\mathsf{D}_{\text{seq}}). \end{aligned}$$

Then, applying the simplification rule S_2 to the occurrences of $\varepsilon \cdot b$, gives the residual:

$$r_1 = (\varepsilon + b) \cdot t \qquad (\mathsf{S}_2).$$

Derive by `b`. The residual r_1 is again a sequence, but now the first factor $\varepsilon + b$ is nullable, so the nullable sequence rule $\mathsf{D}^{n}_{\text{seq}}$ applies when deriving by b:

$$r_2 = r_1 \backslash \mathtt{b} = \Big((\varepsilon + b) \backslash \mathtt{b}\Big) \cdot t + (t \backslash \mathtt{b}) \qquad (\mathsf{D}^{n}_{\text{seq}}).$$

Since the first factor $\varepsilon + b$ is nullable, it can either consume the character itself, leaving t untouched, or match ε and let t consume it. Both are possible, so $\mathsf{D}^{n}_{\text{seq}}$ produces one summand for each, and D_{+} derives the first factor and t in their respective summands:

$$\begin{aligned} (\varepsilon + b) \backslash \mathtt{b} &= (\varepsilon \backslash \mathtt{b}) + (b \backslash \mathtt{b}) = \emptyset + \varepsilon && (\mathsf{D}_{+},\ \mathsf{D}_{\varepsilon},\ \mathsf{D}_{\text{lit}}), \\ t \backslash \mathtt{b} = (b + \varepsilon) \backslash \mathtt{b} &= (b \backslash \mathtt{b}) + (\varepsilon \backslash \mathtt{b}) = \varepsilon + \emptyset && (\mathsf{D}_{+},\ \mathsf{D}_{\text{lit}},\ \mathsf{D}_{\varepsilon}). \end{aligned}$$

Substituting both summands into the two-summand expression:

$$r_1 \backslash \mathtt{b} = (\emptyset + \varepsilon) \cdot t + (\varepsilon + \emptyset).$$

Applying the simplification rules S_3 and S_2 to the first summand and S_3 to the second, gives the final residual after deriving the second character:

$$r_2 = t + \varepsilon \qquad (\mathsf{S}_3,\ \mathsf{S}_2,\ \mathsf{S}_3).$$

Acceptance. The residual $r_2 = (b + \varepsilon) + \varepsilon$ is nullable by N_{+} and N_{ε}, so $\mathtt{ab} \in L(r_0)$.

3.3 Antimirov Partial Derivatives

The derivative of Section 3.2 combines alternative residuals into a single expression using $+$. Antimirov's partial derivatives instead keep them as a *set* of expressions, one per strand of nondeterminism [Antimirov, 1996]. Writing $pd(r, x)$ for this set:

$$\begin{aligned}
pd(\emptyset, x) &= \{\} && (\mathsf{P}_\emptyset)\\
pd(\varepsilon, x) &= \{\} && (\mathsf{P}_\varepsilon)\\
pd(y, x) &= \begin{cases} \{\varepsilon\} & x = y \\ \{\} & x \neq y \end{cases} && (\mathsf{P}_{\text{lit}})\\
pd(r + s, x) &= pd(r, x) \cup pd(s, x) && (\mathsf{P}_+)\\
pd(r^*, x) &= \{\, smartC(r', r^*) \mid r' \in pd(r, x) \,\} && (\mathsf{P}_*)\\
pd(r \cdot s, x) &= \begin{cases} \{\, smartC(r', s) \mid r' \in pd(r, x) \,\} \cup pd(s, x) & n(r) = \text{true} \\ \{\, smartC(r', s) \mid r' \in pd(r, x) \,\} & n(r) = \text{false} \end{cases} && \begin{matrix}(\mathsf{P}^n_{\text{seq}})\\(\mathsf{P}_{\text{seq}})\end{matrix}
\end{aligned} \tag{3.14}$$

The empty language and ε produce the empty set, since neither has a residual with respect to x: where $\mathsf{D}_\emptyset$ and D_ε returned $\emptyset$ as a node in the result, $\mathsf{P}_\emptyset$ and P_ε return no residual at all. A literal y gives $\{\varepsilon\}$ when $x = y$ and $\{\}$ otherwise. Alternation unions the two residual sets, and for r^* one iteration of r consumes x, with each resulting residual placed in front of r^* so the star may continue. None of these alternatives are combined with $+$; each stays a separate element of the set.

Concatenation depends on nullability for the same reason as D_{seq}. If r is not nullable it must consume x, and each of its residuals is concatenated with the untouched s. If r is nullable, r may instead match ε and let s consume x, in which case $pd(s, x)$ joins the set. Where $\mathsf{D}^n_{\text{seq}}$ combined the two cases with $+$ inside one expression, $\mathsf{P}^n_{\text{seq}}$ keeps them as two elements of one set.

The smart constructor. The concatenation in P_{seq} uses an auxiliary constructor *smartC*. Since each element of $pd(r, x)$ is a separate residual rather than a branch of a sum, *smartC* is applied to each independently. The constructor avoids building $\varepsilon \cdot s$:

$$smartC(r', s) = \begin{cases} s & r' = \varepsilon \\ r' \cdot s & \text{otherwise.} \end{cases} \tag{3.15}$$

In Rust. Figure 3.6 implements *smartC* directly as `smart_seq`. Matching on `r` implements the case split of (3.15): the `Eps` arm returns a clone of `s` and never builds $\varepsilon \cdot s$, while the second arm wraps `r` in front of a cloned `s`

```
pub fn smart_seq(r: Regex, s: &Regex) -> Regex {
    match r {
        Regex::Eps => s.clone(),
        r => Regex::seq(r, s.clone()),
    }
}
```

Figure 3.6: `smart_seq`, the smart constructor *smartC*

Figure 3.7 presents the implementation of (3.14), with each `match` arm corresponding to one case. The `Phi`, `Eps`, and `Lit` variants implement $\mathsf{P}_{\emptyset}$, P_{ε}, and P_{lit} directly, returning `Vec::new()` where the equation gives $\{\}$. `Alt` concatenates the two child vectors, implementing the union of P_{+}.

`Star` maps `smart_seq` over r's own residuals, matching P_{*}. The star's body is derived, and each resulting residual is placed in front of a fresh `Regex::star` so the star may continue. `Regex::star(*r.clone())` clones because `Regex::star` needs an owned `Regex` and `r` is only borrowed, once per residual since the closure runs once for each.

`Seq` implements the case split directly. The `map` applies `smart_seq` to each residual of the left operand, matching P_{seq}, and the guarded `out.extend(pderiv(s, x))` adds $pd(s, x)$ for $\mathsf{P}^{n}_{\text{seq}}$. Although the mathematical definition uses sets, the implementation uses `Vec<Regex>` throughout the recursion and defers deduplication to a single `HashSet` insertion at the top, rather than deduplicating at every `Alt` or `Seq` node on the way down.

Every union in `pderiv` is therefore a vector `extend`, not a set operation. Duplicate residuals reached by different strands are only merged once inserted into the `HashSet` used by the matcher below to keep the residual generation separate from deduplication

```
fn pderiv(r: &Regex, x: char) -> Vec<Regex> {
    match r {
        Regex::Phi => Vec::new(),
        Regex::Eps => Vec::new(),
        Regex::Lit(c) => {
            if *c == x { vec![Regex::Eps] } else { Vec::new() }
        }
        Regex::Alt(r, s) => {
            let mut out = pderiv(r, x);
            out.extend(pderiv(s, x));
            out
        }
        Regex::Seq(r, s) => {
            let mut out: Vec<Regex> = pderiv(r, x)
                .into_iter()
                .map(|r_prime| smart_seq(r_prime, s))
                .collect();
            if nullable(r) {
                out.extend(pderiv(s, x));
            }
            out
        }
        Regex::Star(r) => {
            pderiv(r, x)
                .into_iter()
                .map(|r_prime| smart_seq(r_prime, &Regex::star(*r.clone())))
                .collect()
        }
    }
}
```

Figure 3.7: `pderiv`, implementing (3.14) one arm per case of the equation.

Example: No growth under *pd*. Consider $(x+y)^*$ against $\mathtt{xx}$ under pd.

P_* applies with star body $r = x+y$:

$$\begin{aligned} pd((x+y)^*, \mathtt{x}) &= \{\, smartC(r', (x+y)^*) \mid r' \in pd(x+y, \mathtt{x}) \,\} && \mathsf{P}_* \\ &= \{\, smartC(r', (x+y)^*) \mid r' \in \{\varepsilon\} \,\} && \mathsf{P}_+,\ \mathsf{P}_{\text{lit}} \text{ twice} \\ &= \{(x+y)^*\} && \mathsf{SC},\ r' = \varepsilon. \end{aligned} \tag{3.16}$$

The set is $\{(x+y)^*\}$, so deriving by the second $\mathtt{x}$ repeats the computation, and the final partial derivative accepts ε since $(x+y)^*$ is nullable by N_*. So $\mathtt{xx} \in L((x+y)^*)$.

Equation (3.17) traces the derivative and the partial derivative on the same expression:

$$\begin{aligned} \text{Derivative (unsimplified):} \quad & \underbrace{(x+y)^*}_{4} \xrightarrow{\mathtt{x}} \underbrace{(\varepsilon+\emptyset)(x+y)^*}_{8} \xrightarrow{\mathtt{x}} \underbrace{(\emptyset+\emptyset)(x+y)^*+(\varepsilon+\emptyset)(x+y)^*}_{17} \\ \text{Partial derivative:} \quad & \underbrace{\{(x+y)^*\}}_{4} \xrightarrow{\mathtt{x}} \underbrace{\{(x+y)^*\}}_{4} \xrightarrow{\mathtt{x}} \underbrace{\{(x+y)^*\}}_{4} \end{aligned} \tag{3.17}$$

The unsimplified derivative grows at every step, while the partial derivative stays at one element because $\mathsf{P}_\emptyset$ and P_ε contribute no residual, so there is nothing to simplify.

Finiteness. For a regular expression r, the number of distinct partial derivatives reachable by any input is bounded by $\|r\| + 1$ [Antimirov, 1996, Theorem 3.4, p. 309]. This is a ceiling, not a guarantee that all $\|r\| + 1$ are reached. For $(x+y)^*$, $\|r\| = 2$ the ceiling is 3, yet only $(x+y)^*$ is reachable, since deriving by either $\mathtt{x}$ or $\mathtt{y}$ returns it unchanged.

Ordinary derivatives need equivalence classes to be finite [Brzozowski, 1964, Theorem 4.3], giving $d_R \leq 2^{d_P} - 1$. Simplification removes units and redundant $\emptyset$, but not all equivalent forms. In contrast the partial-derivative bound is linear without simplification.

The matcher. The implementation can be summarized by the sequence of state sets produced while consuming the input. For $w = x_1 x_2 \cdots x_n$, the initial state set is $S_0 = \{r\}$, and each subsequent set is obtained by taking pd of every residual in the previous set and unioning the results:

$$S_0 = \{r\}, \qquad S_i = \bigcup_{r' \in S_{i-1}} pd(r', x_i) \quad (1 \leq i \leq n). \tag{3.18}$$

The state set S_i collects every residual reachable after consuming the prefix $x_1 \cdots x_i$. Unlike the derivative matcher's single residual r_i, no element of S_i grows with the input. The number of elements is bounded by $\|r\| + 1$ as shown above. After the final symbol has been consumed, the matcher accepts when some residual in S_n is nullable:

$$w \in L(r) \iff \exists r_n \in S_n.\ n(r_n) = \text{true}.$$

So $S_0 \xrightarrow{x_1} S_1 \xrightarrow{x_2} \cdots \xrightarrow{x_n} S_n$ can be viewed as a path through a lazily constructed NFA. Each state set represents the set of NFA states reachable after reading the corresponding prefix, and $\xrightarrow{x}$ denotes the transition obtained by taking pd with respect to x and unioning. Unlike an explicitly constructed NFA, only the states reached by the input are constructed. A state set is accepting exactly when it contains at least one nullable residual, so acceptance is determined by whether the final state set contains a nullable residual.

In Rust. Figure 3.8 implements (3.18) with a `HashSet<Regex>`, whose `Hash` and `Eq` implementations collapse residuals reached by different paths. The early return when the set empties follows from construction: an empty set has no states to derive. The final `any(nullable)` checks whether some residual is nullable. Where the derivative matcher holds one growing `Regex`, this holds a set of small residuals that never need simplification.

```
pub fn match_pderiv(input: &str, r: &Regex) -> bool {
    let mut states: HashSet<Regex> = HashSet::new();
    states.insert(r.clone());
    for c in input.chars() {
        let mut next: HashSet<Regex> = HashSet::new();
        for state in &states {
            next.extend(pderiv(state, c));
        }
        if next.is_empty() { return false; }
        states = next;
    }
    states.iter().any(nullable)
}
```

Figure 3.8: The partial-derivative matcher.

Worked Example: Applying the partial-derivative-based matcher Applying `match_pderiv` to $r_0 = (a + ab)(b + \varepsilon)$ and $w = \mathtt{ab}$ with the expression written as:

$$r_0 = \mathit{Seq}(s,\, t), \qquad s = a + ab, \qquad t = b + \varepsilon.$$

Derive by `a`. Since s is not nullable, t is untouched and P_{seq} applies to r_0:

$$S_1 = \{\, \mathit{smartC}(r', t) \mid r' \in \mathit{pd}(s, \mathtt{a}) \,\} \qquad (\mathsf{P}_{\text{seq}}).$$

The alternation rule gives the two branches of s separately:

$$\begin{aligned} \mathit{pd}(a, \mathtt{a}) &= \{\varepsilon\} && (\mathsf{P}_{\text{lit}}),\\ \mathit{pd}(ab, \mathtt{a}) &= \{\mathit{smartC}(\varepsilon, b)\} = \{b\} && (\mathsf{P}_{\text{seq}}), \end{aligned}$$

So $\mathit{pd}(s, \mathtt{a}) = \{\varepsilon, b\}$. Applying smartC to each element against t prepends that element to t, with ε absorbed so that $\varepsilon \cdot t$ is never built:

$$S_1 = \{\, t,\ b \cdot t \,\} = \{\, b + \varepsilon,\ \ b\,(b + \varepsilon) \,\}.$$

Derive by `b`. Each state in S_1 is derived separately and the results unioned.

For the first state $t = b + \varepsilon$:

$$\mathit{pd}(b + \varepsilon, \mathtt{b}) = \mathit{pd}(b, \mathtt{b}) \cup \mathit{pd}(\varepsilon, \mathtt{b}) = \{\varepsilon\} \cup \{\} = \{\varepsilon\} \qquad (\mathsf{P}_{+},\ \mathsf{P}_{\text{lit}},\ \mathsf{P}_{\varepsilon}).$$

For the second state $b \cdot t = b\,(b + \varepsilon)$:

$$\mathit{pd}\Big(b\,(b + \varepsilon), \mathtt{b}\Big) = \{\, \mathit{smartC}(\varepsilon,\, b + \varepsilon) \,\} = \{b + \varepsilon\} \qquad (\mathsf{P}_{\text{seq}}).$$

Only the left factor b is derived, as it is a non-nullable literal. The union gives:

$$S_2 = \{\varepsilon\} \cup \{b + \varepsilon\} = \{\, \varepsilon,\ \ b + \varepsilon \,\}.$$

Acceptance. Both states in S_2 are nullable, ε by N_{ε} and $b + \varepsilon$ by N_{+} and N_{ε}, so `match_pderiv("ab",` r_0`)` returns `true` and $\mathtt{ab} \in L(r_0)$, agreeing with Section 3.2.

4 Matching and Parsing

Chapter 3 established matching with a Boolean answer: $w \in L(r)$. A matcher only answers whether the input is accepted by the regular expression. Parsing identifies the evidence for the match and *parse trees* act as their structured witnesses.

4.1 Matching: A Boolean Answer

Consider $r = (a + ab)(b + \varepsilon)$ and $w = \texttt{ab}$. The matcher can determine $w \in L(r)$ using the algorithms of Chapter 3. There are, however, two ways in which r can match w:

$$\underbrace{a}_{a+ab}\ \underbrace{b}_{b+\varepsilon} \quad \text{and} \quad \underbrace{\texttt{ab}}_{a+ab}\ \underbrace{\varepsilon}_{b+\varepsilon} \tag{4.1}$$

Both choices establish the same membership result, but assign the input differently to the subexpressions of r. The Boolean result alone cannot distinguish these cases, so the matcher cannot reveal which decomposition was chosen.

Parsing makes this additional structure explicit, by showing which alternatives were chosen and which parts of the input were consumed by the corresponding subexpressions.

4.2 Parsing: Parse Trees as Matching Evidence

Parse Tree. A *parse tree* provides a structured witness for such a derivation. Sulzmann and Lu view regular expressions as types and *parse trees* as their values [Sulzmann and Lu, 2014]. The judgment $\vdash v : r$ states that v is a valid parse tree for r, recording which alternatives were chosen and how each part of the input was consumed during matching:

$$\frac{}{\vdash () : \varepsilon}\ (\mathsf{T}_\varepsilon) \qquad \frac{\vdash v_1 : r_1}{\vdash \mathit{Left}\ v_1 : r_1 + r_2}\ (\mathsf{T}^L_+)$$

$$\frac{l \in \Sigma}{\vdash l : l}\ (\mathsf{T}_{\text{lit}}) \qquad \frac{\vdash v_2 : r_2}{\vdash \mathit{Right}\ v_2 : r_1 + r_2}\ (\mathsf{T}^R_+) \tag{4.2}$$

$$\frac{\vdash v_1 : r_1 \qquad \vdash v_2 : r_2}{\vdash (v_1, v_2) : r_1 r_2}\ (\mathsf{T}_{\text{seq}}) \qquad \frac{}{\vdash [\,] : r^*}\ (\mathsf{T}^0_*)$$

$$\frac{\vdash v : r \qquad \vdash vs : r^*}{\vdash (v : vs) : r^*}\ (\mathsf{T}_*)$$

In Rust. Figure 4.1 shows the parse-tree type, with one variant for each constructor.

```
#[derive(Debug, Clone, PartialEq, Eq)]
pub enum ParseTree {
    Empty,                                  // ()
    Char(char),                             // l
    Pair(Box<ParseTree>, Box<ParseTree>),   // (v1, v2)
    Left(Box<ParseTree>),                   // Left v
    Right(Box<ParseTree>),                  // Right v
    Star(Vec<ParseTree>),                   // [v1, ..., vn]
}
```

Figure 4.1: The parse tree type, representing the structure of a parse.

The definitions above translate directly to Rust, except for the star case. $\mathsf{T}\varepsilon$ corresponds to `Empty` for the empty parse, T_{lit} to `Char` for a literal, T_{seq} to `Pair` for a concatenation of parse trees, and T_{+}^{L} and T_{+}^{R} to `Left` and `Right`, recording the chosen branch of an alternation. Both star rules are represented by `Star(vec![])`.

Flattening. The typing judgment describes the structure of a match, but a parse tree must also be related to the word that it matches. For this purpose, they define a *flattening* function $|v|$ that maps a parse tree to the word represented by its structure:

$$\begin{aligned} |()| &= \varepsilon, & |(v_1, v_2)| &= |v_1||v_2|, \\ |l| &= l, & |\mathit{Left}\ v| &= |v|, \\ |[]| &= \varepsilon & |\mathit{Right}\ v| &= |v|, \\ & & |v : vs| &= |v||vs|. \end{aligned} \tag{4.3}$$

The flattening function directly connects parse trees to $L(r)$, and thus validates that a parse tree witnesses membership in $L(r)$, serving as an correctness check:

$$L(r) = \{\, |v| \mid\mid\!\vdash v : r \,\}. \tag{4.4}$$

In Rust. Figure 4.2 shows how `flatten` recursively traverses the parse tree and concatenates its leaf characters. The function borrows the tree to allow subsequent inspection and comparison. The `Pair` case concatenates the results of flattening both children, while `Left` and `Right` delegate to their child, and `Star` concatenates the results for each element.

```
pub fn flatten(v: &ParseTree) -> String {
    match v {
        ParseTree::Empty        => String::new(),
        ParseTree::Char(c)      => c.to_string(),
        ParseTree::Pair(v1, v2) => format!("{}{}", flatten(v1), flatten(v2)),
        ParseTree::Left(v)      => flatten(v),
        ParseTree::Right(v)     => flatten(v),
        ParseTree::Star(vs)     => vs.iter().map(flatten).collect(),
    }
}
```

Figure 4.2: The `flatten` function extracts the matched word from a parse tree.

Unambiguous example. Consider $r = ab$ and $w = \texttt{ab}$. Since each subexpression is a literal, the rule T_{lit} gives each literal a unique witness. Then T_{seq} requires the parse tree for a to be followed by the parse tree for b, combining them into a parse tree for ab:

$$\frac{a \in \Sigma}{\vdash a : a}\,(\mathsf{T}_{\text{lit}}) \qquad \frac{b \in \Sigma}{\vdash b : b}\,(\mathsf{T}_{\text{lit}}) \qquad \frac{\vdash a : a \qquad \vdash b : b}{\vdash (a, b) : ab}\,(\mathsf{T}_{\text{seq}}) \tag{4.5}$$

The parse tree $v = (a, b)$ flattens to $|(a, b)| = \texttt{ab}$.

Ambiguous Example. Recall $r = (a + ab)(b + \varepsilon)$ and $w = \texttt{ab}$ from Section 4.1. The word admits two distinct decompositions, corresponding to different parse trees.

The first decomposition uses the right alternatives of both subexpressions:

$$\underbrace{\frac{\dfrac{a \in \Sigma}{\vdash a : a}\,(\mathsf{T}_{\text{lit}}) \qquad \dfrac{b \in \Sigma}{\vdash b : b}\,(\mathsf{T}_{\text{lit}})}{\vdash (a, b) : ab}\,(\mathsf{T}_{\text{seq}})}{\vdash \mathit{Right}(a, b) : a + ab}\,(\mathsf{T}^R_+)}_{a + ab} \qquad \underbrace{\frac{\dfrac{b \in \Sigma}{\vdash b : b}\,(\mathsf{T}_{\text{lit}}) \qquad \vdash () : \varepsilon\,(\mathsf{T}_\varepsilon)}{\vdash \mathit{Right}() : b + \varepsilon}\,(\mathsf{T}^R_+)}_{b + \varepsilon}$$

By T_{seq}, these derivations combine into the parse tree:

$$\frac{\dfrac{\vdash a : a \qquad \vdash b : b}{\vdash (a, b) : ab}}{\vdash \mathit{Right}(a, b) : a + ab} \qquad \frac{\vdash () : \varepsilon}{\vdash \mathit{Right}() : b + \varepsilon} \tag{4.6}$$

$$v_1 = \Big(\mathit{Right}(a, b), \mathit{Right}()\Big), \qquad |v_1| = \texttt{ab}.$$

The second decomposition uses the left alternatives of both subexpressions:

$$\underbrace{\frac{\dfrac{a \in \Sigma}{\vdash a : a}\,(\mathsf{T}_{\text{lit}})}{\vdash \mathit{Left}\, a : a + ab}\,(\mathsf{T}^L_+)}_{a + ab} \qquad \underbrace{\frac{\dfrac{b \in \Sigma}{\vdash b : b}\,(\mathsf{T}_{\text{lit}})}{\vdash \mathit{Left}\, b : b + \varepsilon}\,(\mathsf{T}^L_+)}_{b + \varepsilon}$$

By T_{seq}, these derivations combine into the second parse tree:

$$\frac{\vdash a : a}{\vdash \mathit{Left}\, a : a + ab} \qquad \frac{\vdash b : b}{\vdash \mathit{Left}\, b : b + \varepsilon} \tag{4.7}$$

$$v_2 = \Big(\mathit{Left}\, a, \mathit{Left}\, b\Big), \qquad |v_2| = \texttt{ab}.$$

Both trees are valid parse trees for the same expression and word, yet they differ in structure. The typing judgment alone does not determine which parse tree should be selected. A *disambiguation policy* specifies this choice.

5 Derivative-Based Parser

Chapter 3 showed how a Brzozowski-derivative matcher determines whether $w \in L(r)$ and Chapter 4 showed that a matching word may admit multiple valid parse trees. A parser must therefore select a single parse tree according to a disambiguation policy.

Following Sulzmann and Lu [Sulzmann and Lu, 2014], this chapter presents two derivative-based methods for constructing POSIX parse trees. The direct method performs a forward derivative pass followed by a backward pass for tree reconstruction. The bit-coded method records parse-tree information during the forward pass, avoiding a separate backward pass.

5.1 The Standard Derivative-Based Parser

Consider the example $r = (a + ab)(b + \varepsilon)$ with input $w = \texttt{ab}$. Chapter 4 showed that it is ambiguous, as the word admits two parse trees. The question here is whether a parse tree can be recovered from the final derivative:

$$\begin{aligned} r &\xrightarrow{a} (\varepsilon + \varepsilon b)(b + \varepsilon) \\ &\xrightarrow{b} (\emptyset + (\emptyset b + \varepsilon))(b + \varepsilon) + (\varepsilon + \emptyset). \end{aligned} \tag{5.1}$$

The derivation was already computed in Section 3.2, shown here unsimplified. The final derivative is nullable, so $w \in L(r)$ and both disjuncts accept the empty word. Its first disjunct, $(\emptyset + (\emptyset b + \varepsilon))(b + \varepsilon)$, reflects $ab \cdot \varepsilon$, and its second, $(\varepsilon + \emptyset)$, reflects $a \cdot b$.

Since a derivative step only records the remaining input language and not how consumed characters were matched, the final derivative alone does not show which decomposition was selected, and cannot be used to recover the parse tree. However, the derivation sequence itself contains this information. Each step consumed one character and is invertible given that character, so reversing the steps recovers the tree, as shown in Figure 5.1:

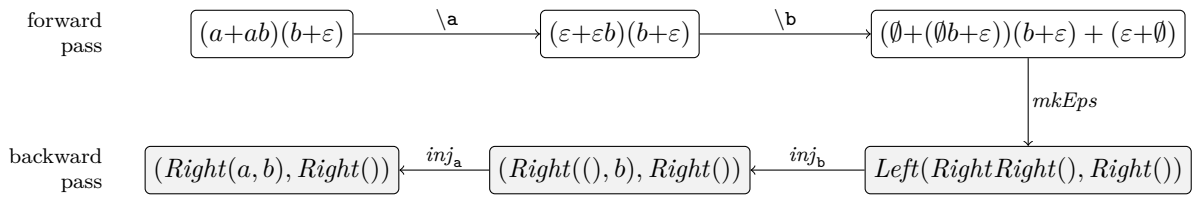


Figure 5.1: Overview of the standard derivative-based parser on the ambiguous example. The forward pass computes derivatives, while the backward pass uses *mkEps* to construct a parse tree for the final nullable derivative and *inj* to reconstruct the original parse tree.

The two-pass pipeline. The direct construction uses two passes. The forward pass applies `deriv` (Figure 3.3) and `nullable` (Figure 3.2) to process the input.

The backward pass applies two new functions to walk the derivation sequence in reverse and reconstruct the parse tree. First *mkEps* builds a parse tree for the empty word from the final nullable expression. Then injection steps are applied, with *inj* reinserting one consumed character into the parse tree, moving in w from right to left. Each injection step takes the current parse tree, the corresponding derivative step, and its consumed character, and returns the parse tree for the preceding expression.

As shown in Figure 5.1 after the first injection of b, the tree is $(\mathit{Right}((), b), \mathit{Right}())$, where $()$ marks the position still to be filled. After the second injection of a, that position is replaced by the subtree $\mathit{Right}(a, b)$, giving the complete parse tree $(\mathit{Right}(a, b), \mathit{Right}())$.

In the following the operations for the standard derivative parser are shown in more detail.

5.1.1 Building the Empty-Word Tree

The backward pass begins with *mkEps*, which reads a parse tree for the empty word off the final nullable expression. The function follows [Sulzmann and Lu, 2014, Fig. 3] and is defined recursively:

$$\begin{aligned}
\mathit{mkEps}_{\varepsilon} &= () && (\mathsf{E}_{\varepsilon}) \\
\mathit{mkEps}_{r^*} &= [\,] && (\mathsf{E}_{*}) \\
\mathit{mkEps}_{r_1 r_2} &= (\mathit{mkEps}_{r_1}, \mathit{mkEps}_{r_2}) && (\mathsf{E}_{\mathrm{seq}}) \\
\mathit{mkEps}_{r_1+r_2} &= \begin{cases} \mathit{Left}\ \mathit{mkEps}_{r_1} & \text{if } n(r_1) \quad (\mathsf{E}^L_+) \\ \mathit{Right}\ \mathit{mkEps}_{r_2} & \text{otherwise} \quad (\mathsf{E}^R_+) \end{cases}
\end{aligned} \tag{5.2}$$

The first three rules follow from the structure of the expression: $(\mathsf{E}_{\varepsilon})$ gives ε an empty parse tree, (E_{*}) treats a star matching the empty word as zero iterations, and $(\mathsf{E}_{\mathrm{seq}})$ matches a concatenation against ε only when both factors do. No rules are needed for $\emptyset$ or literals, since neither can match the empty word. The alternation rule is where disambiguation occurs. If exactly one of r_1 and r_2 is nullable, that branch is the only possible witness. If both are, (E^L_+) breaks the tie by preferring the left one, following POSIX's leftmost-longest policy [Sulzmann and Lu, 2014]. In the example, the final expression is:

$$(\emptyset + (\emptyset b + \varepsilon))(b + \varepsilon) + (\varepsilon + \emptyset).$$

At top-level $+$, both summands are nullable, so (E^L_+) applies and picks the left one. Within the left summand's first factor $\emptyset + (\emptyset b + \varepsilon)$, only the right alternative is nullable, so (E^R_+) applies, and within it only the right sub-alternative ε is nullable, so (E^R_+) applies again, contributing $\mathit{RightRight}()$. The second factor $b + \varepsilon$ contributes $\mathit{Right}()$ via (E^R_+), giving:

$$\mathit{Left}(\mathit{RightRight}(), \mathit{Right}()).$$

Since *mkEps* handles disambiguation, the forward pass cannot simplify. It must keep the branch that consumed a, or injection cannot recover $\mathit{Right}(a, b)$. Simplifying gives:

$$r \xrightarrow{a} (\varepsilon + \varepsilon b)(b + \varepsilon) \xrightarrow{\text{simp}} (\varepsilon + b)(b + \varepsilon) \xrightarrow{b} (\emptyset + \varepsilon)(b + \varepsilon) + (\varepsilon + \emptyset) \xrightarrow{\text{simp}} (b + \varepsilon) + \varepsilon,$$

This still witnesses $w \in L(r)$, but $(\emptyset + \varepsilon) \to \varepsilon$ and $(\varepsilon + \emptyset) \to \varepsilon$ erase the pair structure holding both characters. *mkEps* on $(b + \varepsilon) + \varepsilon$ gives $\mathit{Left}(\mathit{Right}(()))$ and injecting b fills that one empty leaf, leaving none for a, so $\mathit{Right}(a, b)$ is unrecoverable.

In Rust. The recursive definition of *mkEps* (5.2) is implemented directly in Rust, as shown in Figure 5.2, following the same cases for each expression form.

```
pub fn mk_eps(r: &Regex) -> ParseTree {
    match r {
        Regex::Eps          => ParseTree::Empty,
        Regex::Star(_)      => ParseTree::Star(Vec::new()),
        Regex::Seq(r1, r2) => ParseTree::Pair(
                               Box::new(mk_eps(r1)), Box::new(mk_eps(r2))),
        Regex::Alt(r1, r2) =>
            if nullable(r1)   {ParseTree::Left(Box::new(mk_eps(r1)))}
            else              {ParseTree::Right(Box::new(mk_eps(r2)))},
        Regex::Lit(c)       => panic!("mk_eps called on Lit('{}')", c),
        Regex::Phi          => panic!("mk_eps called on Phi"),
    }
}
```

Figure 5.2: `mk_eps`: construct a parse tree for the empty word.

The expression r is borrowed rather than consumed because the backward pass needs it again during injection. The alternation arm encodes disambiguation as a conditional, examining `r2` only when `nullable(r1)` is false. The panic arms enforce the precondition that *mkEps* applies only to nullable expressions, rejecting the non-nullable $\emptyset$ and literals.

5.1.2 Reinserting Characters via Injection

After *mkEps* constructed the parse tree v for the empty input, the reconstruction starts from this initial tree and proceeds backwards through the derivation sequence. Each application of *inj* takes the current tree, the corresponding derivative step, and its consumed character, returning a tree for the preceding expression. *inj* recursively follows the structure of the residual until it reaches the subexpression containing the literal that consumed the character. It reinserts the character there and then unwinds to rebuild the surrounding tree. After all characters have been reinserted, the parse tree for the original expression and input is recovered. The injection function is defined by the following rules [Sulzmann and Lu, 2014, Fig. 3]:

$$
\begin{aligned}
inj_{x\backslash x}() &= x && (\mathsf{I}_{\text{lit}}) \\
inj_{(r_1+r_2)\backslash x}(\mathit{Left}\ v_1) &= \mathit{Left}\ (inj_{r_1\backslash x}\ v_1) && (\mathsf{I}^L_+) \\
inj_{(r_1+r_2)\backslash x}(\mathit{Right}\ v_2) &= \mathit{Right}\ (inj_{r_2\backslash x}\ v_2) && (\mathsf{I}^R_+) \\
inj_{r^*\backslash x}(v, vs) &= (inj_{r\backslash x}\ v) : vs && (\mathsf{I}_*) \\
inj_{(r_1 r_2)\backslash x}(v_1, v_2) &= (inj_{r_1\backslash x}\ v_1,\ v_2) && (\mathsf{I}_{\text{seq}}),\ n(r_1) = \text{false} \\
inj_{(r_1 r_2)\backslash x}(\mathit{Left}(v_1, v_2)) &= (inj_{r_1\backslash x}\ v_1,\ v_2) && (\mathsf{I}^L_{\text{seq}}),\ n(r_1) = \text{true} \\
inj_{(r_1 r_2)\backslash x}(\mathit{Right}\ v_2) &= (mkEps_{r_1},\ inj_{r_2\backslash x}\ v_2) && (\mathsf{I}^R_{\text{seq}}),\ n(r_1) = \text{true}
\end{aligned}
\tag{5.3}
$$

Rules $(\mathsf{I}_{\text{lit}})$ and (I_*) invert their derivative rules directly, reconstructing the consumed character and the current iteration, the latter prepended to the remaining ones. Since the consumed character lies in exactly one branch of an alternation, the incoming tree is either *Left* v_1 or *Right* v_2. Rule (I^L_+) recurses into v_1 and rebuilds the *Left* node, and (I^R_+) recurses

into v_2 and rebuilds the *Right* node. The concatenation rules cover the three cases of the sequence derivative. If $n(r_1) = \text{false}$, r_1 cannot match ε and must consume x, so (I_{seq}) recurses into v_1. If $n(r_1) = \text{true}$, the derivative recorded which component consumed x as a tag: ($\mathsf{I}^L_{\text{seq}}$) handles an incoming $\mathit{Left}(v_1, v_2)$ tag and recurses into v_1, while ($\mathsf{I}^R_{\text{seq}}$) handles a *Right* v_2 tag by reconstructing r_1's empty match with mkEps_{r_1} and recursing into v_2. Since $\emptyset$ consumes no characters, *inj* is never applied with $\emptyset$ as the residual.

```
pub fn inject(r: &Regex, l: char, v: ParseTree) -> ParseTree {
    match r {
        Regex::Lit(c) => { assert!(*c == l);
            ParseTree::Char(l) }
        Regex::Alt(r1, r2) => match v {
            ParseTree::Left(v1) =>
                ParseTree::Left(Box::new(inject(r1, l, *v1))),
            ParseTree::Right(v2) =>
                ParseTree::Right(Box::new(inject(r2, l, *v2))),
            _ => panic!("inject/Alt: want Left|Right"),
        },
        Regex::Star(r1) => match v {
            ParseTree::Pair(v1, vs) => {
                let mut it = vec![inject(r1, l, *v1)];
                if let ParseTree::Star(rest) = *vs { it.extend(rest); }
                ParseTree::Star(it)
            }
            _ => panic!("inject/Star: want Pair"),
        },
        Regex::Seq(r1, r2) => if !nullable(r1) {
            match v {
                ParseTree::Pair(v1, v2) => ParseTree::Pair(
                    Box::new(inject(r1, l, *v1)), v2),
                _ => panic!("inject/Seq: want Pair"),
            }
        } else {
            match v {
                ParseTree::Left(vp) => match *vp {
                    ParseTree::Pair(v1, v2) => ParseTree::Pair(
                        Box::new(inject(r1, l, *v1)), v2),
                    o => panic!("inject/SeqL: want Pair, got {:?}", o),
                },
                ParseTree::Right(v2) => ParseTree::Pair(
                    Box::new(mk_eps(r1)),
                    Box::new(inject(r2, l, *v2))),
                _ => panic!("inject/Seq: want Left|Right"),
            }
        },
        Regex::Eps | Regex::Phi => panic!("inject called on Eps/Phi"),
    }
}
```

Figure 5.3: `inject`: reinsert a character into a parse tree.

In Rust. `inject(r, l, v)` takes three arguments because *inj* is indexed by both an expression and a character. Figure 5.3 shows the complete implementation.

The function matches on the expression `r` first and on the tree `v` second, following the order in which (5.3) indexes its rules. This order matters because a tree tag is interpreted relative to the expression that produced it. Under an alternation, `Left` selects the left disjunct, while under a nullable concatenation it indicates that the derivative continued inside the left operand. The `Lit` arm reconstructs a leaf after asserting that the consumed character matches, and the `Star` arm inverts (I_*), reinserting the character into the current iteration and prepending it to the remaining ones.

The expression `r` is borrowed and read only, while the tree `v` is consumed and decomposed. In the concatenation arms where the right operand was not derived, `v2` moves directly into the new pair, since (5.3) leaves it unchanged. The `Right` arm of concatenation is the only concatenation case that synthesizes content, calling `mk_eps(r1)` because the derivative skipped `r1` and the forward pass produced no tree for that half of the pair.

Calling `nullable(r1)` in the `Seq` arm recomputes a decision the forward pass already made when it chose between D_{seq} and $\mathsf{D}^n_{\text{seq}}$. However, carrying this as a flag at every step would require additional storage, while `nullable` is cheap to evaluate on demand.

5.1.3 Parser

The standard parser has two passes. The forward pass stores each residual and the backward pass applies *mkEps* then *inj* in reverse. The split is needed because *inj* requires the pre-character residual, which only the stored sequence provides:

$$\begin{aligned} r_0 &= r, & r_i &= r_{i-1}\backslash x_i & (1 \le i \le n),\\ v_n &= \mathit{mkEps}(r_n), & v_i &= \mathit{inj}_{r_{i-1}\backslash x_i}(x_i,\ v_{i+1}) & (0 \le i < n). \end{aligned} \tag{5.4}$$

Every r_i is kept because each injection step needs r_{i-1}; re-deriving it would duplicate the forward pass. The vector holds the entire sequence $r_0, \ldots, r_n$, not just the final residual.

In Rust. Figure 5.4 implements (5.4). The forward loop fills the `Vec expressions` with $r_0, \ldots, r_n$, the nullability check returns early if r_n is not nullable, and the backward loop walks the vector in reverse, applying `inject` to each residual and its character.

```
pub fn parse_deriv_std_loop(input: &str, r: &Regex) -> Option<ParseTree>{
    let chars: Vec<char> = input.chars().collect();
    let mut expressions = Vec::with_capacity(chars.len() + 1);
    expressions.push(r.clone());
    for &c in &chars {
        expressions.push(deriv(expressions.last().unwrap(), c));
    }
    if !nullable(expressions.last().unwrap()) { return None; }
    let mut tree = mk_eps(expressions.last().unwrap());
    for i in (0..chars.len()).rev() {
        tree = inject(&expressions[i], chars[i], tree);
    }
    Some(tree)
}
```

Figure 5.4: `parse_deriv_std_loop`: the standard parser.

5.1.4 Worked Example: Applying the Standard Parser

Apply the standard parser to the ambiguous example $r_0 = (a + ab)(b + \varepsilon)$ on input $\mathtt{ab}$.

Forward pass. There are two derivative steps, one per input symbol. After deriving $\mathtt{a}$, the left factor $\varepsilon + \varepsilon b$ is nullable, so the second derivation on $\mathtt{b}$ uses $\mathsf{D}^n_{\text{seq}}$ rather than D_{seq}:

$$\begin{aligned} r_0 = (a + ab)(b + \varepsilon) &\xrightarrow[\mathsf{D}_{\text{seq}}]{a} r_1 = (\varepsilon + \varepsilon b)(b + \varepsilon) \\ &\xrightarrow[\mathsf{D}^n_{\text{seq}}]{b} r_2 = \underbrace{(\emptyset + (\emptyset b + \varepsilon))(b + \varepsilon)}_{X} + \underbrace{(\varepsilon + \emptyset)}_{Y}. \end{aligned}$$

The final expression $r_2 = X + Y$ is nullable, and both disjuncts accept the empty word. X records $ab \cdot \varepsilon$ (the first factor matched ab, the second ε); Y records $a \cdot b$ (the first matched a, the second b).

Backward pass: mkEps. The backward pass begins by building the parse tree for the empty word, $mkEps(r_2)$. Since $r_2 = X + Y$ and both disjuncts are nullable, E^L_+ selects the left one:

$$mkEps(r_2) = Left\Big(mkEps(X)\Big) \qquad (\mathsf{E}^L_+),$$

where $X = (\emptyset + (\emptyset b + \varepsilon))(b + \varepsilon)$. Since X is a concatenation, E_{seq} combines the trees for its two factors:

$$mkEps(X) \xrightarrow{\mathsf{E}_{\text{seq}}} \Big(mkEps(\emptyset + (\emptyset b + \varepsilon)),\ mkEps(b + \varepsilon)\Big).$$

The first factor selects ε by applying E^R_+ twice, since $\emptyset$ and $\emptyset b$ are not nullable; the second factor selects ε by E^R_+, since b is not nullable. These give

$$\begin{aligned} mkEps(\emptyset + (\emptyset b + \varepsilon)) &\xrightarrow{\mathsf{E}^R_+} Right\ mkEps(\emptyset b + \varepsilon) \\ &\xrightarrow{\mathsf{E}^R_+} Right\ Right\ mkEps(\varepsilon) \\ &\xrightarrow{\mathsf{E}_\varepsilon} Right\ Right\ () \end{aligned} \qquad \begin{aligned} mkEps(b + \varepsilon) &\xrightarrow{\mathsf{E}^R_+} Right\ mkEps(\varepsilon) \\ &\xrightarrow{\mathsf{E}_\varepsilon} Right\ () \end{aligned}$$

and hence

$$v_2 = mkEps(r_2) = Left\Big(Right\ Right\ (), Right\ ()\Big).$$

Backward pass: Inj. Reinsertion begins with $\mathtt{b}$ and maps v_2 to a tree for $r_1 = (\varepsilon + \varepsilon b)(b + \varepsilon)$. The *Left* tag of v_2 sends $\mathsf{I}^L_{\text{seq}}$ into the first factor $\varepsilon + \varepsilon b$, leaving the second factor's tree $Right\ ()$ unchanged. Inside $\varepsilon + \varepsilon b$, I^R_+ selects the ab branch; since ε is nullable, $\mathsf{I}^R_{\text{seq}}$ supplies the empty tree for ε before I_{lit} reinserts $\mathtt{b}$:

$$\begin{aligned} inj_{r_1 \backslash \mathtt{b}}(v_2) = inj_{r_1 \backslash \mathtt{b}}\Big(\underbrace{Left(Right\ Right\ (),\ Right\ ())}_{v_2}\Big) \\ \xrightarrow{\mathsf{I}^L_{\text{seq}}} \Big(inj_{(\varepsilon + \varepsilon b) \backslash \mathtt{b}}(Right\ Right\ ()),\ Right\ ()\Big) \\ \xrightarrow{\mathsf{I}^R_+} \Big(Right\ inj_{\varepsilon b \backslash \mathtt{b}}(Right\ ()),\ Right\ ()\Big) \\ \xrightarrow{\mathsf{I}^R_{\text{seq}}} \Big(Right\ (mkEps(\varepsilon),\ inj_{b \backslash \mathtt{b}}()),\ Right\ ()\Big) \\ \xrightarrow{\mathsf{E}_\varepsilon,\ \mathsf{I}_{\text{lit}}} \underbrace{\Big(Right((), b),\ Right\ ()\Big)}_{v_1}. \end{aligned}$$

Reinsertion of $\mathtt{a}$ maps v_1 to a tree for $r_0 = (a + ab)(b + \varepsilon)$. The outer sequence's first factor $a+ab$ is not nullable, so I_{seq} fires directly on v_1's first component, leaving the second component $\mathit{Right}\,()$ untouched. That first component, $\mathit{Right}((), b)$, carries a Right tag, so I^R_+ selects ab. Inside $ab = \mathit{Seq}(a, b)$ the first factor a is again not nullable, so I_{seq} fires once more and recurses into a, where I_{lit} reinserts the character:

$$
\begin{aligned}
\mathit{inj}_{r_0\backslash\mathtt{a}}(v_1) &= \mathit{inj}_{r_0\backslash\mathtt{a}}\Big(\underbrace{\big(\mathit{Right}((), b),\ \mathit{Right}\,()\big)}_{v_1}\Big) \\
&\xrightarrow{\mathsf{I}_{\text{seq}}} \Big(\mathit{inj}_{(a+ab)\backslash\mathtt{a}}(\mathit{Right}((), b)),\ \mathit{Right}\,()\Big) \\
&\xrightarrow{\mathsf{I}^R_+} \Big(\mathit{Right}\,\mathit{inj}_{ab\backslash\mathtt{a}}(((), b)),\ \mathit{Right}\,()\Big) \\
&\xrightarrow{\mathsf{I}_{\text{seq}}} \Big(\mathit{Right}\,(\mathit{inj}_{a\backslash\mathtt{a}}(()),\ b),\ \mathit{Right}\,()\Big) \\
&\xrightarrow{\mathsf{I}_{\text{lit}}} \underbrace{\big(\mathit{Right}(a, b),\ \mathit{Right}\,()\big)}_{v_0}.
\end{aligned}
$$

The implementation stores r_0, r_1, and r_2 in its vector. One entry per character, plus the initial expression, kept for the backward pass to reconstruct from. The tree it returns is $(\mathit{Right}(a, b), \mathit{Right}())$, which flattens to $\mathtt{ab}$ and matches the input.

5.2 The Bit-Coded Derivative-Based Parser

The direct construction of Sulzmann and Lu [Sulzmann and Lu, 2014] reconstructs the tree only after the full input, since each injection step needs the preceding expression. Their bit-coded optimization records this information during a single pass instead.

Each subexpression carries the bit-sequence it has committed to so far, written $bs@r$. P and Q denote the annotated factors for $r_0 = (a + ab)(b + \varepsilon)$. Deriving $w = \mathtt{ab}$ gives:

$$
\begin{aligned}
ri_0 &= [\,]@\Big(\underbrace{[0]@a \oplus [1]@(a\,b)}_{P} \cdot \underbrace{[0]@b \oplus [1]@\varepsilon}_{Q}\Big) \\
&\xrightarrow{a} ri_1 = [\,]@\Big(\underbrace{[0]@\varepsilon \oplus [1]@b}_{P'} \cdot Q\Big) \\
&\xrightarrow{b} ri_2 = [\,]@\Big([1]@Q \ \oplus\ [0, 0]@\varepsilon\Big).
\end{aligned}
\tag{5.5}
$$

Figure 5.5 shows this derivation together with the extraction step it feeds into:

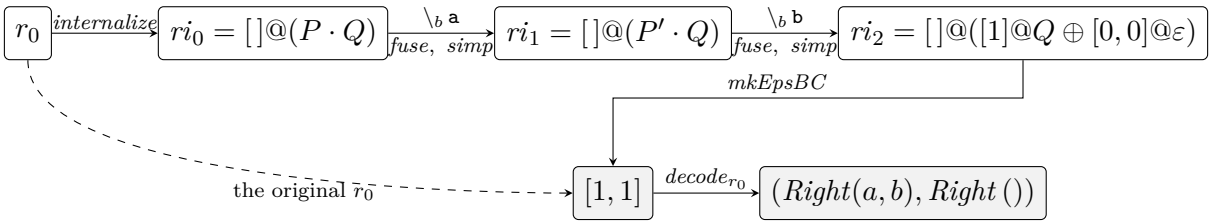


Figure 5.5: The bit-coded parser on the ambiguous example. *internalize* introduces the bits, derivative threads them forward, and *mkEpsBC* and $decode_{r_0}$ extract and read them against r_0.

The bit-coded pipeline. *internalize* tags each alternation branch with a bit that records its preference, producing $ri_0 = []@(P \cdot Q)$ with $P = [0]@a \oplus [1]@(a\,b)$ and $Q = [0]@b \oplus [1]@\varepsilon$, where 0 selects the left branch and 1 the right.

Deriving by `a` extracts P's leading a, giving P' while Q is untouched; deriving by `b` consumes the rest. Since P''s ε branch is nullable, both decompositions survive: in the first, the derivative continues inside P', which finishes and folds its bit 1 onto Q; in the second, Q matches `b` instead, and P''s completion bit fuses with Q's match into $[0,0]@\varepsilon$. This yields $ri_2 = []@([1]@Q \oplus [0,0]@\varepsilon)$.

Unlike the standard construction, the bit-coded construction requires simplification to prevent unbounded growth of annotated expressions. The function *simp*, from [Sulzmann and Lu, 2014, Fig. 6], discards dead branches and collapses finished subexpressions, while *fuse* preserves bits on surviving nodes, as when P collapses into P' and then onto Q.

Once `ab` is exhausted, *mkEpsBC* walks ri_2 the way *mkEps* walks a plain expression, taking the leftmost nullable branch at each remaining choice: $[1]@Q$ is nullable, so it contributes its 1, and inside Q the ε branch is nullable, contributing a second 1, giving $bits = [1,1]$. $decode_{r_0}$ then replays these two bits against the original r_0. The first 1 selects $a+ab$'s right branch and the second selects $b+\varepsilon$'s right branch, giving the tree $(Right(a,b), Right\,())$.

This shows that the bit-coded construction reconstructs the parse tree in a single step, while the direct construction reverses residuals step by step.

5.2.1 Annotated Expressions

Building parse-tree information during the forward pass extends the derivative computation with bit-sequence annotations, propagated by the bit-threaded derivative $ri\backslash_b l$. An annotated expression ri extends r by associating each non-$\emptyset$ node with a bit-sequence bs recording the reconstruction choices made during the forward pass.

$$ri ::= \emptyset \mid (bs@\varepsilon) \mid (bs@x) \mid (bs@ri_1 \oplus ri_2) \mid (bs@ri_1\,ri_2) \mid (bs@ri^*). \tag{5.6}$$

In Rust. Figure 5.6 defines `ARegex`, the annotated counterpart of `Regex`. Every variant except `Phi` carries a leading `Bits` field (cf. (5.6)). `Bits` is a persistent, `Rc`-backed cons list, not a `Vec<bool>`: prepending a prefix costs only the prefix, independent of the accumulated history. A `Vec<bool>` would copy the entire prefix on every derivative step, making an n-character run cost $O(n^2)$. the persistent representation keeps it $O(n)$.

```
pub enum ARegex {
    Phi,
    Eps(Bits), Lit(Bits, char),
    Alt(Bits, Box<ARegex>, Box<ARegex>),
    Seq(Bits, Box<ARegex>, Box<ARegex>),
    Star(Bits, Box<ARegex>),
}
```

Figure 5.6: The annotated-expression type extends `Regex` with a leading bit-sequence, `Bits`.

All following operations act on this `ARegex`. Since derivative and simplification steps remove nodes while the bits attached to them must survive, each operation needs a way to transfer those bits onto the node that remains. The helper *fuse* provides this transfer.

5.2.2 Fusing Bit-Lists

fuse, defined in [Sulzmann and Lu, 2014, Fig. 5], transfers bits from a node that is about to be removed onto the node that survives in its place. It prepends bs to the top node's annotation bs' without descending into its children, so outer reconstruction choices precede those already recorded in the subtree.

$$\begin{aligned} \mathit{fuse}(bs, \emptyset) &= \emptyset \\ \mathit{fuse}(bs, (bs'@\varepsilon)) &= (bs +\!\!+ bs')@\varepsilon \\ \mathit{fuse}(bs, (bs'@x)) &= (bs +\!\!+ bs')@x \\ \mathit{fuse}(bs, (bs'@ri_1 \oplus ri_2)) &= (bs +\!\!+ bs')@(ri_1 \oplus ri_2) \\ \mathit{fuse}(bs, (bs'@ri_1\, ri_2)) &= (bs +\!\!+ bs')@(ri_1\, ri_2) \\ \mathit{fuse}(bs, (bs'@ri^*)) &= (bs +\!\!+ bs')@ri^*. \end{aligned} \tag{5.7}$$

In Rust. Figure 5.7 implements the six rules of (5.7), one per variant. The helper `Bits::prepend` corresponds to the concatenation $+\!\!+$ and places `bs` before the node's existing bits:

```
pub fn fuse(bs: &[bool], ri: ARegex) -> ARegex {
    if bs.is_empty() { return ri; }
    match ri {
        ARegex::Phi => ARegex::Phi,
        ARegex::Eps(p) => ARegex::Eps(p.prepend(bs)),
        ARegex::Lit(p, c) => ARegex::Lit(p.prepend(bs), c),
        ARegex::Alt(p, r1, r2) => ARegex::Alt(p.prepend(bs), r1, r2),
        ARegex::Seq(p, r1, r2) => ARegex::Seq(p.prepend(bs), r1, r2),
        ARegex::Star(p, r) => ARegex::Star(p.prepend(bs), r),
    }
}

impl Bits {
    // prefix ++ self, O(prefix.len()) regardless of how long self
    // already is -- the existing chain is shared, not copied.
    pub fn prepend(&self, prefix: &[bool]) -> Bits {
        if prefix.is_empty() { return self.clone(); }
        Bits(Rc::new(Node::Prepend(Rc::from(prefix), self.0.clone())))
    }
}
```

Figure 5.7: `fuse`: prepend a bit-list to the annotation of the top node.

The `Phi` arm is the only case that returns without calling `prepend`, because `Phi` represents the empty language and carries no annotation slot, so there is no bit-list for `fuse` to prepend to. Every other arm rebuilds the top node with a combined annotation and moves its two children, or single child in the star case, unchanged. Since `fuse` prepends `bs` only to the top node's annotation, the bits carried by its children are not touched, and the operation rewrites exactly one node regardless of how large the subtree below it is.

The function is therefore non-recursive, and no subtree needs to be copied. The implementation realizes this directly, since `ri` is consumed by value and its untouched children

move into the rebuilt node without a clone. `Bits::prepend` wraps the node's existing chain in a new `Rc`, sharing rather than copying it: the allocation is proportional only to `bs`'s length, independent of how many bits the node already carries. This is why `Bits` replaces a plain `Vec<bool>` here: with a `Vec`, `prepend` would have to copy the existing content on every call, a cost proportional to the combined length of the two bit-lists – the node's own history can grow with every character consumed so far, which would make a single `fuse` call as expensive as the whole parse up to that point. An empty `bs` is the one case that still skips any work, since prepending an empty prefix returns the node's existing chain unchanged.

5.2.3 Internalizing Expressions

The forward pass operates on annotated expressions, but the input is a plain expression, so r must first be converted into its annotated counterpart ri. The function *internalize*, following [Sulzmann and Lu, 2014, Fig. 5], performs this conversion recursively, and it is the only step that introduces new bits:

$$\begin{aligned}
\mathit{internalize}(\emptyset) &= \emptyset \\
\mathit{internalize}(\varepsilon) &= [\,]@\varepsilon \\
\mathit{internalize}(x) &= [\,]@x \\
\mathit{internalize}(r_1 + r_2) &= [\,]@\Big(\mathit{fuse}([0], \mathit{internalize}(r_1)) \oplus \mathit{fuse}([1], \mathit{internalize}(r_2))\Big) \\
\mathit{internalize}(r_1\, r_2) &= [\,]@\Big(\mathit{internalize}(r_1)\ \mathit{internalize}(r_2)\Big) \\
\mathit{internalize}(r^*) &= [\,]@\mathit{internalize}(r)^*.
\end{aligned} \tag{5.8}$$

The tags [0] and [1] on the two branches of $r_1 + r_2$ mark the left and right branch respectively, so the decoder can later recover which one was taken. Every other case introduces no new bit and uses only the empty list $[\,]$. The two calls to *fuse* in (5.8) are the only points in the construction where new bits are introduced.

In Rust. Figure 5.8 directly maps the constructors of `Regex` to their annotated counterparts. Each recursive call applies `internalize` to the corresponding subexpression, preserving the structure of the original expression in the annotated representation.

```
pub fn internalize(r: &Regex) -> ARegex {
    match r {
        Regex::Phi => ARegex::Phi,
        Regex::Eps => ARegex::Eps(Bits::empty()),
        Regex::Lit(c) => ARegex::Lit(Bits::empty(), *c),
        Regex::Alt(r1, r2) => {
            let ri1 = fuse(&[false], internalize(r1));
            let ri2 = fuse(&[true], internalize(r2));
            ARegex::Alt(Bits::empty(), Box::new(ri1), Box::new(ri2))
        }
        Regex::Seq(r1, r2) => ARegex::Seq(Bits::empty(),
            Box::new(internalize(r1)), Box::new(internalize(r2))),
        Regex::Star(r1) => ARegex::Star(Bits::empty(),
            Box::new(internalize(r1))),
    }
}
```

Figure 5.8: `internalize` converts a `Regex` into an `ARegex`. Each alternation branch receives a tag recording its side, and the two `fuse` calls are the only points where new bits are introduced.

The `Alt` case calls `fuse` with a non-empty bit-list, tagging the left operand with `[false]` and the right with `[true]` before wrapping them in a `Alt` with an empty outer bit-list. Every later bit originates from one of these two tags and is propagated during derivation.

The `Phi`, `Eps`, `Lit`, `Seq`, and `Star` cases use an empty bit-list, since they introduce no disambiguation choice at this stage. Star iteration choices are instead determined during derivative computation.

5.2.4 Extracting Bits

mkEpsBC, from [Sulzmann and Lu, 2014, Fig. 5], is the bit-list counterpart of *mkEps*. It keeps the left preference and star closing bit, but extracts bits instead of building a tree. Each rule prepends the current node's bits to those returned recursively.

$$
\begin{aligned}
mkEpsBC_{(bs@\varepsilon)} &= bs \\
mkEpsBC_{(bs@ri^*)} &= bs +\!\!+ [1] \\
mkEpsBC_{(bs@ri_1\, ri_2)} &= bs +\!\!+ mkEpsBC_{ri_1} +\!\!+ mkEpsBC_{ri_2} \\
mkEpsBC_{(bs@ri_1 \oplus ri_2)} &= bs +\!\!+ \begin{cases} mkEpsBC_{ri_1} & \varepsilon \in L(ri_1) \\ mkEpsBC_{ri_2} & \text{otherwise.} \end{cases}
\end{aligned}
\tag{5.9}
$$

As with *mkEps*, the rules cover only the four nullable forms. $\emptyset$ and literals cannot match the empty word, so *mkEpsBC* is called only on nullable expressions and both are handled as preconditions.

In Rust. Figure 5.9 implements `mk_eps_bc` alongside `mk_eps` (Figure 5.2), with one difference. Where `mk_eps` builds a `ParseTree`, `mk_eps_bc` concatenates bit-lists, calling `bs.to_vec()` where `mk_eps` would borrow or clone: this is the point where a node's `Bits` chain (Section 5.2.1) is finally walked and materialized into a concrete `Vec<bool>`,

something `fuse` deliberately avoids doing on every derivative step. The `Star` arm also does not recurse, since a star matching the empty word uses zero iterations, so it only appends the closing `true` and never examines the child expression:

```
pub fn mk_eps_bc(ri: &ARegex) -> Vec<bool> {
    match ri {
        ARegex::Phi => panic!("mk_eps_bc called on Phi"),
        ARegex::Lit(_, c) => panic!("mk_eps_bc called on Lit('{}')", c),
        ARegex::Eps(bs) => bs.to_vec(),
        ARegex::Alt(bs, r1, r2) => {
            let mut result = bs.to_vec();
            if nullable_bc(r1) {
                result.extend(mk_eps_bc(r1));
            } else {
                result.extend(mk_eps_bc(r2));
            }
            result
        }
        ARegex::Seq(bs, r1, r2) => {
            let mut result = bs.to_vec();
            result.extend(mk_eps_bc(r1));
            result.extend(mk_eps_bc(r2));
            result
        }
        ARegex::Star(bs, _) => {
            let mut result = bs.to_vec();
            result.push(true);
            result
        }
    }
}
```

Figure 5.9: `mk_eps_bc` implements (5.9). `Phi` and `Lit` panic, since neither can be nullable.

5.2.5 Annotated Derivatives

$ri\backslash_b l$ has the same case structure as the plain derivative (3.6), but extended with bits threaded through the rules. Each rule corresponds to its unannotated counterpart:

$$
\begin{aligned}
\emptyset\backslash_b x &= \emptyset && (\mathsf{B}_\emptyset)\\
(bs@\varepsilon)\backslash_b x &= \emptyset && (\mathsf{B}_\varepsilon)\\
(bs@c)\backslash_b x &= \begin{cases} bs@\varepsilon & c = x \\ \emptyset & \text{otherwise} \end{cases} && (\mathsf{B}_{\text{lit}})\\
(bs@ri_1 \oplus ri_2)\backslash_b x &= (bs@(ri_1\backslash_b x) \oplus (ri_2\backslash_b x)) && (\mathsf{B}_+)\\
(bs@ri_1\ ri_2)\backslash_b x &= (bs@(ri_1\backslash_b x)\ ri_2) && (\mathsf{B}_{\text{seq}}),\ \varepsilon \notin L(ri_1)\\
(bs@ri_1\ ri_2)\backslash_b x &= bs@\Big(([\,]@(ri_1\backslash_b x)\ ri_2) \oplus \mathit{fuse}(\mathit{mkEpsBC}_{ri_1}, ri_2\backslash_b x)\Big) && (\mathsf{B}^n_{\text{seq}}),\ \varepsilon \in L(ri_1)\\
(bs@ri^*)\backslash_b x &= (bs@\mathit{fuse}([0], ri\backslash_b x)\ ([\,]@ri^*)) && (\mathsf{B}_*).
\end{aligned}
\tag{5.10}
$$

The rules B_{+}, B_{seq}, and B_{*} preserve the outer bs, since derivation changes the residual but not the choices already recorded above it.

The nullable sequence case $\mathsf{B}^{n}_{\text{seq}}$ is the exception. Since ri_1 is nullable, x may be matched by either factor, so the rule introduces a new alternation. Its left branch continues into ri_1, while its right branch recurses into ri_2, with the empty match's bits extracted by *mkEpsBC* and fused onto ri_2's derivative, so the choice inside ri_1 survives. This performs during the forward pass what $\mathsf{I}^{R}_{\text{seq}}$ performs during injection. The outer bs attaches to the new alternation, since the alternation as a whole is what $ri_1\ ri_2$ derived into.

In Rust. Figure 5.10 implements (5.10), following the case structure of plain `deriv` with bits threaded through the constructors. The base cases `Phi`, `Eps`, and `Lit` match plain `deriv`, except that `Lit` carries its bit-list onto the resulting `Eps`.

The `Alt` arm derives both operands and rebuilds the alternation with the outer `bs` on the wrapper, not pushed into either branch. This matches B_{+}, where the recorded choice belongs to the alternation itself and derivation does not change it, so the bit stays on the outer constructor until the decoder reads it.

The `Seq` arm reflects B_{seq} and $\mathsf{B}^{n}_{\text{seq}}$. In the non-nullable case, `r1` must consume the character, so only its derivative is needed and `r2` moves directly into the rebuilt sequence. In the nullable case, both alternatives are possible. The bits of `r1`'s empty match are extracted by `mk_eps_bc` *before* `r1` is consumed by `deriv_bc`, since after that call the original `r1` is gone. Those bits are then fused onto the derivative of `r2`, so the choice made inside `r1` reaches the surviving branch. The left branch is wrapped in a fresh `Seq` with an empty bit-list, since the outer `bs` moves to the new `Alt` wrapper, matching $\mathsf{B}^{n}_{\text{seq}}$. Two clones are therefore needed. `r2.clone()` provides one copy for the left branch and one for its own derivative, and `nullable_bc` borrows `r1` before `deriv_bc` consumes it.

The `Star` arm follows the same pattern as the nullable case of `Seq`. The body is needed both for its derivative and for the reconstructed star, so `r.clone()` provides the second copy, and the reconstructed star is wrapped in a fresh `Star` with an empty bit-list for the same reason as the left branch above. The leading `[false]` prepended to the derivative marks one more iteration, matching the outermost bit in B_{*}

```
pub fn deriv_bc(ri: ARegex, l: char) -> ARegex {
    match ri {
        // ...
        ARegex::Alt(bs, r1, r2) => {
            let d1 = deriv_bc(*r1, l);
            let d2 = deriv_bc(*r2, l);
            ARegex::Alt(bs, Box::new(d1), Box::new(d2))
        }
        ARegex::Seq(bs, r1, r2) => {
            if nullable_bc(&r1) {
                let eps_bits = mk_eps_bc(&r1);
                let d1 = deriv_bc(*r1, l);
                let d2 = deriv_bc(*r2.clone(), l);
                let left_branch = ARegex::Seq(Bits::empty(), Box::new(d1), r2);
                let right_branch = fuse(&eps_bits, d2);
                ARegex::Alt(bs, Box::new(left_branch), Box::new(right_branch))
            } else {
                let d1 = deriv_bc(*r1, l);
                ARegex::Seq(bs, Box::new(d1), r2)
            }
        }
        ARegex::Star(bs, r) => {
            let d = deriv_bc(*r.clone(), l);
            let fused = fuse(&[false], d);
            let new_star = ARegex::Star(Bits::empty(), r);
            ARegex::Seq(bs, Box::new(fused), Box::new(new_star))
        }
    }
}
```

Figure 5.10: Annotated derivative `deriv_bc`, implementing $\mathsf{B}_{\emptyset}$, B_{ε}, B_{lit}, B_{+}, $\mathsf{B}_{\text{seq}}/\mathsf{B}^{n}_{\text{seq}}$, B_{*}.

5.2.6 Decoding Bits

$decode_r$ reconstructs a `ParseTree` from the bit sequence of $mkEpsBC$, walking r with one bit per alternation to choose a branch. Its worker $decodeInner$ returns the current tree and the remaining bits, so the recursion continues at the correct position:

$$
\begin{aligned}
decode_r(bs) &= v \quad \text{if } decodeInner_r(bs) = (v, [\,]) \\[2ex]
decodeInner_{\varepsilon}(bs) &= (Empty,\ bs) \\
decodeInner_{x}(bs) &= (Char(x),\ bs) \\
decodeInner_{r_1+r_2}(0{:}bs) &= (Left(v),\ bs_1) \quad \text{if } decodeInner_{r_1}(bs) = (v, bs_1) \\
decodeInner_{r_1+r_2}(1{:}bs) &= (Right(v),\ bs_1) \quad \text{if } decodeInner_{r_2}(bs) = (v, bs_1) \\
decodeInner_{r_1 r_2}(bs) &= (Pair(v_1, v_2),\ bs_2) \\
&\qquad \text{if } decodeInner_{r_1}(bs) = (v_1, bs_1) \text{ and } decodeInner_{r_2}(bs_1) = (v_2, bs_2) \\
decodeInner_{r^*}(1{:}bs) &= (Star([\,]),\ bs) \\
decodeInner_{r^*}(0{:}bs) &= (Star(v{:}vs),\ bs_2) \\
&\qquad \text{if } decodeInner_{r}(bs) = (v, bs_1) \text{ and } decodeInner_{r^*}(bs_1) = (Star(vs), bs_2)
\end{aligned}
\tag{5.11}
$$

In Rust. Figure 5.11 shows `decode_inner`, the worker behind `decode`. The public entry point `decode` wraps it and asserts that no bits remain. This assertion is the side condition under which $decode_r$ is defined. The worker must consume the entire bit-string.

The worker consumes the `&[bool]` one bit per step using Rust's slice patterns: `[false, rest @ ..]` matches a bit-string whose first element is `false` and binds the remainder to `rest`, and `[true, rest @ ..]` does the same for `true`.

```
fn decode_inner<'a>(r: &Regex, bs: &'a [bool]) -> (ParseTree, &'a [bool]) {
    match r {
        Regex::Eps => (ParseTree::Empty, bs),
        Regex::Lit(c) => (ParseTree::Char(*c), bs),
        Regex::Alt(r1, r2) => match bs {
            [false, rest @ ..] => {
                let (v, remaining) = decode_inner(r1, rest);
                (ParseTree::Left(Box::new(v)), remaining)
            }
            [true, rest @ ..] => {
                let (v, remaining) = decode_inner(r2, rest);
                (ParseTree::Right(Box::new(v)), remaining)
            }
            [] => panic!("decode: unexpected end of bits at Alt"),
        },
        Regex::Seq(r1, r2) => {
            let (v1, after_r1) = decode_inner(r1, bs);
            let (v2, after_r2) = decode_inner(r2, after_r1);
            (ParseTree::Pair(Box::new(v1), Box::new(v2)), after_r2)
        }
        Regex::Star(r1) => {
            let mut iterations = Vec::new();
            let mut remaining = bs;
            loop {
                match remaining {
                    [true, rest @ ..] => {
                        remaining = rest;
                        break;
                    }
                    [false, rest @ ..] => {
                        let (v, after) = decode_inner(r1, rest);
                        iterations.push(v);
                        remaining = after;
                    }
                    [] => panic!("decode: unexpected end of bits inside Star"),
                }
            }
            (ParseTree::Star(iterations), remaining)
        }
        Regex::Phi => panic!("decode: called on Phi"),
    }
}
```

Figure 5.11: `decode_inner`: the worker behind `decode`, one arm per expression form.

The `Alt` arm matches the leading bit with `[false, rest @ ..]` or `[true, rest @ ..]` and recurses into the branch it names, returning `rest` as the remaining bits, since only the leading bit belongs to the alternation.

The `Seq` arm shows why the worker returns a pair. It recurses into the first factor, receives both a tree and the bits the first factor left unconsumed, and passes those bits to the second factor. The second factor then returns its own tree and the bits that follow it, which are the bits for any enclosing expression. Returning only the tree would leave the second recursion with no way to know where to start.

The `Star` arm loops, since the number of iterations is not known in advance and is itself encoded in the bits. Each leading `false` is matched by `[false, rest @ ..]` and corresponds to one iteration, so the arm recurses into the body and continues the loop from whatever bits the body left behind. The loop ends at the leading `true` matched by `[true, rest @ ..]`, which `mk_eps_bc` appended.

5.2.7 Parser

The bit-coded parser performs three steps on the input. First, *internalize* converts the pattern r into an annotated expression ri. Then, for each input character c, the parser takes the annotated derivative $ri\backslash_b c$ and simplifies it, replacing ri with $simp(ri\backslash_b c)$. After the last character, the parser checks whether the final ri is nullable. If it is, the parser extracts the bits with $mkEpsBC$ and decodes them against the original r:

$$\begin{aligned} ri_0 &= internalize(r), \\ ri_i &= simp(ri_{i-1}\backslash_b x_i) \quad (1 \le i \le n), \\ result &= \begin{cases} decode_r(mkEpsBC(ri_n)) & n(ri_n) \\ \text{no match} & \text{otherwise.} \end{cases} \end{aligned} \tag{5.12}$$

The loop holds only the current ri. Unlike the standard derivative parser, which keeps a growing vector of residuals for the backward reconstruction, the bit-coded parser needs no history. The bits record everything needed to decode, so the forward pass can discard each intermediate expression as soon as the next one is computed.

In Rust. Figure 5.12 implements (5.12).

```
pub fn parse_bitcoded_loop(input: &str, r: &Regex) -> Option<ParseTree> {
    let mut ri = internalize(r);
    for l in input.chars() { ri = simp(deriv_bc(ri, l)); }
    if !nullable_bc(&ri) { return None; }
    Some(decode(r, &mk_eps_bc(&ri)))
}
```

Figure 5.12: `parse_bitcoded_loop`: the bit-coded parser implementation.

The loop matches the three cases of (5.12) one for one. `internalize` initialises `ri`, the `for` loop applies `simp(deriv_bc(ri, l))` to each character and the nullability check gates the final `decode(mk_eps_bc(ri))`.

5.2.8 Worked Example: Applying the Bitcoded Parser

Consider the same input $\mathtt{ab}$ and regular expression $r_0 = (a + ab)(b + \varepsilon)$ as in Section 5.1.4.

Internalize. Before consuming input, *internalize* tags each alternative with [0] for the left branch and [1] for the right branch:

$$\mathit{internalize}(r_0) = ri_0 = [\,]@\Big([\,]@([0]@a \oplus [1]@(a\,b)) \cdot [\,]@([0]@b \oplus [1]@\varepsilon)\Big). \tag{5.13}$$

Bit-Threaded Derivatives. The two input symbols are consumed by successive bit-threaded derivatives with *simp* applied after each step. The first derivative uses B_{seq}. After consuming $\mathtt{a}$, the first factor becomes nullable, so the second derivative uses $\mathsf{B}^n_{\text{seq}}$:

$$\begin{aligned}
ri_0 &\xrightarrow[\mathsf{B}_{\text{seq}}]{a} ri_1 = [\,]@\Big([\,]@([0]@\varepsilon \oplus [1]@b) \cdot [\,]@([0]@b \oplus [1]@\varepsilon)\Big) \\
ri_1 &\xrightarrow[\mathsf{B}^n_{\text{seq}}]{b} ri_2' = [\,]@\Big(\underbrace{[\,]@(\emptyset \oplus [1]@\varepsilon) \cdot [\,]@([0]@b \oplus [1]@\varepsilon)}_{X} \oplus \underbrace{[0]@\Big([0]@\varepsilon \oplus \emptyset\Big)}_{Y} \Big) \\
&\xrightarrow{\;\mathit{simp}\;} ri_2 = [\,]@\Big([1]@([0]@b \oplus [1]@\varepsilon) \oplus [0,0]@\varepsilon\Big).
\end{aligned} \tag{5.14}$$

Each disjunct of ri_2' has a dead branch. In X, the first factor's derivative by $\mathtt{b}$ is $\emptyset \oplus [1]@\varepsilon$: the [0] side cannot consume $\mathtt{b}$, the [1] side owes one. In Y, the second factor's derivative by $\mathtt{b}$ is $[0]@\varepsilon \oplus \emptyset$: the [0] side matches, the [1] side cannot. The outer [0] on Y is the bit that $\mathsf{B}^n_{\text{seq}}$ fused on from the first factor's empty match.

simp drops the dead branches. From X, the survivor is $[1]@\varepsilon$, and since ε contributes nothing to the concatenation, the [1] is carried onto the second factor, giving $[1]@([0]@b \oplus [1]@\varepsilon)$. From Y, the survivor is $[0]@\varepsilon$; the outer [0] is merged with the inner [0] and the resulting $[0,0]$ is carried onto the ε, giving $[0,0]@\varepsilon$. The two results are the disjuncts of ri_2, so both bits are recorded there and the tree can later be decoded from the bits alone.

Extracting the bits. Both disjuncts of ri_2 are nullable, so EB^L_+ selects the left one. Inside, b is not nullable while ε is, so EB^R_+ contributes its [1] tag:

$$\mathit{mkEpsBC}(ri_2) \xrightarrow{\mathsf{EB}^L_+} \mathit{mkEpsBC}\Big([1]@([0]@b \oplus [1]@\varepsilon)\Big) \xrightarrow{\mathsf{EB}^R_+} [1,1]. \tag{5.15}$$

Unlike *mkEps*, which constructs a partial parse tree, *mkEpsBC* extracts only the committed choices as a bit-string.

Decoding the bits. The decoder interprets $[1,1]$ against the original r_0: the first bit selects the right branch of $a + ab$ (producing ab), and the second selects the right branch of $b + \varepsilon$:

$$\mathit{decode}_{r_0}([1,1]) = \Big(\mathit{Right}(a,b), \mathit{Right}(\,)\Big) = v_0. \tag{5.16}$$

The decoder consumes the bits left to right, one per alternation it encounters, in the same order that *internalize* introduced them. Because *mkEpsBC* preserved that order when it fused and prepended bits, the first bit always belongs to the outermost alternation, and so on down.

Flattening gives $\mathit{flatten}(v_0) = a \cdot b \cdot \varepsilon = \mathtt{ab}$, confirming the parse. The bit-coded construction produces the same tree as the direct construction, but stores only ri_2, a single expression bounded by the pattern instead of one residual per character.

6 Partial-Derivative-Based Parser

Chapter 5 presented two derivative-based methods for constructing POSIX parse trees. The standard parser stores the derivation sequence and reconstructs the tree in a backward pass using *mkEps* and the injections. The bit-coded parser records the choices as bits during the forward pass and decodes them against the original expression at the end.

This chapter presents the partial-derivative alternative of Sulzmann and Lu [Sulzmann and Lu, 2012]. Instead of one residual per step, the partial derivative produces a set of states, each pairing a residual expression with the reconstruction information for one path through the original expression. The parser maintains this set as an NFA state set, and at the end walks it in generation order, taking the first accepting state. Because partial derivatives break apart the structure of the original expression, this walk yields the leftmost alternative at every choice rather than the longest one, and therefore a non-POSIX parse tree.

As in Chapter 5, two variants are presented. The standard parser attaches an injection closure to each state. The bit-coded parser replaces these closures with bit-sequences. Both compute the same parse tree.

6.1 The Standard Partial-Derivative Parser

Consider again $r_0 = (a + ab)(b + \varepsilon)$ with input $w = \mathtt{ab}$. Where *deriv* reduces r_0 to a single residual at each step, Antimirov's partial derivative *pderiv* reduces it to a *set* of states, one per way the next character can be consumed. Each state pairs a residual expression with an injection that rebuilds a parse tree for r_0 from a parse tree for that residual. The injection is attached to its state as soon as the state is created, rather than recovered afterwards from a stored sequence the way *inj* was for the standard derivative parser in Section 5.1.

Deriving r_0 by $\mathtt{a}$ produces two states, one for each way $\mathtt{a}$ can be consumed:

$$\begin{aligned} \mathit{pderiv}(r_0, \mathtt{a}) = \big\{ &(\ \varepsilon \cdot (b + \varepsilon), \ \lambda(_, v_2).\ (\mathit{Left}\, a, \ v_2) \ ), \\ &(\ (\varepsilon \cdot b)\,(b + \varepsilon), \ \lambda((v_1, v_2), v_2').\ (\mathit{Right}(a, v_2), \ v_2') \ ) \big\}. \end{aligned} \tag{6.1}$$

A state is a pair: a residual expression describing what is left to match, and an injection rebuilding a parse tree of the original expression from a parse tree of the residual. The two states above correspond to the two branches of the alternation $a + ab$ in the first factor. The second factor $b + \varepsilon$ is untouched in both and appears as the trailing component of each residual.

Each residual has the shape $r' \cdot (b + \varepsilon)$: the derivative of the first factor, followed by the untouched second factor, never simplified. The leading ε in each r' records that the consumed character matched a leading literal and left nothing behind in that position; it is not removed, because the injection below pattern-matches on exactly this pair shape.

The first state's residual $\varepsilon \cdot (b + \varepsilon)$ is a sequence, so its tree is a pair (v_1, v_2): v_1 is the tree of the leading ε, which is necessarily *Empty*, and v_2 is the tree of the untouched second factor $b + \varepsilon$. The closure returns $(\mathit{Left}(a), v_2)$. It discards v_1 because the tree of ε carries no information, and replaces the first position with $\mathit{Left}(a)$ to record that the left branch of the alternation $a + ab$ was taken. The second factor's tree v_2 is passed through unchanged.

The second state's residual $(\varepsilon \cdot b)\,(b + \varepsilon)$ is a sequence whose first component is itself a sequence, so its tree is a pair $((v_1, v_2), v_2')$: the inner pair (v_1, v_2) is the tree of $\varepsilon \cdot b$, with v_1 the tree of the leading ε and v_2 the tree of the trailing b; the outer v_2' is the tree of the untouched second factor. The closure returns $(\mathit{Right}(a, v_2), v_2')$. It rebuilds the consumed a (which the leading ε records) and the still-owed b (which v_2 records) as the pair (a, v_2), places that pair inside a right injection to record that the right branch ab of the alternation was taken, and passes the second factor's tree v_2' through unchanged. The binder v_1 is bound but not used in the body, for the same reason as the first state's `_`: the tree of ε carries no information.

Transitioning both states on the second character `b` gives a new state set:

$$\begin{aligned} \mathit{pderiv}(r_0, \mathtt{a}, \mathtt{b}) = \big\{&(\varepsilon,\ \lambda_.\ (\mathit{Left}\, a,\ \mathit{Left}\, b)\ ),\\ &(\varepsilon \cdot (b + \varepsilon),\ \lambda(_, v_2).\ (\mathit{Right}(a, b),\ v_2)\ )\big\}. \end{aligned} \tag{6.2}$$

The first state now has residual ε, meaning every input character has been consumed, and is accepting; the second has residual $\varepsilon \cdot (b + \varepsilon)$ – the second factor is untouched by `b`, since it was the first factor's leftover b that matched – which is also accepting. Selection walks the state set in generation order and returns the first accepting state. Applying its injection to $\mathit{mkEps}_{\varepsilon} = ()$ yields $(\mathit{Left}\, a, \mathit{Left}\, b)$, the leftmost alternative at every choice (Figure 6.1).

Two properties of the state set determine which tree is selected. First, the order of states in the set encodes the preference. When a state produces several successors, they are appended in the order the rules generate them, with the left alternative first, so the leftmost path appears earliest. Second, when two states share a residual, *nub* keeps the first and discards the later one with its injection, so the earlier path wins ties. Both mechanisms favour the leftmost alternative at every choice over the longest one, which is what distinguishes the tree this parser returns from the POSIX tree of Chapter 5.

6.1.1 Partial Derivatives with Injections

A partial derivative of an expression r with respect to a character x is not a single expression but a set of pairs. Each pair consists of a residual expression r' whose language is one component of the left quotient, together with an injection closure ι that maps a parse tree for r' back to a parse tree for r. Stripped of the injection, the residual side of this definition is exactly *pd*, the Antimirov partial derivative already used by the matcher of Section 3.3 [Antimirov, 1996], so *pderiv* reuses its rule names $\mathsf{P}_{\emptyset}, \mathsf{P}_{\varepsilon}, \mathsf{P}_{\mathrm{lit}}, \mathsf{P}_{+}, \mathsf{P}_{\mathrm{seq}}, \mathsf{P}^{n}_{\mathrm{seq}}, \mathsf{P}_{*}$

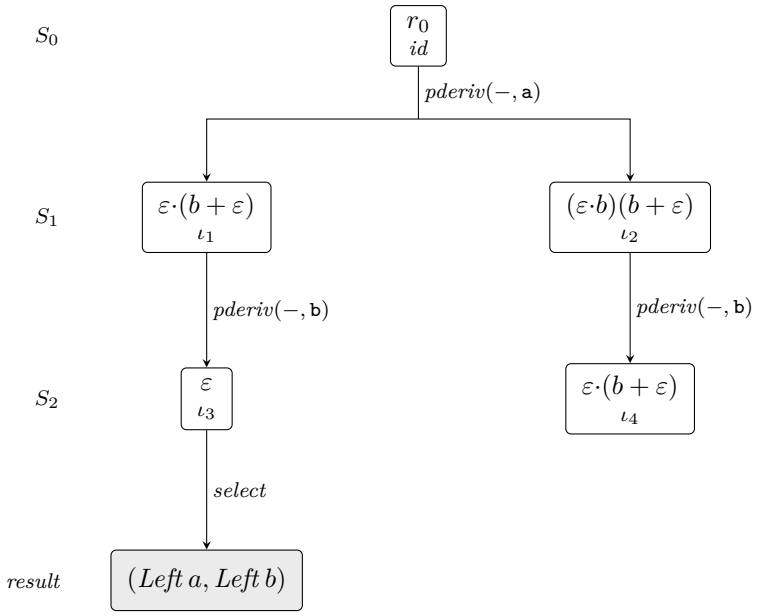


Figure 6.1: The standard partial-derivative parser on $r_0 = (a + ab)(b + \varepsilon)$ with input `ab`. Each box is a state (r, ι): a residual and an injection, both shown without any simplification. $pderiv(-, x)$ advances every state by one character; *select* takes the first accepting state and applies its injection to *mkEps* of its residual.

from Equation 3.14, each now additionally threading an injection alongside its residual. The idea of pairing a residual with a reconstruction function is, in turn, in the style of Sulzmann and Lu's pattern partial derivative [Sulzmann and Lu, 2012, Fig. 11], which pairs a residual *pattern* with a function building a variable-binding environment; here the same idea builds a *ParseTree* instead.

The rules that do not depend on whether their operand is nullable are:

$$
\begin{aligned}
pderiv(\emptyset, x) &= \{\} && (\mathsf{P}_\emptyset)\\
pderiv(\varepsilon, x) &= \{\} && (\mathsf{P}_\varepsilon)\\
pderiv(y, x) &= \begin{cases} \{(\varepsilon,\ \lambda_.\, y)\} & x = y \\ \{\} & x \neq y \end{cases} && (\mathsf{P}_{\text{lit}})\\
pderiv(r + s, x) &= \{\, (r',\ \lambda v.\, Left\,(\iota\, v)) \mid (r', \iota) \in pderiv(r, x) \,\}\\
&\quad \cup\ \{\, (s',\ \lambda v.\, Right\,(\iota\, v)) \mid (s', \iota) \in pderiv(s, x) \,\} && (\mathsf{P}_+)\\
pderiv(r^*, x) &= \{\, (r'\, r^*,\ \lambda(v, Star(vs)).\, Star([\iota\, v] +\!\!+ vs)) \mid (r', \iota) \in pderiv(r, x) \,\} && (\mathsf{P}_*)
\end{aligned}
\tag{6.3}
$$

exactly as in *pd*, with the recursive call's injection wrapped by *Left*, *Right*, or prepended to the iteration list. No rule is needed for $\emptyset$ or ε beyond their empty results, since neither can consume a character.

Concatenation alone must branch on $n(r)$, so it is shown separately:

$$
pderiv(r \cdot s, x) = \begin{cases}
\{\, (r'\, s,\ \lambda(v_1, v_2).\, (\iota\, v_1,\ v_2)) \mid (r', \iota) \in pderiv(r, x) \,\} & n(r) \quad (\mathsf{P}^n_{\text{seq}})\\
\cup\ \{\, (s',\ \lambda v.\, (mkEps_r,\ \iota\, v)) \mid (s', \iota) \in pderiv(s, x) \,\} & \\
\{\, (r'\, s,\ \lambda(v_1, v_2).\, (\iota\, v_1,\ v_2)) \mid (r', \iota) \in pderiv(r, x) \,\} & \neg\, n(r) \quad (\mathsf{P}_{\text{seq}})
\end{cases}
\tag{6.4}
$$

After expansion, each rule's result is deduplicated by residual, keeping the first occurrence; this is *nub*, applied wherever a transition can produce two states with the same residual, namely P_+, $\mathsf{P}^n_{\text{seq}}$, and P_*. P_{seq} has only one source of states and needs no deduplication of its own, since its single recursive call is already duplicate-free. This is the step that breaks apart the original expression: where the derivative of $r + s$ produces a single alternation, the partial derivative produces separate states for each branch, and the outer alternation disappears from the residual.

Rule (P_{seq}) – the only one present regardless of nullability – passes the outer pair through unchanged, with the recursive injection applied to the first component only. When r is nullable, rule ($\mathsf{P}^n_{\text{seq}}$) additionally unions in the states that skip r entirely, using $mkEps_r$ to synthesise its empty match for the first component. Rule (P_+) wraps the recursive injection in a *Left* or *Right* node, reinserting the alternation choice. Rule (P_*) reconstructs the current iteration by applying the recursive injection and prepending it to the remaining iterations.

Unlike *pd*, which applies the smart constructor *smartC* in P_{seq} and P_* to avoid building $\varepsilon \cdot s$ (Equation 3.15), *pderiv* never simplifies: $r' \cdot s$ above is built with the plain pairing constructor, even when $r' = \varepsilon$. A simplified residual would collapse exactly the pair-shaped node that the injection above pattern-matches on, so *pderiv* trades the smaller residuals *pd* produces for residuals whose shape the injection can always follow. This is the same tradeoff the standard derivative parser makes by never simplifying (Section 5.1), for the same reason: *inj* there pattern-matches on the derivative's structure just as ι does here.

In Rust. The recursive definition in Equations 6.3 and 6.4 is implemented directly in Rust, with two departures from a literal reading of the equations, both for performance rather than semantics.

First, `pderiv_tree` itself builds only *one* transition's closure, type `InjStep`, a reference-counted closure exactly as before. The accumulated injection a state carries across many characters – what the equations call ι once composition has happened several times – is a separate type, `Inj`, described in Section 6.1.2: a persistent list of `InjStep`s rather than a nested closure, so that applying or dropping a long-lived `Inj` does not recurse one native stack frame per character.

Second, the `Seq` and `Star` arms use `smart_seq` – the Rust counterpart of *smartC* used by `pderiv` in Figure 3.7 – wherever the equations build $\varepsilon \cdot s$, except in `Seq`'s nullable branch, which still pairs its continue-strand with the plain `Regex::seq` constructor. Collapsing $\varepsilon \cdot s$ to s changes the shape the next transition's closure must match against, so each arm that uses `smart_seq` branches on whether the residual collapsed (`r1_ == Regex::Eps`) and builds the closure for that shape specifically, rather than assuming a `Pair` is always present. Without this, two strands that independently derive to the same residual but reached it through different wrapper shapes would not dedup against each other, and the frontier can grow unboundedly on an ambiguous pattern even though the number of distinct residuals stays bounded.

```
pub(crate) type InjStep = Rc<dyn Fn(ParseTree) -> ParseTree>;

pub(crate) fn pderiv_tree(r: &Regex, x: char) -> Vec<(Regex, InjStep)> {
    match r {
        Regex::Eps | Regex::Phi => vec![],

        Regex::Lit(c) => {
            if *c == x {
                vec![(Regex::Eps,
                      Rc::new(move |_| ParseTree::Char(x))) as InjStep]
            } else { vec![] }
        }

        Regex::Alt(r1, r2) => {
            let mut out: Vec<(Regex, InjStep)> = pderiv_tree(r1, x)
                .into_iter()
                .map(|(r1_, inj1)| {
                    let f: InjStep = Rc::new(move |v| {
                        ParseTree::Left(Box::new(inj1(v)))
                    });
                    (r1_, f)
                })
                .collect();
            out.extend(pderiv_tree(r2, x).into_iter().map(|(r2_, inj2)| {
                let f: InjStep = Rc::new(move |v| {
                    ParseTree::Right(Box::new(inj2(v)))
                });
                (r2_, f)
            }));
            nub_tree(out)
        }

        Regex::Seq(r1, r2) => {
            if nullable(r1) {
                // Continue-strand: not smart-constructed, matching
                // the reference literally -- the skip-strand below
                // needs r1's own closure untouched to inject into.
                let mut out: Vec<(Regex, InjStep)> = pderiv_tree(r1, x)
                    .into_iter()
                    .map(|(r1_, inj1)| {
                        let r2c = (**r2).clone();
                        let f: InjStep = Rc::new(move |v| match v {
                            ParseTree::Pair(v1, v2) =>
                                ParseTree::Pair(Box::new(inj1(*v1)), v2),
                            _ => panic!("pderiv_tree/Seq: want Pair"),
                        });
                        (Regex::seq(r1_, r2c), f)
                    })
                    .collect();

                let v1_eps = mk_eps(r1);
                out.extend(pderiv_tree(r2, x).into_iter().map(|(r2_, inj2)| {
                    let v1c = v1_eps.clone();
                    let f: InjStep = Rc::new(move |v2| {
                        ParseTree::Pair(Box::new(v1c.clone()),
                                        Box::new(inj2(v2)))
                    });
```

The `Phi`, `Eps`, and `Lit` arms implement $\mathsf{P}_\emptyset$, P_ε, and P_{lit} directly, matching the same arms of `pderiv` in Figure 3.7, except that each returns a closure alongside its residual instead of nothing. The `Alt` arm implements P_+ and is where the partial derivative visibly breaks apart the expression: each branch produces its own states, tagged by *Left* or *Right* in the injection, then deduplicated by `nub_tree`.

The `Seq` arm's `if nullable(r1)` matches the case split between $\mathsf{P}^n_{\text{seq}}$ and P_{seq}. The nullable branch's continue-strand is built exactly as the equations describe, with the plain pairing constructor, since the skip-strand alongside it needs `r1`'s own closure to still expect a `Pair`; the nullable branch additionally computes `r1`'s empty tree once with `mk_eps` and captures a clone of it in every skip-strand closure, then deduplicates the combined list. The non-nullable branch, by contrast, does use `smart_seq`, with the closure split on whether the residual collapsed to `Eps`, as explained above.

The `Star` arm implements P_*, with the closure appending the reconstructed iteration to the star's list (left-to-right order, reversed once at the end by `select` rather than paying `insert(0, ..)`'s cost on every iteration), again split on whether `smart_seq` collapsed the residual.

6.1.2 State Sets and Selection

The parser maintains a state set of (residual, injection) configurations. For each input character, every configuration in the state set is transitioned with `pderiv_tree`, and the resulting configurations are collected into a new state set, replacing the old one – unlike the standard derivative parser, nothing from earlier state sets is retained once the next one has been computed. A configuration's parent injection is composed with the transition injection, so that at the end the state set's injections map residuals all the way back to the original expression.

Conceptually the composition is direct: when a parent configuration (r, inj_p) transitions to (r', inj_s), the new configuration's injection is $\lambda v.\, inj_p(inj_s(v))$. Composing this way as nested closures, however, makes both calling the result and dropping it recurse one native stack frame per character – fine at tens of thousands of characters, a real stack-overflow risk beyond that. The `Inj` type instead stores a configuration's accumulated steps as a persistent cons-list and applies them in an explicit loop:

```
enum InjNode { Identity, Step(InjStep, Rc<InjNode>) }
pub(crate) struct Inj(Rc<InjNode>);

impl Inj {
    pub(crate) fn then(&self, step: InjStep) -> Self {
        Inj(Rc::new(InjNode::Step(step, self.0.clone())))
    }
    pub(crate) fn apply(&self, v: ParseTree) -> ParseTree {
        let mut v = v;
        let mut cur = &self.0;
        loop {
            match &**cur {
                InjNode::Identity => return v,
                InjNode::Step(step, rest) => { v = step(v); cur = rest; }
```

```
            }
        }
    }
}
```

`then` adds one more character's step in O(1), just wrapping the existing chain in a new node, the same way `Bits::prepend` (Section 5.2.2) wraps an existing bit chain; `apply` walks the whole chain in one iterative pass, applying each step in the order it was added, with no recursive call regardless of how long the chain is. `Inj` also has its own iterative `Drop`, for the same reason `Bits`'s does: the derived recursive drop glue would walk the chain one stack frame per character too.

```
pub(super) fn step_frontier(
    frontier: Vec<(Regex, Inj)>, x: char
) -> Vec<(Regex, Inj)> {
    let mut out = Vec::new();
    for (r, inj_parent) in frontier {
        for (r_next, inj_step) in pderiv_tree(&r, x) {
            out.push((r_next, inj_parent.then(inj_step)));
        }
    }
    nub_tree(out)
}
```

Figure 6.3: `step_frontier`: one transition of the NFA state set on a single character. The function name retains the older "frontier" terminology; it computes the successor state set $\delta(S, x)$. The final `nub_tree` deduplicates *across* parent configurations, not just within one parent's own expansion (which `pderiv_tree` already does on its own output) – without it, two strands that independently derive to the same residual both survive, and the state set can double every character on an ambiguous pattern even though the number of distinct residuals stays bounded by the pattern's own size.

After the input is exhausted, selection walks the state set in priority order and returns the tree for the first accepting configuration:

```
pub(super) fn select(frontier: &[(Regex, Inj)]) -> Option<ParseTree> {
    for (r, inj) in frontier {
        if nullable(r) {
            return Some(reverse_stars(inj.apply(mk_eps(r))));
        }
    }
    None
}
```

Figure 6.4: `select`: return the tree for the first accepting configuration in the state set. `reverse_stars` undoes, in one O(total output size) pass, the left-to-right-via-push construction order `pderiv_tree`'s `Star` arm uses (Figure 6.2).

Because the state set lists configurations in the order the rules generated them, and because `nub_tree` keeps the first of each duplicate residual, selection favours the configuration that committed earliest to the leftmost alternative. This is the non-POSIX leftmost

policy. Unlike the standard derivative parser, there is no separate backward pass here: by the time `select` runs, every configuration's injection is already the full composition back to the original expression, so selection only has to apply it once.

6.1.3 Parser

The parser has a single pass: every input character advances the whole state set at once via *step_frontier*, and the final state set is handed directly to *select*:

$$S_0 = \{(r, id)\}, \qquad S_i = step_frontier(S_{i-1}, x_i) \;\; (1 \le i \le n), \qquad result = select(S_n). \tag{6.5}$$

Unlike the standard derivative parser's two-pass driver (Equation 5.4), no backward pass is needed here: each configuration in S_i already carries its injection composed all the way back to r_0 (Section 6.1.2), so *select* only ever applies one injection, to the single accepting configuration it finds in S_n.

In Rust. Figure 6.5 implements (6.5).

```
pub fn parse_pderiv_std(input: &str, r: &Regex) -> Option<ParseTree> {
    let mut frontier: Vec<(Regex, Inj)> =
        vec![(r.clone(), Inj::identity())];
    for c in input.chars() {
        frontier = step_frontier(frontier, c);
        if frontier.is_empty() {
            return None;
        }
    }
    select(&frontier)
}
```

Figure 6.5: The standard partial-derivative parser.

The loop computes $S_0, \ldots, S_n$ in place, overwriting `frontier` on every iteration rather than appending to a growing `Vec` the way `parse_deriv_std_loop` (Figure 5.4) does – nothing from an earlier state set is retained once the next one exists. The early return on an empty frontier corresponds to $S_i = \{\}$: no configuration can consume the current character, so the input cannot be matched. The final `select` returns `None` if no configuration in S_n is accepting, and otherwise the tree for the first one.

6.1.4 Worked Example: Standard Partial-Derivative Parser

Recall the ambiguous example $r_0 = (a + ab)(b + \varepsilon)$ on input `ab`.

Forward pass. The initial state set holds a single configuration:

$$S_0 = \big[(r_0,\ id)\big].$$

Transitioning with respect to `a`:

$$pderiv(r_0, \mathtt{a}) = pderiv\Big(Alt(a, ab), \mathtt{a}\Big) \cdot Alt(b, \varepsilon).$$

Inside the first factor:

$$\begin{aligned} \mathit{pderiv}(a, \mathtt{a}) &= \Big[(\varepsilon,\ \lambda_.\, a)\Big] \\ \mathit{pderiv}(ab, \mathtt{a}) &= \Big[(\mathit{Seq}(\varepsilon, b),\ \lambda(v_1, v_2).\, (a, v_2))\Big]. \end{aligned}$$

After the alternation rule wraps each state with *Left* or *Right*:

$$\begin{aligned} \mathit{pderiv}(a + ab, \mathtt{a}) = \Big[&(\varepsilon,\ \lambda v.\, \mathit{Left}\,(\lambda_.\, a)(v)), \\ &(\mathit{Seq}(\varepsilon, b),\ \lambda(v_1, v_2).\, \mathit{Right}\,(a, v_2))\Big]. \end{aligned}$$

The outer sequence rule combines this with the untouched second factor:

$$\begin{aligned} S_1 = \Big[&(\mathit{Seq}(\varepsilon, b + \varepsilon),\ \lambda(v_1, v_2).\, (\mathit{Left}\, a,\ v_2)), \\ &(\mathit{Seq}(\mathit{Seq}(\varepsilon, b), b + \varepsilon),\ \lambda((v_1, v_2), v_2').\, (\mathit{Right}(a, v_2),\ v_2'))\Big]. \end{aligned}$$

Transitioning with respect to `b` steps both configurations. For the first, the left factor ε is nullable, so the sequence rule produces the state that jumps directly into the second factor:

$$\mathit{pderiv}(b + \varepsilon, \mathtt{b}) = \Big[(\varepsilon,\ \lambda v.\, \mathit{Left}\, b)\Big].$$

The second factor's injection inserts the character at the leftmost position of the pair:

$$\mathit{inj}_1(\mathit{Left}\, b) = (\mathit{Left}\, a,\ \mathit{Left}\, b).$$

For the second configuration, the leading $\mathit{Seq}(\varepsilon, b)$ is not nullable (ε is, but b is not), so rule (P_{seq}) fires and the derivative continues inside b:

$$\mathit{pderiv}(\mathit{Seq}(\varepsilon, b), \mathtt{b}) = \Big[(\varepsilon,\ \lambda_.\, ((), b))\Big],$$

and the outer sequence rule rebuilds the pair *without* simplifying the leading ε away, giving the second configuration's residual $\varepsilon \cdot (b + \varepsilon)$, not $b + \varepsilon$. The two resulting state-set entries are:

$$\begin{aligned} S_2 = \Big[&(\varepsilon,\ \lambda_.\, (\mathit{Left}\, a,\ \mathit{Left}\, b)), \\ &(\varepsilon \cdot (b + \varepsilon),\ \lambda(_, v_2).\, (\mathit{Right}(a, b),\ v_2))\Big]. \end{aligned}$$

Selection. The first configuration's residual is ε, which is nullable, and selection returns immediately:

$$\mathit{select}(S_2) = (\lambda_.\, (\mathit{Left}\, a, \mathit{Left}\, b))(\mathit{mkEps}_{\varepsilon}) = (\mathit{Left}\, a, \mathit{Left}\, b).$$

Flattening confirms the result:

$$\mathit{flatten}\Big((\mathit{Left}\, a, \mathit{Left}\, b)\Big) = \mathtt{ab}.$$

The second configuration is never inspected: its residual $\varepsilon \cdot (b + \varepsilon)$ is also nullable, and would likewise decode to a valid tree, but generation order already placed the first configuration ahead of it, and *select* stops at the first match.

This is the non-POSIX parse tree. The derivative-based parser from Chapter 5 returns $(\mathit{Right}(a, b), \mathit{Right}())$ for the same input, the POSIX tree. The two trees agree on which word they spell out, but differ on how the input is decomposed: the non-POSIX tree commits to the shorter first alternative a in the first factor, while the POSIX tree commits to the longer alternative ab.

6.2 The Bit-Coded Partial-Derivative Parser

The standard partial-derivative parser carries a closure per configuration, which accumulates the reconstruction path during the forward pass. The bit-coded variant replaces closures with bit-sequences: each configuration's annotation records the choices made so far, and decoding the final bit-sequence against the original expression yields the parse tree.

Transitioning $w = \texttt{ab}$ from $r_0 = (a + ab)(b + \varepsilon)$ gives:

$$\begin{aligned} S_0 &= \Big[(r_0,\ [\,])\Big] \\ &\xrightarrow{\texttt{a}} S_1 = \Big[(b + \varepsilon,\ [\mathit{false}]),\ (b\,(b + \varepsilon),\ [\mathit{true}])\Big] \\ &\xrightarrow{\texttt{b}} S_2 = \Big[(\varepsilon,\ [\mathit{false}, \mathit{false}]),\ (b + \varepsilon,\ [\mathit{true}])\Big]. \end{aligned} \tag{6.6}$$

The first configuration of S_1 records that the character `a` was matched by the left branch of the first factor's alternation. The second configuration records that it was matched by the right branch, leaving the residual b to be consumed later. After the second character, only the first configuration is accepting, and its bits $[\mathit{false}, \mathit{false}]$ decode to $(\mathit{Left}\, a, \mathit{Left}\, b)$.

Figure 6.6 shows this transition together with the selection and decode steps.

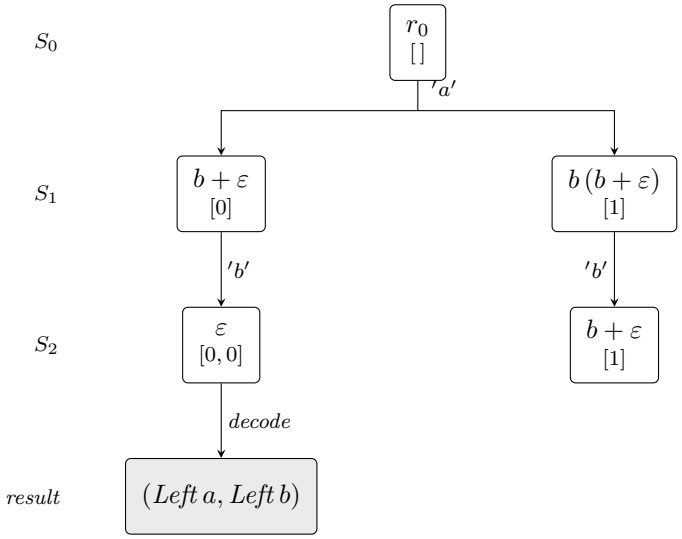


Figure 6.6: Overview of the bit-coded partial-derivative parser on the ambiguous example $r_0 = (a + ab)(b + \varepsilon)$ with input `ab`. Each box is a configuration: a residual expression and a bit-sequence, written 0 for *false* and 1 for *true*. Selection takes the first accepting configuration in S_2 and appends *mkEpsBC* of its residual, yielding the bit-string $[0, 0]$, which decode_{r_0} reads against the original expression.

The bit-coded parser keeps the same state-set structure as the standard one. Only the configuration's annotation changes, from a closure to a sequence of bits, and the final step changes from applying an injection to calling *decode*.

6.2.1 Bit-Coded Partial Derivatives

The bit-coded partial derivative $\mathit{pderivBC}(r, x)$ returns a list of (residual, bit-sequence) pairs. It follows the same case structure as (**??**), with one difference: where the standard

version builds an injection closure, the bit-coded version prepends a fixed bit-sequence to the configuration's existing bits.

The bit conventions are inherited from Chapter 5.2. Each alternation contributes a leading *false* for the left branch or a leading *true* for the right branch. Each star iteration contributes a leading *false*, and each star termination contributes a *true*. Concatenation contributes no bits of its own.

$$
\begin{aligned}
\mathit{pderivBC}(\phi, x) &= \emptyset \\
\mathit{pderivBC}(\varepsilon, x) &= \emptyset \\
\mathit{pderivBC}(c, x) &= \begin{cases} \{(\varepsilon,\, [\,])\} & c = x \\ \emptyset & \text{otherwise} \end{cases} \\
\mathit{pderivBC}(r_1 + r_2, x) &= \{\, (r_1',\, [\mathit{false}] +\!\!+\, bs) \mid (r_1', bs) \in \mathit{pderivBC}(r_1, x) \,\} \\
&\quad \cup \{\, (r_2',\, [\mathit{true}] +\!\!+\, bs) \mid (r_2', bs) \in \mathit{pderivBC}(r_2, x) \,\} \\
\mathit{pderivBC}(r_1\, r_2, x) &= \{\, (\mathit{seq}(r_1', r_2),\, bs) \mid (r_1', bs) \in \mathit{pderivBC}(r_1, x) \,\} \\
&\quad \cup \begin{cases} \{\, (r_2',\, \mathit{mkEpsBC}_{r_1} +\!\!+\, bs) \mid (r_2', bs) \in \mathit{pderivBC}(r_2, x) \,\} & n(r_1) \\ \emptyset & \text{otherwise} \end{cases} \\
\mathit{pderivBC}(r^*, x) &= \{\, (\mathit{seq}(r', r^*),\, [\mathit{false}] +\!\!+\, bs) \mid (r', bs) \in \mathit{pderivBC}(r, x) \,\}.
\end{aligned}
\tag{6.7}
$$

As in the standard version, the result is deduplicated by residual, keeping the first occurrence.

The bit-coded sequence rule is the counterpart of the injection rule in the standard version. When the first factor is nullable, the bits recorded inside its empty match are extracted with $\mathit{mkEpsBC}_{r_1}$ and prepended to the second factor's bits, so the choice made inside the first factor survives in the surviving configuration.

In Rust. Figure 6.7 implements (6.7) directly. The base cases and the alternation case mirror the standard version with bit-lists in place of closures. The sequence case inlines both the non-nullable and nullable paths. The star case prepends the leading *false* that marks one more iteration.

```
pub fn pderiv_bc(r: &Regex, x: char) -> Vec<(Regex, Vec<bool>)> {
    match r {
        Regex::Phi | Regex::Eps => vec![],

        Regex::Lit(y) => {
            if *y == x { vec![(Regex::Eps, vec![])] } else { vec![] }
        }

        Regex::Alt(r1, r2) => {
            let mut out: Vec<(Regex, Vec<bool>)> = pderiv_bc(r1, x)
                .into_iter()
                .map(|(r1_, mut bs)| {
                    let mut b = vec![false]; b.append(&mut bs);
                    (r1_, b)
                })
                .collect();
            out.extend(pderiv_bc(r2, x).into_iter().map(|(r2_, mut bs)| {
                let mut b = vec![true]; b.append(&mut bs);
                (r2_, b)
            }));
            nub2(out)
        }

        Regex::Seq(r1, r2) => {
            if nullable(r1) {
                let mut out: Vec<(Regex, Vec<bool>)> = pderiv_bc(r1, x)
                    .into_iter()
                    .map(|(r1_, bs)| (Regex::seq(r1_, *r2.clone()), bs))
                    .collect();

                let eps_bits = mk_eps_bits(r1);
                out.extend(pderiv_bc(r2, x).into_iter().map(|(r2_, bs)| {
                    let mut b = eps_bits.clone(); b.extend(bs);
                    (r2_, b)
                }));
                nub2(out)
            } else {
                pderiv_bc(r1, x)
                    .into_iter()
                    .map(|(r1_, bs)| (smart_seq(r1_, r2), bs))
                    .collect()
            }
        }

        Regex::Star(r1) => {
            let out: Vec<(Regex, Vec<bool>)> = pderiv_bc(r1, x)
                .into_iter()
                .map(|(r1_, mut bs)| {
                    let mut b = vec![false]; b.append(&mut bs);
                    (smart_seq(r1_, &Regex::star(*r1.clone())), b)
                })
                .collect();
            nub2(out)
        }
    }
}
```


Figure 6.7: `pderiv_bc`: one bit-coded partial-derivative transition

The `Seq` arm keeps the outer sequence constructor in the nullable case, matching the standard version's behaviour. It uses `smart_seq` only in the non-nullable case, where the derivative of the first factor is guaranteed to be a proper expression and no outer pair needs to be preserved. This asymmetry mirrors the paper's reference implementation and is needed so that the residual's shape stays compatible with the decoder.

6.2.2 Extracting Bits

When a configuration's residual is nullable, its accumulated bits must be extended with the bits of the residual's empty match. This is the role of $mkEpsBC$, which mirrors $mkEps$ from Chapter 5 but produces a bit-sequence instead of a parse tree.

$$\begin{aligned} mkEpsBC_{\varepsilon} &= [\,] \\ mkEpsBC_{r^*} &= [true] \\ mkEpsBC_{r_1 r_2} &= mkEpsBC_{r_1} +\!\!+ mkEpsBC_{r_2} \\ mkEpsBC_{r_1 + r_2} &= \begin{cases} [false] +\!\!+ mkEpsBC_{r_1} & n(r_1) \\ [true] +\!\!+ mkEpsBC_{r_2} & \text{otherwise.} \end{cases} \end{aligned} \tag{6.8}$$

The alternation rule reproduces the left preference of $mkEps$, emitting *false* when the left branch is nullable and *true* otherwise. The star rule emits a single terminating *true*, since a star matching the empty word uses zero iterations.

```
pub fn mk_eps_bits(r: &Regex) -> Vec<bool> {
    match r {
        Regex::Phi => panic!("mk_eps_bits called on Phi"),
        Regex::Lit(c) => panic!("mk_eps_bits called on Lit('{}')", c),
        Regex::Eps => Vec::new(),
        Regex::Alt(r1, r2) => {
            if nullable(r1) {
                let mut bits = vec![false];
                bits.extend(mk_eps_bits(r1));
                bits
            } else {
                let mut bits = vec![true];
                bits.extend(mk_eps_bits(r2));
                bits
            }
        }
        Regex::Seq(r1, r2) => {
            let mut bits = mk_eps_bits(r1);
            bits.extend(mk_eps_bits(r2));
            bits
        }
        Regex::Star(_) => vec![true],
    }
}
```

Figure 6.8: `mk_eps_bits`: the bit-string analogue of `mk_eps` for plain `Regex`.

The `Phi` and `Lit` arms panic, matching the precondition that *mkEpsBC* is applied only to nullable expressions. Neither empty language nor a literal can match the empty word.

6.2.3 State Sets and Selection

State-set transition for the bit-coded parser mirrors the standard version, with bit-list concatenation replacing closure composition. Each configuration's bits accumulate across the whole input so far, so a plain `Vec<bool>` clone-then-extend on every transition is O(current length) per step, $O(n^2)$ over n characters – the same cost `Bits`/`Inj` avoid elsewhere in this and the previous chapter. `AppendBits` is the append-optimized mirror of `Bits` (Section 5.2.1): it stores the reverse of the real sequence internally, so appending a suffix becomes prepending its reverse, reusing `Bits::prepend`'s O(suffix length) guarantee, and materializes the sequence back into a plain `Vec<bool>` only once, when the bits are actually consumed:

```
pub(super) fn step_frontier(
    frontier: &[(Regex, AppendBits)], x: char
) -> Vec<(Regex, AppendBits)> {
    let mut out = Vec::new();
    for (r, bits) in frontier {
        for (r_next, extra) in pderiv_bc(r, x) {
            out.push((r_next, bits.append(&extra)));
        }
    }
    nub2(out)
}
```

Figure 6.9: `step_frontier` (bit-coded): as with the standard parser's version, `nub2` deduplicates across the whole combined state set, not just within one configuration's own expansion (which `pderiv_bc` already does via its own `nub2` call).

Selection walks the state set in priority order and returns the bits of the first accepting configuration, extended with *mkEpsBC* of the residual:

```
pub(super) fn select_bits(
    frontier: &[(Regex, AppendBits)]
) -> Option<Vec<bool>> {
    for (r, bits) in frontier {
        if nullable(r) {
            return Some(bits.append(&mk_eps_bits(r)).to_vec());
        }
    }
    None
}
```

The resulting bit-sequence is interpreted by the shared `decode` function from Chapter 5.2, which walks the original expression and consumes one bit per alternation and per star iteration.

6.2.4 Parser

The parser composes `step_frontier` over the input, checks that the state set is non-empty, and calls `select_bits` followed by `decode`.

```
pub fn parse_pderiv_bc(input: &str, r: &Regex) -> Option<ParseTree> {
    let mut frontier = vec![(r.clone(), AppendBits::empty())];
    for c in input.chars() {
        frontier = step_frontier(&frontier, c);
        if frontier.is_empty() {
            return None;
        }
    }
    let bits = select_bits(&frontier)?;
    Some(decode(r, &bits))
}
```

Figure 6.10: The bit-coded partial-derivative parser.

The driver has the same structure as the standard one. Only the configuration type and the final step differ: bits and `decode` in place of closures and injection application.

6.2.5 Worked Example: Ambiguous Expression

Consider again the ambiguous example $r_0 = (a + ab)(b + \varepsilon)$ on input `ab`.

Forward pass. The initial state set holds a single configuration with empty bits:

$$S_0 = \Big[(r_0,\ [\,])\Big].$$

Transitioning by `a`, the first factor is not nullable, so *pderivBC* recurses into it:

$$\begin{aligned} \mathit{pderivBC}(a + ab, \mathtt{a}) &= \mathit{pderivBC}(a, \mathtt{a}).\mathit{map}(+[\mathit{false}]) \cup \mathit{pderivBC}(ab, \mathtt{a}).\mathit{map}(+[\mathit{true}]) \\ &= \Big[(\varepsilon,\ [\mathit{false}]),\ (b,\ [\mathit{true}])\Big]. \end{aligned}$$

The sequence rule then places the second factor after each derivative:

$$S_1 = \Big[(b + \varepsilon,\ [\mathit{false}]),\ (b\,(b + \varepsilon),\ [\mathit{true}])\Big].$$

Transitioning by `b`, both configurations are stepped. The first configuration reaches the second factor directly:

$$\mathit{pderivBC}(b + \varepsilon, \mathtt{b}) = \Big[(\varepsilon,\ [\mathit{false}])\Big],$$

giving bits $[\mathit{false}, \mathit{false}]$. The second configuration's derivative comes from b inside the leading sequence:

$$\mathit{pderivBC}(b, \mathtt{b}) = \Big[(\varepsilon,\ [\,])\Big],$$

and the outer sequence rule reconstructs the residual $b + \varepsilon$ with the bits unchanged:

$$S_2 = \Big[(\varepsilon,\ [\mathit{false}, \mathit{false}]),\ (b + \varepsilon,\ [\mathit{true}])\Big].$$

Selection and decoding. The first configuration is accepting, so *selectBits* returns its bits extended with $mkEpsBC_{\varepsilon} = [\,]$:

$$selectBits(S_2) = [false, false].$$

Decoding these bits against r_0:

$$decode_{r_0}\Big([false, false]\Big) = \Big(Left(a), Left(b)\Big).$$

The first *false* selects the left branch of the first factor's alternation, committing to the shorter alternative a. The second *false* selects the left branch of the second factor's alternation, committing to the shorter alternative b. The resulting tree is the non-POSIX parse tree, identical to the one produced by the standard partial-derivative parser.

Flattening validates the result:

$$flatten\Big((Left(a), Left(b))\Big) = \texttt{ab}.$$

Both partial-derivative parsers therefore produce the same tree on this example, and both differ from the derivative-based parsers of Chapter 5, which commit to the longer first alternative and return $(Right(a, b), Right())$.

7 Translation: ExtPat to Regex

The four parsers of Chapters 5 and 6 operate on the six-constructor `Regex` of Chapter 2. A real pattern uses a richer surface language. A pattern author writes shorthand character classes, bounded repetition, alternation over ranges, and sometimes a PCRE flag suffix, none of which appear in the core language. The frontend reads such a pattern and produces the `Regex` that the parsers consume.

Rejection on constructs that cannot lower. When a construct cannot be translated without changing what the pattern denotes, it is rejected with an error.

Full-string matching. A translated pattern matches the whole input or nothing. The frontend has no search padding, no substring search, and no anchor handling.

A 98-character working alphabet. Shorthand classes and negated classes enumerate a fixed set of 98 characters, printable ASCII (95) plus tab, newline, and carriage return (3). A literal character in the pattern is not translated through that set. It becomes a `Regex::Lit` that carries the full Unicode `char`. A literal `é` therefore matches `é` even though `é` is not one of the 98, while `\W` enumerates only those 98 minus the word characters and cannot match `é`. The contrast is visible in `café`. Every letter is a literal, so the pattern matches the string `"café"` as-is, including the `é`.

Two functions form the public interface. `parse_pattern` reads a bare pattern and the `parse_pcre_rule` reads a `/PATTERN/FLAGS` wrapper and applies the flag suffix. Both return either a translated `Regex` or a rejection reason.

Translation proceeds in two passes. A recursive-descent parser reads the pattern string into an intermediate type, `ExtPat`. A second pass, `translate`, lowers `ExtPat` to `Regex`. PCRE flags are applied as structural rewriting between the two passes. Figure 7.1 shows the flow.

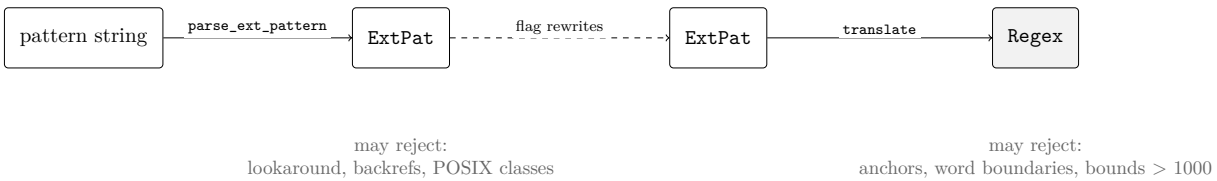


Figure 7.1: The frontend's three stages. `parse_ext_pattern` produces `ExtPat`. Dashed arrow is the optional flag-rewrite step (`i` case-folds, `s` keeps newline in `.`). `translate` lowers to `Regex`. Grey text marks each stage's rejection causes.

7.1 The External AST

The parser produces an intermediate type, `ExtPat`, that sits between the surface string and the core `Regex`. It carries precedence, grouping, quantifiers, and character classes as units, without committing to a particular lowering. The 19 variants fall into four groups.

Group	Variant	Surface syntax
Structural	`Empty`	empty branch of `a|` or `(|b)`
	`Concat`	`abc`
	`Or`	`a|b|c`
	`Group`	`( ... )`
	`GroupNonMarking`	`(?: ... )`
Quantifier	`Opt`	`r?`
	`Plus`	`r+`
	`Star`	`r*`
	`Bound`	`r{m,n}`
Class	`Any`	`[abc]`
	`NoneOf`	`[^abc]`
	`Escape`	`\d`, `\w`, `\s`, ...
	`Char`	a single literal
Atomic / special	`Carat`	`^`
	`Dollar`	`$`
	`Dot`	`.`
	`WordBoundary`	`\b`, `\B`
	`Eps`	ε
	`Never`	$\emptyset$

Table 7.1: The 19 `ExtPat` variants, grouped by role.

The quantifier variants are single-level. `r*` carries its atom but not a chain, since PCRE treats chained quantifiers as syntax errors and the parser produces none. The anchor and word-boundary variants are first-class. The parser produces `ExtPat::Carat` and `ExtPat::WordBoundary` and leaves their rejection to `translate`, so the rejection message can name the construct.

7.2 Parsing: String to ExtPat

Parsing is a single recursive-descent pass with one function per grammar level. Precedence is fixed by the call chain, from loosest to tightest.

1. alternation (`|`), left-associated;
2. concatenation (implicit), left-associated;
3. postfix quantifiers (`?`, `+`, `*`, `{m,n}`), at most one per atom.

So `abc*` parses as `ab(c*)` and `a|b|c` as `(a|b)|c`. The association is observable in the resulting `Regex` tree, and the growth measurements of Chapter 8 depend on it.

A trailing `?` after a quantifier (`r*?`, `r+?`, `r??`, `r{m,n}?`) is accepted and discarded. Laziness governs the order in which a backtracking engine explores matches. The frontend has

no backtracking engine, no capture structure, and no operation that returns a substring. So `r*` and `r*?` produce the same `ExtPat::Star`.

7.2.1 Accepted surface syntax

Table 7.2 groups every accepted surface form by how it lowers. Three mechanisms appear. *Structural* forms map to one `Regex` constructor (or a one-line rewrite of one). *Unrolling* forms clone the translated body once per repetition. *Alphabet expansion* forms lower to an `Alt` chain over characters.

Surface construct	Lowers to	Note
Structural		
literal `a`	`Lit('a')`	full Unicode `char`
`ab`	`Seq(a, b)`	
`a|b|c`	`Alt(a, Alt(b, c))`	left fold
`(...)`, `(?:...)`	body	wrapper dropped
ε	`Eps`	
$\emptyset$	`Phi`	
`r?`	`Alt(r, Eps)`	
`r+`	`Seq(r, Star(r))`	
`r*`	`Star(r)`	
`\n`, `\t`, `\r`	`Lit(ch)`	one control character
`\xHH`, `\0`–`\7`	`Lit(ch)`	octal clamped to 255
Unrolling		
`r{m,n}`	`m` copies + `n-m` optional copies	`m,n` $\leq$ 1000 (`MAX_BOUND`)
`r{m,}`	`m` copies + `Star`	`m` $\leq$ 1000
Alphabet expansion		
`.`	`Alt` chain, 98 leaves	97 unless `/s`
`\d`, `\w`, `\s`	`Alt` chain, 10, 63, 4 leaves	ASCII shorthands
`\D`, `\W`, `\S`	`Alt` chain, 88, 35, 94 leaves	alphabet minus positive set
`[abc]`, `[a-z]`	`Alt` chain over listed chars	
`[^abc]`	`Alt` chain over complement	alphabet minus listed chars

Table 7.2: Surface syntax accepted by the frontend, grouped by lowering mechanism.

Two parts of the table deserve a note.

Character classes with ranges. A leading or trailing `-` is a literal member, as is a leading `]`. Shorthand escapes inside a class expand inline as separate members and cannot be range endpoints, so `[a-\d]` is rejected.

Octal escapes inside a class. Inside a class, `\` followed by an octal digit reads up to three octal digits. A pattern such as `[\042\223\224\262\263\271]` denotes a class of codepage punctuation bytes. Reading `\042` as the three literal characters `0`, `4`, `2` would denote a different language.

Named groups. `(?<name>...)` and `(?P<name>...)` are accepted with the name dropped. The name is not recoverable from the resulting `Regex`, since `Regex` carries no capture structure.

7.2.2 Rejected surface syntax

Table 7.3 lists every rejection the frontend performs, with the pass that rejects it.

Construct	Rejected by	Reason
Lookahead `(?=`, `(?!`, lookbehind `(?<=`, `(?<!`	`parse_ext_pattern`	Not regular. No `ExtPat` constructor
Backreferences `\1`–`\9`, `(?P=name)`	`parse_ext_pattern`	Not regular. No `ExtPat` constructor
POSIX classes `[[:alpha:]]`	`parse_ext_pattern`	Silent reading as literal chars would denote a different language
Anchors `^`, `$`	`translate`	Position-dependent. No `Regex` lowering
Word boundary `\b`, `\B`	`translate`	Position-dependent. No `Regex` lowering
Bounds above 1000 (`MAX_BOUND`)	`translate`	Unrolling prohibitive
Flags `m`, `x`	`parse_pcre_rule`	`m` requires anchors this frontend lacks. `x` makes whitespace non-literal

Table 7.3: Surface syntax rejected by the frontend, with the pass that rejects it.

The split between the two passes is deliberate. Constructs with no `ExtPat` representation reject at parse time. Constructs the AST can represent but `translate` cannot lower reject at translation, so the rejection message can name the construct. One example error message for the second class is in Figure 7.2.

```
a\bb        -- word boundary '\b'/'\B' is position-dependent, not
               a regular-language construct -- unsupported
```

Figure 7.2: Example rejection message emitted by `translate` for a word boundary in the pattern.

7.2.3 One worked example

Take the pattern `[a-c]+\d{2}` (no flags). Parsing produces the `ExtPat` tree in Figure 7.3.

```
Concat [
    Plus(Any(['a', 'b', 'c'])),
    Concat [
        Escape('d'),
        Escape('d'),
    ],
]
```

Figure 7.3: The `ExtPat` produced from `[a-c]+\d{2}`.

Translation lowers each piece by the mechanism its row in Table 7.2 names.

- `Any(['a','b','c'])` alphabet-expands to a 3-leaf `Alt` chain.
- `Plus` rewrites structurally to `Seq(r, Star(r))`.

- `Escape('d')` alphabet-expands to a 10-leaf `Alt` chain (one per digit), applied twice.
- The two `Concat` layers fold left into one `Seq` chain.

The resulting `Regex` is 54 nodes, the third entry in Table 7.5. The four remaining digit leaves contribute 40 of those nodes; the rest is `Plus`'s `Seq-Star` wrapper (3 nodes) and the surrounding `Seq` structure (11 nodes).

```
Seq(                                                    -- outer Concat
  Seq(                                                  -- [a-c]+
    Alt(a, Alt(b, c)),                                  -- [a-c], 5 nodes
    Star(Alt(a, Alt(b, c)))),                           -- Plus desugar
  Seq(                                                  -- \d{2}
    Seq(                                                -- first copy
      Eps,                                              -- unroll seed
      Alt(0, Alt(1, Alt(2, Alt(3, Alt(4,
          Alt(5, Alt(6, Alt(7, Alt(8, 9)))))))))),      -- \d, 19 nodes
    Alt(0, Alt(1, Alt(2, Alt(3, Alt(4,
        Alt(5, Alt(6, Alt(7, Alt(8, 9)))))))))          -- \d, 19 nodes
```

Figure 7.4: The full `Regex` produced by translating `[a-c]+\d{2}`. 54 nodes, matching the third entry of Table 7.5. The two `Eps` leaves come from `repeat_exact` seeding its unrolling at `Eps`; `translate` does not simplify them away.

7.3 PCRE Flags

A pattern passed to `parse_pcre_rule` may be wrapped in a `/PATTERN/FLAGS` form. The flag suffix is read and sorted into three groups.

Flag	Group	Effect
`i`	honoured	case-fold every literal letter
`s`	honoured	`.` includes newline
`m`	rejected	would change anchor meaning this frontend lacks
`x`	rejected	would change whitespace meaning this parser lacks
`R`	ignored	names a search this frontend does not perform
`A`	ignored	names an anchor this frontend does not perform
`E`	ignored	tunes a `$` this frontend rejects
`U`, `H`, `P`, `C`, ...	ignored	names a buffer or normalisation strategy

Table 7.4: The three flag groups. An ignored flag leaves the parsed pattern exactly as it would be without the flag.

A flag that changes what the pattern denotes is honoured. A buffer selector such as `U` names which region of the input to test against, not what the pattern itself matches. Under full-string matching there is only one region. A relative-position flag `R` names how to position a subsequent search, which does not exist in this frontend. Both leave the language unchanged. An honoured flag such as `i` changes the language directly and is applied.

Honoured flags run between parse and translate as structural rewrites over `ExtPat`. `case_fold` turns every literal letter into a two-element `Any`. `strip_dot_newline` turns

every `Dot` into `NoneOf(\n)`.

7.4 Translation: ExtPat to Regex

`translate` is a structural pass over `ExtPat`. Every accepted form lowers by one of the three mechanisms introduced in Section 7.2's accepted-syntax table. Three parts need more care than a single row can convey.

7.4.1 Character classes and the working alphabet

An `ExtPat::Any(cs)` lowers to an alternation over `cs`, one `Alt` node per element folded left. `NoneOf(cs)` first takes the complement of `cs` against the working alphabet and then does the same. `Dot` lowers to `Any` over the full alphabet when the `s` flag is set, and has already been rewritten to `NoneOf(\n)` otherwise. Shorthand escapes (`\d`, `\w`, `\s` and their capitalised forms) lower through the same path.

`Dot`, the shorthand classes, and any negated class therefore cannot match a character outside the working alphabet. A pattern with a non-ASCII literal is unaffected, since a direct literal lowers to a single `Regex::Lit` that carries the full `char`. The alphabet constrains only the constructs that need to enumerate a set.

7.4.2 Bounds and unrolling

`ExtPat::Bound` is the only quantifier without a direct core construct, since `Regex` has no `{m,n}` form. It lowers by unrolling. `r{m,n}` becomes `m` exact copies of the translated body, followed by `n-m` optional copies. `r{m,}` becomes `m` exact copies followed by a `Star`. Both forms clone the body, so the resulting `Regex` has size proportional to the bound.

Two caps guard this unrolling. A bound whose upper or lower endpoint exceeds 1000 is rejected rather than unrolled. `a{1001}` rejects with a message naming the cap, before any translation runs. A second cap is applied by `core_regex_tagged`, the function Chapter 8's dataset preparation calls (`src/data/prepare.rs`). That cap rejects a pattern if the translated `Regex` would exceed 20,000 nodes or an estimated recursion depth of 2000. `translate` itself has no node-count limit. The 20,000-node cap is that pipeline's own decision.

7.4.3 Anchor recovery

Translation is where anchors are rejected, with one exception. After unwrapping any top-level `(...)` or `(?:...)` shell, if the pattern is a `Concat` whose first element is `Carat` and last element is `Dollar`, those two anchors are stripped before translation. Under full-string matching they are provably redundant. Consuming the whole input already requires both endpoints, so removing them changes nothing about what the pattern denotes. Both `^abc$` and `(^abc$)` strip to the same `ExtPat`, and both are accepted. `^$` alone strips to `Eps`.

Every other anchor placement reaches `translate` and is rejected there. Four shapes appear.

- Start-only, `^abc`. The opening anchor is present but nothing closes the wrap.

- End-only, `abc$`. The closing anchor is present but nothing opens the wrap.
- Nested, `(a^|b)c` or `(a^b)*c`. An anchor inside a group, alternation branch, or quantifier. After shell unwrapping the top level is not a `Concat` whose first element is `Carat` and last element is `Dollar`, so nothing is stripped.
- Multiple pairs, `^a$b$`. The outermost pair strips, but the inner `Dollar` remains in the stripped expression and `translate` rejects it.

`parse_pattern` and `parse_pcre_rule` strip no anchors at all. Both reject every anchor they encounter, the recoverable wrap included. A caller that has reason to recover the redundant wrap does so before invoking `translate`.

7.5 Translation Size

Classes expand to alternations and bounds unroll, so the translated `Regex` can be much larger than the source pattern. The demo `demo_translate_sizes` measures this directly.

Pattern	Nodes	Pattern	Nodes
`a*`	2	`\d`	19
`abc`	5	`\d{2}`	41
`a{2,4}`	15	`[a-c]+\d{2}`	54
`a+`	4	`\w`	125
`a?`	3	`[^a]`	193
`a{2}`	5	`.`	195
`a{2,}`	8	`\w+`	252

Table 7.5: Translated size in `Regex` nodes for representative patterns.

The table has two halves. The left seven patterns translate without expanding through the alphabet, and each grows only with its own structure. The right seven use at least one alphabet-wide construct (`\d`, `\w`, `[^a]`, `.`), so their node counts are dominated by the 98-character expansion.

Three reference points. `\d` is 19 nodes: 10 digits plus 9 `Alt` wrappers. `\w` is 125 nodes: 63 characters plus 62 wrappers. `[^a]` is 193 nodes: 97 characters plus 96 wrappers.

Two facts follow. A class costs roughly two orders of magnitude more than one literal, so `[abc...]` and `a|b|c|...` translate to the same tree but with very different source size. Case folding doubles the literal count, since each letter becomes a two-element class, so `/abc/i` translates to 11 nodes against `/abc/`'s 5.

8 Evaluation

This chapter evaluates the four parser variants implemented in Chapters 5 and 6. The four candidates are:

- the standard derivative-based parser (`deriv_std`, direct and loop variants),
- the bit-coded derivative-based parser (`deriv_bc`),
- the standard partial-derivative parser (`pderiv_std`),
- the bit-coded partial-derivative parser (`pderiv_bc`).

Three aspects are covered. Section 8.1 describes the two data sources the rest of the chapter draws on: a real, human-authored corpus with its own given examples, and a hand-curated synthetic set built specifically to stress the parsers' asymptotic behaviour. Neither source needs a candidate-input generator of the kind a corpus without given examples would require; the two differ in how their inputs come to exist, not in whether they are real measurements. Section 8.2 then checks, on the prepared corpus, how often the POSIX parsers and the non-POSIX parsers agree on the shape of the parse tree they build for the same pattern and input. The main focus of the chapter, however, is Section 8.3, which compares the four variants on execution time and memory consumption, on both the synthetic set and the real corpus.

The chapter is organised as follows. Section 8.1 describes both datasets and the pipeline that prepares the real one. Section 8.2 verifies parser agreement. Section 8.3 presents timing and memory results. Section 8.4 interprets them.

All measurements are produced by binaries under `benches/` and `examples/`, run with a fixed `-release` profile. The corpus and synthetic measurements are plain reporting binaries run once per source, for the reasons given in `BENCHMARKS.md`, not Criterion targets.

8.1 Dataset Preparation

8.1.1 Two datasets, not one

The evaluation draws on exactly two pattern sources, chosen for two different jobs.

S2ELab is a real, human-authored benchmark: 762 patterns, each with a natural-language description and the author's own labeled `matches`/`non_matches`.[1] It supplies

[1]Published as *Re(gEx|DoS)Eval: Evaluating Generated Regular Expressions and their Proneness to DoS Attacks*, ICSE-NIER 2024 (DOI: 10.1145/3639476.3639757); dataset at https://github.com/s2e-lab/RegexEval. (TODO: replace this footnote with a `references.bib` entry and a proper citation

realistic measurements — how the four parser variants behave on patterns a person actually wrote, with inputs that person actually judged to match or not. Section 8.1.2 explains why it was chosen over the other candidate sources, and Section 8.1.4 describes the pipeline that turns it into benchmark input.

The pathological stress set is the opposite kind of dataset: 113 hand-curated patterns, organised into categories by the specific complexity mechanism each one is built to trigger, with no given examples at all. Its job is *controlled* measurement — holding every variable fixed except the one a category exists to stress (input length, pattern size, or both) and sweeping it by a chosen, reproducible amount. Section 8.1.9 describes it.

Both are read-only inputs to the benchmarks; neither is synthesised at benchmark time. The distinction that matters for the rest of this chapter is not "real versus fake" — every pattern in both sets is a real regular expression, benchmarked with the real parsers — but *where the inputs come from*: S2ELab's inputs are given by a human author and kept as given; the pathological set's inputs are derived mechanically from a pattern's own stress parameter, because no corpus of real-world patterns reliably contains an example of, say, `(a+)+` swept across ten input lengths on request. Every table in Sections 8.2–8.3 is labeled by which of the two it was measured on, and the two are never pooled into one figure.

Four other sources — Snort/Suricata, SpamAssassin, RegexLib, and Innovatorved — were extracted and translated at an earlier stage of this project and remain readable by the frontend's own diagnostic tooling (`dataset_stats`), but no longer feed either benchmark dataset: Snort and SpamAssassin ship no given examples at all, and RegexLib's and Innovatorved's labels carry no author guarantee. Using either would have reintroduced exactly the heuristic input-generation machinery Section 8.1.2 explains this evaluation now avoids.

8.1.2 Why S2ELab

A benchmark corpus for a membership-deciding parser needs, for every pattern, both a string that is claimed to match and one that is claimed not to. A source that does not ship such pairs has to manufacture them — walk the pattern's own language and hope the walk is representative, or corrupt a positive and hope the corruption is a meaningful negative. Either way, the resulting input reflects the sampler's own bias, not a judgement any person made about the pattern. S2ELab is the one source available to this project that avoids that problem entirely: every one of its 762 patterns ships with the original author's own `matches` and `non_matches`, so the benchmark corpus is built from real judgements end to end, with no generation step in between.

The second reason is specific to the full-string-matching model fixed in Chapter 7. Table 8.1 compares all five candidate sources by what fraction of their accepted patterns carry an unambiguous top-level `^...$` wrap — the one anchor shape Chapter 7's frontend can recover losslessly under full-string matching (Section 7.4.3). S2ELab is the clear outlier: 80.1% of its accepted patterns carry that wrap, against 59.4% for RegexLib and under a third for the three search-oriented sources. A source's own authors write for a search-based engine by default and mark full-match intent only when they mean it; S2ELab's patterns are written to be tested against committed labels, so authors mark

once the full author list is confirmed from the paper directly.)

intent far more often. That difference is exactly what this project's full-string-matching model needs most.

Source	Full-match wrap	%
Snort	459	22.0
SpamAssassin	23	14.8
RegexLib	1291	59.4
Innovatorved	1445	30.1
S2ELab	**463**	**80.1**

Table 8.1: Of each source's accepted patterns, how many carry an explicit top-level `^...$` wrap. S2ELab's patterns are written against committed labels and mark full-match intent far more consistently than the four search-oriented sources.

8.1.3 Anchors and full-match intent

Chapter 7 already fixed how anchors are treated: a pattern is rejected if it contains `^` or `$` anywhere except as a single top-level `^...$` wrap around the whole pattern, in which case the wrap is provably redundant under full-string matching and is stripped before translation (Section 7.4.3). Nothing in this chapter changes that mechanism; what this section adds is which of its two outcomes a given pattern reaches, and what that means for which results get reported.

S2ELab. Regular-expression authors commonly signal "this whole string is what I mean" by wrapping a pattern in `^...$`, and many S2ELab entries follow that convention — the worked example in Section 8.1.5 is one. Every accepted S2ELab pattern therefore falls into one of two groups: 463 carry the top-level wrap and have it stripped by Chapter 7's recovery mechanism, landing on the engine's native full-match reading with no ambiguity about what was intended; the remaining 115 carry no anchor at all and are translated unchanged, still matched under this project's only semantics (full-string), but with no way to tell from the pattern itself whether the author meant a full match or a search that this project does not implement. Both groups are prepared and written to the corpus, since dropping either at preparation time would foreclose a methodological choice that belongs in the benchmarks, not the pipeline.

That choice is made once, uniformly: every benchmark in this chapter defaults to measuring the 463 full-match-wrapped patterns only, because reporting a number for the remaining 115 would silently assume a reading of the pattern its own author never committed to. (Both groups remain fully available on opt-in, since the choice is a default, not a dataset deletion.) Section 8.2 and Section 8.3 therefore report S2ELab figures over 463 patterns, not 578.

The pathological stress set. The same question does not arise there in the first place. Every pathological pattern is hand-written directly against the engine's own full-string-matching model — there is no external authoring convention to interpret, because the patterns were authored for this project. None of them contains an anchor: verified directly against Chapter 7's own frontend, `^`, `$`, word boundaries, backreferences, lookaround, and POSIX classes all hard-reject at parse or translate time, so none of those constructs appear in the set at all. The pathological set complies with Chapter 7's model by construction, not by a recovery step applied after the fact.

8.1.4 The S2ELab pipeline

No input is generated, derived, or heuristically assembled for S2ELab. `run_pipeline` turns the raw JSON into the benchmark corpus in seven steps.

1. **Extract.** Each entry's pattern and its author's own `matches`/`non_matches` become one `ExtractedRule`.
2. **Deduplicate.** A pattern byte-identical to one already seen this run is skipped; its fate is already decided.
3. **Translate.** `core_regex_tagged` (Chapter 7) parses and translates, applying the one anchor recovery described in Section 8.1.3. A pattern the frontend rejects stops here.
4. **Deduplicate examples.** A string appearing in both `matches` and `non_matches` is kept once, as a positive — the stronger claim.
5. **Check labels.** Every surviving example is re-checked against the real translated `Regex`. A disagreement is dropped, not corrected: the dataset's claim did not hold under this project's own semantics.
6. **Balance per pattern.** Described in Section 8.1.7.
7. **Write.** Each surviving pair becomes one line of `prepared.jsonl` under `data/processed/s2-elab` tagged with which of the two anchor groups it came from.

8.1.5 A worked example through the pipeline

One entry is traced end to end, so the later tables have a concrete referent. S2ELab entry 100 (fields trimmed):

```
{
  "expression":
    "^(0[1-9]|1[0-2])\/((0[1-9]|2\d)|3[0-1])\/(19\d\d|200[0-3])$",
  "matches": ["12/31/2003", "01/01/1900", "11/31/2002", ...],
  "non_matches": ["1/1/2002", "01/01/02", "01/01/2004",
                  "12/123/4322", ...]
}
```

The natural-language description behind this entry asks for a date between January 1900 and December 2003 in `MM/DD/YYYY` form; the pattern is the author's own attempt at that, anchors included.

Extract. One `ExtractedRule` carrying the raw expression and the two given lists unchanged.

Translate. The pattern's top level is a single `Concat` opening with `^` and closing with `$`, so Section 7.4.3's recovery applies: both anchors are stripped before translation, and the entry is tagged as a full-match pattern.

Check labels. Every given example is run through the translated `Regex`. The check is against the pattern's own semantics, not against real-world correctness: `"11/31/2002"` is listed as a `match`, and *is* one under the translated pattern, since the day group (`(0[1-9]|2\d)|3[0-1]`) accepts `31` regardless of which month group matched — the pattern never correlates the two, so it silently accepts a November 31st that the calendar does not have. The label check keeps this example, correctly: the author's claim and the pattern's own language agree, and whether the pattern itself encodes a real calendar is a question the check cannot and does not answer. A different S2ELab entry's claim can still disagree with its own translated pattern — that is what `inputs_label_mismatch` counts — but this one does not illustrate it.

Balance. The entry's surviving matches and non-matches are truncated to the shorter side's count, as Section 8.1.7 describes for the whole corpus.

Write. The surviving pairs are appended to `prepared.jsonl`, each line carrying the pattern, the input, the claimed outcome, and the full-match tag.

This worked example is the entry the rest of the chapter refers to when it discusses a label check in concrete terms, and it is also why the check exists: a real, human-authored corpus contains real human mistakes, and benchmarking against an uncorrected label would silently fold that mistake into every parser's measured accuracy.

8.1.6 Corpus overview

Chapter 7 (Table **??**, Table **??**) already gives the extraction and acceptance counts: 762 extracted, 7 duplicate, 177 rejected, 578 accepted (463 full-match wrap recovered, 115 no anchor at all), with the rejection causes broken down there. What follows is specific to this chapter's own pipeline, downstream of acceptance.

Stage	Count
Accepted (from Chapter 7)	578
Usable (at least one match and one non-match surviving the label check)	572
Given examples written to the corpus	13,248
dropped, label disagreed with the real `Regex`	255
dropped, per-pattern balance (Section 8.1.7)	779

Table 8.2: S2ELab corpus preparation downstream of acceptance.

8.1.7 Per-pattern balance

A pattern with 40 given matches and 2 given non-matches would, left as given, contribute 40 positive measurements and 2 negative ones — not because matching is 20 times more representative for that pattern, but because its author happened to write more positive examples. After the label check, each usable pattern's longer side (matches or non-matches) is truncated to its shorter side's count, keeping the first surviving examples in their original order. A pattern with fewer than one example on either side is excluded entirely rather than balanced against zero.

Patterns are *not* balanced against each other: a pattern with 50 surviving examples on each side keeps all 50, even though another pattern in the same corpus has only one

of each. An earlier version of this pipeline did also truncate every pattern down to the single rarest pattern's own count, so that one thinly-labeled pattern dragged every other pattern in the corpus down to its level; on the real S2ELab data this meant almost every pattern lost nearly all of its given examples to equalize against a single outlier. That stage is removed. Section 8.1.8 explains what replaced it, and why the replacement needs no truncation at all.

8.1.8 Why macro-averaging, not more truncation

Removing the dataset-wide truncation reintroduces the imbalance it was built to avoid, just one level up: once patterns are allowed to keep however many examples they actually have, a pattern with 50 surviving examples contributes 50 times as many individual measurements to a flat, pooled average as a pattern with 1. A flat mean over every measured input would then be dominated by whichever patterns happen to be the most heavily labeled, which has nothing to do with how hard any of them is to parse.

The fix is to change what is averaged, not to throw inputs away to force every pattern to the same count. Section 8.3.2 and Section 8.3.3 compute each pattern's own mean first, then take the unweighted average of those per-pattern means — a *macro-average*. Every pattern counts once in that average, regardless of how many given examples it happened to ship with, and no example is discarded to make the counts line up. The median, 99th-percentile, and maximum figures reported alongside the mean are left as flat statistics over every individual measured input, since they answer a different question on purpose: not "how costly is the typical *pattern*", which the macro-average answers, but "how bad does the single worst *input* get", where weighting by pattern is not the relevant correction.

8.1.9 The pathological stress set

Section 8.1.1 introduced the second dataset as a hand-curated, template-generated set rather than a corpus. This section describes how it is built and what it is for.

A real corpus answers "how do the parsers behave on patterns people actually write"; it does not reliably answer "how does `deriv_std`'s state size grow as input length doubles", because no real author sets out to write ten variants of `(a+)+` at ten chosen lengths. The pathological set exists to answer exactly that second kind of question: each row is a single, deliberately chosen regular expression, built to isolate one specific complexity mechanism (nested quantifiers, overlapping alternation, bounded repetition over a star, and so on), together with a rule for turning one swept parameter — an input length n, a pattern size p, or both — into a concrete (pattern, input) pair at any scale the evaluator chooses.

For most rows, only the input scales: the regular expression itself is fixed (`(a+)+`, say), and n repeats of a character are fed through it. A smaller number of rows need the *pattern* itself to scale — the family that forces full enumeration of a near-miss tail, for instance, is only pathological because its quantified group is repeated some number of times, so that repeat count has to be the swept parameter, substituted directly into the pattern template rather than only into the input. Both cases reuse the same small token language: `{N}`/`{P}` inside an input specification is a bare number; inside a pattern template, `{{N}}` keeps the regular expression's own literal brace pair around that number so that, for example, a template written as `(a?){{N}}a{{N}}` becomes the concrete pattern `(a?){20}a{20}` at $n = 20$.

113 patterns are organised into 12 categories (Table 8.3), named by the specific mechanism each one stresses rather than by source. Each category carries its own recommended sweep, calibrated against a real measured run rather than chosen a priori: a routine, un-overridden benchmark pass is meant to actually cross from "every parser measures in microseconds" to "the standard derivative parser is gone" within that category's own default points, not stop short of the transition the way an arbitrarily small sweep would.

Cat.	Name	Mechanism	Patterns
A	Nested quantifiers	Quantified group inside another quantifier	16
B	Overlapping alternation	Alternatives sharing a prefix under a quantifier	15
C	Bounded rep over star	`{m,n}` wrapping a `*`/`+` body, unrolled at translate time	7
D	Anchored-failure style	Must fully consume input; near-miss tail forces full enumeration	4
E	Prefix-overlap alternation	One alternative is a strict prefix of another	8
F	Alternation-star + literal	`(alt)*`/`(alt)+` then a literal that fails late	7
H	Many adjacent stars/groups	Product of `.+`, `.*`, `\w+` etc. in sequence	4
I	Trivial pattern, large input	Simple pattern, very long input (throughput baseline)	10
J	Large pattern, trivial input	Thousands of literals/groups, short input	5
K	Large pattern, large input	Both axes at once	4
L	Trivial pattern, trivial input	Baseline/sanity check	33
Total			113

Table 8.3: The 12 pathological-set categories (`A`–`L`, skipping `G`). `I`/`J`/`K`/`L` form a 2x2 matrix of (pattern size: trivial/large) × (input size: trivial/large), with `L` the all-trivial baseline; the remaining seven categories are the ones with a genuine blow-up mechanism to stress.

One category deserves a specific note, since it is the chapter's own reproduction of a published result. Two rows in category E (`(a|b|ab)*`/`(a|b|ab)+`, swept into the millions of characters) are a literal reproduction of Sulzmann and Lu [2014]'s own Figure 7, and are deliberately excluded from category E's routine default sweep — at that scale they would slow down every unrelated run through the same category — in favour of an explicit, individually-invoked sweep reported in Section 8.3.2.

8.1.10 Real and synthetic, kept separate

The two datasets described above answer different questions and are never pooled into one figure. S2ELab (572 usable patterns, 13,248 real given inputs, all of it a real author's own judgement) answers how the four parser variants behave on regular expressions people actually write. The pathological set (113 hand-curated patterns, 913 test cases under each category's own default sweep, scalable to millions of characters for a specific reproduction) answers how each variant's cost actually grows as a chosen stress parameter is swept, something no corpus of real patterns reliably exercises on request. Every table in the sections that follow names which of the two it was measured on.

8.2 Parser Agreement

Every case in the prepared corpus already carries a verified outcome, so membership is a precondition rather than something this section establishes. The question here is the shape

of the parse tree. On a matching case, the POSIX parsers and the non-POSIX parsers may build different trees — the same input can admit more than one valid decomposition — but some patterns and inputs leave neither policy a choice, forcing both onto the same structure. This section measures how often that happens across the corpus.

8.2.1 Method

The check is implemented in `demo_parser_agreement`. For every prepared case, all five parsers — `deriv_std_rec`, `deriv_std_loop`, `deriv_bc`, `pderiv_std`, `pderiv_bc` — run on the same (pattern, input) pair.

Two conditions are checked as preconditions. First, every parser returns the case's verified membership outcome. Second, within each family the parsers agree with each other on the tree: the three POSIX variants build the same tree, and the two non-POSIX variants build the same tree. A violation of either condition is a bug, not a measurement.

The measured quantity is cross-family tree agreement: of the cases where both families matched and were internally consistent, how many produced the same tree. The two families apply different disambiguation policies, so a difference is expected where the input admits several decompositions; a match where the input admits only one forces both families onto the same structure.

Each parser call runs in a subprocess so that a stack overflow or an infinite loop is contained. A 2-second wall-clock timeout kills the process. Timeouts and crashes are counted separately from tree-shape outcomes, since neither produces a result to compare.

The two datasets are run through this check separately, with separate binaries, and never pooled into one row: `demo_parser_agreement` against the 463 full-match-wrapped S2ELab patterns (the same default scope as every other benchmark in this chapter, Section 8.1.3), and `demo_parser_agreement_pathological` against the 113-row pathological set.

```
cargo build --release --example demo_parser_agreement_worker
cargo run --release --example demo_parser_agreement
cargo run --release --example demo_parser_agreement_pathological
```

8.2.2 Results

Table 8.4 reports a real run of both checks, one row per dataset (pathological broken out by category in Table 8.5). *Match agr.* is the membership precondition (every parser's verdict against the case's verified outcome); *Shape agr./disagr.* is the measured cross-family quantity, reported only over cases where both preconditions held. Within-family disagreement (*POSIX*/*non-POSIX*) is a hard precondition the tool asserts rather than a percentage it reports: a violation prints as a bug line in the run's output, not a count in these columns.

8.2.3 Discussion

Both preconditions hold everywhere in the final data: every parser's membership verdict matches the verified outcome (100% on both datasets), and no within-family shape violation was printed by either run. That second claim came with one real correction along the

Dataset	Entries	Timed out	Crashed	Match agr.	Shape agr.	Shape disagr.
S2ELab (full-match)	13 248	82	0	100.0%	99.7%	0.3%
Pathological (total)	869	443	1	100.0%	89.6%	10.4%

Table 8.4: Parser agreement, measured once per dataset and never pooled. S2ELab: 6624 matching / 6624 non-matching cases, 6527 of 6544 compared cases built the same tree across families (17 *Diff*). Pathological: category breakdown in Table 8.5; 182 of 202 compared cases agree (one category, B, accounts for 18 of the 21 *Diff*). The single pathological crash is `deriv_std_rec` stack-overflowing on a 10-million-character input in category I — the same failure mode as Section 8.3.2's FIG7A/FIG7B finding, confirmed here on a trivial pattern, so it is a property of input length alone, not of `(a|b|ab)*`'s specific ambiguity.

Cat.	Entries	Timed out	Match agr.	Shape agr.	Shape disagr.	Compared
A	160	112	100%	100%	0%	27
B	150	95	100%	35.7%	64.3%	28
C	56	32	100%	100%	0%	12
D	40	12	100%	100%	0%	14
E	80	45	100%	87.5%	12.5%	16
F	70	46	100%	91.7%	8.3%	12
H	28	20	100%	100%	0%	3
I	60	44	100%	100%	0%	3
J	16	4	100%	100%	0%	6
K	20	14	100%	100%	0%	3
L	189	19	100%	100%	0%	78

Table 8.5: Pathological-set agreement by category (one crashed entry, category I, omitted from this breakdown's totals; see Table 8.4). Category H's numbers are post-fix (see Section 8.3.2's discussion of `P29`/`P31`).

way, worth recording rather than silently fixing: an earlier pass of the pathological check flagged category H at only 75% match agreement, with two rows, `P29` (`(.+){10}`) and `P31` (`(.*){10}`), where all five parsers unanimously returned *match* against an input the dataset's own stored label called a non-match. Hand-checking both patterns confirmed the parsers were right and the dataset was wrong: the non-match input for both rows was built by appending a trailing character to an otherwise-matching string, a construction that only forces a non-match when the pattern bounds the total length it can consume. `.` matches any single character including the appended one, and ten unbounded `.+`/`.*` groups impose no such bound, so any sufficiently long input matches regardless of what follows it — `P31` in fact has no genuine non-match at *any* length, since ten `.*` groups accept the empty string. `P29`'s non-match input was replaced with a fixed 9-character string, one short of the ten-group minimum the pattern actually needs to match (a constraint `P29`'s own pre-existing notes already stated, just not in the input construction); `P31`'s non-match side was removed outright rather than replaced with another wrong one. Table 8.5 reports category H after this fix: 100% match agreement, consistent with every other category. No other row or category was affected.

Where a cross-family tree-shape disagreement does occur, it concentrates sharply rather than spreading evenly: S2ELab's 17 *Diff* cases are 0.3% of 6544 compared, but the pathological set's are 10.4% of 202, and within the pathological set, category B alone accounts for 18 of the 21 (64.3% of its own 28 compared cases) — exactly the category built around alternatives sharing a prefix under a quantifier (Table 8.3), the single most reliable way to construct genuine ambiguity on purpose. Categories E and F, also alternation-heavy by design, show smaller but still nonzero disagreement (12.5% and 8.3%); every other category — A, C, D, H, I, J, K, L, none of them built around alternative-sharing

structure — shows exactly 0% disagreement. This is the expected reading, not a surprise: *Diff* reflects a real difference in disambiguation policy between the POSIX and non-POSIX families, visible exactly on the cases the pathological set was deliberately shaped to make ambiguous, and absent everywhere a pattern leaves only one way to decompose a match.

8.3 Performance Analysis

The four parsers are compared on two axes: execution time (Section 8.3.2) and memory consumption (Section 8.3.3). Both axes are measured on synthetic inputs first, since synthetic families isolate a single cost, and on the corpus second, since the corpus is the environment the parsers will encounter in practice.

8.3.1 Cost drivers

Before the measurements, it is worth separating the properties of a corpus that make the four parsers cost something. Three are distinguishable, and each is exercised to a different degree by the corpus of Section 8.1 and by the synthetic families of Section 8.3.2.

Input length. The two standard derivative parsers retain every intermediate expression until the backward pass reconstructs the tree, so their live state grows with the number of characters consumed. The bit-coded and partial-derivative parsers do not carry that history. A corpus of long inputs separates the two groups; a corpus of short inputs does not, however many entries it contains.

Pattern structure. Translation size and derivative cost scale with the pattern's own node count. Deeply nested groups, alternation over many branches, and repeated classes inflate the translated `Regex`, and every parser pays for that size on every input. This driver is independent of input length: a short input against a large pattern can cost more than a long input against a small one. It also splits into two phases this evaluation measures separately, since they are paid at different times and by different code: *translation* (the PCRE string into the translated `Regex`, `core_regex` in Chapter 7), measured once per distinct pattern by `bench_corpus_translate`/`bench_pathological_translate`; and the *parser's own derivative loop* over an already-translated `Regex`, measured per (pattern, input) pair by the timing and memory benches below. A pattern can be cheap to translate and expensive to derive, or the reverse — the pathological categories **C**, **D**, **J**, and **K** are specifically built around the first phase (bounded-repetition unrolling and large-tree construction in `core_regex` itself), while **A**, **B**, **E**, **F**, and **H** are built around the second.

Ambiguity. When a single input admits more than one valid decomposition of the pattern, the parsers keep more state alive and must eventually choose among the surviving trees. The choice is governed by the parser family's disambiguation policy (Chapter 5 versus 6), and the two families are visible on exactly those (pattern, input) pairs where the policies diverge. Ambiguity is a property of a pattern, not of a corpus; the pathological categories built around overlapping or prefix-sharing alternation (**B**, **E**, **F**) are what isolate it at a controlled scale.

Not every cost is a driver. There is also a fixed per-call cost paid once per (pattern, input) pair regardless of length, structure, or ambiguity: allocating the initial residual, setting up the derivation loop, and constructing the parse tree from the final nullable expression.

This cost is why the four parsers differ even on the corpus's own shortest, simplest entries, where none of the three drivers above is active enough to matter.

The pathological categories of Section 8.1.9 are each shaped to isolate one of these costs at a scale no real corpus reliably reaches; the corpus tables that follow mix all three, and reading them without this separation makes the numbers hard to interpret. S2ELab's patterns are short, human-authored rules with modest structural complexity and short given inputs, so it exercises all three drivers only mildly — which is exactly why a second, deliberately extreme dataset exists at all (Section 8.1.1).

8.3.2 Timing

Pathological benchmarks

Synthetic timing is measured by `bench_pathological_time` against the 113-row stress set of Section 8.1.9, grouped by category (Table 8.3). Selection and sweep are controlled by this file's own `PATHO_*` variables — `PATHO_SECTION`, `PATHO_CATEGORY`, `PATHO_ROW_ID` to narrow to one or a few rows, and `PATHO_N`/`PATHO_P` to override a row's default sweep — plus the same `BENCH_PATTERN_LIMIT`/`BENCH_ENTRY_TIMEOUT_MS` used by the corpus benches below. There is no `BENCH_MATCH_INTENT` here: these patterns are authored directly in full-match form, with no anchor to strip (Section 8.1.3).

```
cargo bench --bench bench_pathological_time
PATHO_CATEGORY=A,B cargo bench --bench bench_pathological_time
```

Table 8.6 reports a real routine, un-overridden pass per category, each row's own default sweep from Table 8.3. `Timed out` is the share of that category's entries that hit the 2-second per-entry cap; the `mean` column is computed only over the entries that completed, same convention as Table 8.9 — a low mean next to a high timeout share describes the easy entries that survived, not category-wide speed.

		std_rec		std_loop		deriv_bc		pderiv_bc		pderiv_std	
Cat.	Entries	mean	to%	mean	to%	mean	to%	mean	to%	mean	to%
A	160	189.5	70%	156.8	71%	159.7	1%	15.4	0%	17.1	0%
B	150	167.3	63%	165.8	63%	98.3	4%	95.7	0%	98.4	0%
C	56	88.1	57%	87.9	57%	240.1	25%	230.3	18%	218.5	18%
D	40	253.5	20%	250.7	20%	137.4	0%	192.4	15%	204.3	15%
E	80	293.4	56%	288.0	56%	78.8	0%	57.5	0%	61.7	0%
F	70	392.8	66%	366.7	66%	42.9	40%	104.1	3%	102.1	3%
H†	32	290.8	75%	290.4	75%	289.8	13%	265.7	22%	189.7	25%

Table 8.6: Mean per-entry time (ms, completed entries only) and timeout share, one real routine pass per category at each category's own default sweep. `std_rec`/`std_loop` abbreviate `deriv_std_rec`/`deriv_std_loop`; `to%` is the percentage of that category's entries that timed out (0 crashes in A–H). †H's row is from the run that surfaced the `P29`/`P31` labeling bug of Section 8.2 (32 entries, one row's non-match side not yet removed); the fix drops one input but does not change how long any of the remaining ones take to parse, so the timing figures stand, just against a since-corrected entry count (28 after the fix).

Three things stand out against the design intent of Section 8.1.9. First, **A**, **B**, **E** confirm the intended shape exactly: both standard variants time out on roughly 56–71% of entries while `deriv_bc` and both partial-derivative variants stay near 0%, so the routine default sweep already shows the standard construction failing without stressing the other three. Second, **F** is the one routine category where `deriv_bc` itself times out at a real rate

(40%, against under 4% for the partial-derivative pair) — by design, per Table 8.3's own `suggested_n_notes`: F is calibrated specifically to show the F-family mechanism defeating bit-coding too, not only the standard construction, which the routine pass now confirms directly rather than only in the individually-swept `FIG_F` sweep below. Third, **C**'s timeout rate is close to uniform across *all five* parsers (57% for both standard variants, 18–25% for the other three) rather than concentrated on the standard pair — consistent with Section 8.3.1's prediction that C's cost is dominated by `core_regex`'s own bound-unrolling at translation time, a cost every parser pays identically before its own derivative loop ever runs, rather than by a difference in derivative-loop efficiency. Fourth, **H** is the one category where the usual ranking inverts: `deriv_bc` times out *less* often (13%) than either partial-derivative variant (22%, 25%), the opposite of every other category in this table. H's mechanism — several unbounded groups or stars placed adjacently rather than nested — is exactly the frontier-of-residuals shape the partial-derivative construction is usually best at, so this is not evidence against the general finding of Section 8.4.3; it is a reminder that "usually best" is a property of the common case, not a guarantee, and one small (4-pattern) category is enough to show an exception. It is also, independently, the category where the `P29`/`P31` labeling bug above was caught, which is not a coincidence: H is the smallest and least-tested category, exactly where a dataset bug is most likely to survive unnoticed.

Category H's timing numbers are reported separately in the discussion above rather than in Table 8.6, since this run also caught a real labeling bug in two of H's rows (`P29`, `P31`) via the agreement check of Section 8.2 — see that section for the full account and the fix.

Individually swept representatives. A routine per-category pass uses each category's conservative default sweep, calibrated to keep the whole routine run fast; it is not where any individual pattern's own asymptotic behaviour is best seen. For that, six rows are singled out for a dedicated, individually invoked sweep that pushes well past the routine default, using `PATHO_ROW_ID` to isolate one (or a few related) rows and `PATHO_N` to choose the swept points directly (full commands and the reasoning behind each choice in `docs/BENCHMARKS.md`):

- **P60**/**P61** (category E): the literal Sulzmann and Lu [2014] Figure 7 reproduction, `(a|b|ab)*` swept into the millions of characters.
- **P20** (category E, labelled `FIG7C`): the same family's `+` variant, pushed to the same paper scale.
- **P01** (category A, labelled `FIG_A`): the single most-cited ReDoS example, `(a+)+`.
- **P09** (category A, labelled `FIG_B`): the same nested-quantifier family, swept further than P01's own default before `deriv_bc` gives way.
- **P25**, **P27**, **P28** (category F, labelled `FIG_F`, run together): the alternation-star-then-literal family, where unlike the E-family rows above, `deriv_bc` itself — not only `deriv_std` — eventually times out.

Categories **C**, **D**, and **H** have no row singled out this way: their calibrated default sweep already reaches each category's own interesting regime within the routine pass above (a lower ceiling than A/B/E/F, documented in each row's own `suggested_n_notes`, `docs/DATASET.md`), so a separate, individually invoked sweep would not show anything the routine table does not already.

Table 8.7 reports both P60 and P61 from a real run (`BENCH_ENTRY_TIMEOUT_MS=120000`, no memory cap); the three remaining representatives (`FIG7C`/P20, `FIG_A`/P01, `FIG_B`/P09, `FIG_F`/P25+P27+P28) are still pending and skeletonised in Table 8.8.

Row	Parser	Wall time (match side)		
P60 ($(a\|b\|ab)^*$, 'a'-repeated; 1 M/5 M/10 M)	`deriv_std_rec`	crashed at 1 M, 5 M, and 10 M (every point tested)		
	`deriv_std_loop`	crashed at 1 M, 5 M, and 10 M (every point tested)		
	`deriv_bc`	3.85 s	19.45 s	39.85 s
	`pderiv_bc`	3.26 s	16.88 s	33.86 s
	`pderiv_std`	2.72 s	13.68 s	27.62 s
P61 ($(a\|b\|ab)^*$, 'ab'-repeated; 500 K/2.5 M/5 M)	`deriv_std_rec`	crashed at 500 K, 2.5 M, and 5 M (every point tested)		
	`deriv_std_loop`	crashed at 500 K, 2.5 M, and 5 M (every point tested)		
	`deriv_bc`	2.84 s	14.40 s	28.25 s
	`pderiv_bc`	2.35 s	11.96 s	23.73 s
	`pderiv_std`	2.50 s	12.71 s	24.61 s

Table 8.7: Figure 7 reproduction, real run. Wall time is the match-side entry's own reported `micros`; the non-match side is within a few percent throughout and omitted for space. Peak memory is not reported here: this run used `bench_pathological_time` (crash/timeout plus wall time only) — a dedicated `bench_pathological_memory` pass at this same scale is still pending and would be needed for a real peak-bytes figure at 1 M+ characters; earlier informal estimates for that figure have been removed rather than presented as measured.

Both standard-derivative variants crash — not time out — at *every* point tested on both rows, including the smallest (500 K characters for P61). The 120-second timeout was never reached; each crashed entry's own wall time averages 14–16 seconds across both rows and both variants (92.7–96.6 s totalled over six crashed entries per parser per row), uniformly regardless of whether N was 500 K or 10 M, which is itself informative: a crash that takes the same time at every scale tested is a crash that happens early relative to the full input, not one that depends on reaching the far end of it.

The two variants fail by two different, independently verifiable mechanisms, both visible directly in `src/parsers/deriv_std/`. `parse_deriv_std_rec` (`parse.rs`) recurses once per input character before injecting — `parse_recursive_helper` calls itself at `pos+1` and only then calls `inject` on the way back out — so its native call-stack depth equals the number of characters consumed so far. At 500,000 to 10,000,000 characters this exceeds `bench_worker`'s per-call stack budget long before either the heap or the 120-second cap becomes relevant: a textbook stack overflow, not a resource exhaustion the OS arbitrates. `parse_deriv_std_loop` does not share that recursion — both its forward derivation loop and its backward reconstruction loop over `inject` are flat `for` loops with no per-character native recursion — yet it crashes on exactly the same inputs at exactly the same rate. Its failure is a heap one: the forward loop pushes every intermediate derivative onto a single `Vec<Regex>` and keeps all of them alive until the backward pass finishes, and `deriv` (`src/regex/deriv.rs`) is called with no simplification step in between — confirmed by grep, no call to `simplify` anywhere in `deriv_std`'s own hot path — so nothing bounds how large the stored sequence of expressions can get as the derivation proceeds on a genuinely ambiguous pattern like `(a|b|ab)*`. Table 8.14's much smaller canonical example already shows this: `deriv_std` reaches 9179 nodes at just $n = 16$ characters on `(a+(b+ab))*`, a pattern simpler than the Figure 7 one. At 500,000+ characters on a pattern built specifically to maximise this growth, the stored sequence exhausts available memory well inside the 120-second cap, independent of whatever the recursion-depth problem above does to the other variant. The two failures are therefore both real, both architectural, and both independent of each other — fixing the stack overflow alone (e.g.

by making the backward pass iterative, Task #78 of this project's own backlog) would not rescue `deriv_std_loop`, since its own failure is never a stack problem to begin with.

Row	PATHO_N				
P20 (FIG7C)	1 000 000	5 000 000	10 000 000	–	–
P01 (FIG_A)	10 000	30 000	50 000	200 000	1 000 000
P09 (FIG_B)	100 000	500 000	1 000 000	5 000 000	–
P25/P27/P28 (FIG_F)	5 000	10 000	30 000	50 000	200 000

Table 8.8: Skeleton for the four remaining representative sweeps — column headers are the already-chosen `PATHO_N` points; every cell (per-parser wall time, as in Table 8.7) is pending the dedicated run. Given P60/P61's result above, `FIG7C` is expected to show the same two-mechanism crash for both standard variants at every point, since P20 is the same family's `+` variant.

Corpus benchmarks

Corpus timing is measured by `bench_corpus_time`, a plain reporting binary rather than a Criterion target. It runs each parser once over the admitted S2ELab corpus (no source selection — S2ELab is the only source with real prepared data, Section 8.1.2), timing every entry individually and reporting the total, mean, median, p99, and maximum per-entry cost. One pass per parser is what makes the measurement feasible: a full corpus pass takes minutes per parser, so Criterion's sample-count model would multiply the runtime for no statistical gain over a single pass.

```
cargo bench --bench bench_corpus_time
BENCH_PATTERN_LIMIT=5   cargo bench --bench bench_corpus_time
BENCH_SPLIT_OUTCOME=1   cargo bench --bench bench_corpus_time
BENCH_MATCH_INTENT=both cargo bench --bench bench_corpus_time
```

`BENCH_MATCH_INTENT` chooses which of the two groups described in Section 8.1.3 to measure; every table in this chapter uses the default (`explicit`, the 463 full-match-wrapped patterns) unless stated otherwise, for the reason given there. Setting `BENCH_SPLIT_OUTCOME` prints a second table splitting each parser's per-entry durations by the entry's verified outcome (match or non-match), which shows whether a parser's total is skewed by one side of the corpus rather than spread evenly across it.

Table 8.9 reports a full run over the default 463-pattern, 10 888-entry scope. `mean` is the macro-average of Section 8.1.8: each pattern's own mean is taken first, and those per-pattern means are then averaged unweighted, so a pattern with many given examples does not outweigh one with few.

Parser	Entries	Timed out	Mean (ms)	Median (ms)	Max (ms)
`deriv_std_rec`	10 888	67	7.379	0.211	1 811.961
`deriv_std_loop`	10 888	66	6.902	0.226	1 909.783
`deriv_bc`	10 888	0	1.653	0.316	270.844
`pderiv_bc`	10 888	0	0.613	0.097	100.659
`pderiv_std`	10 888	0	0.598	0.094	104.699

Table 8.9: Corpus timing over the full default scope (463 patterns, 10 888 entries), one pass per parser. `Timed out` entries (a 2-second per-entry cap, both standard variants only) are excluded from `mean`/`median`/`max`, so those columns are a lower bound on the standard variants' true cost, not an overstatement of it.

Three observations follow. First, the two standard derivative variants are the slowest by a wide margin — roughly 4–12× `deriv_bc`'s mean and over an order of magnitude past both partial-derivative variants — and this is *before* accounting for the 66–67 entries each one timed out on entirely, which the mean excludes. Second, within the bit-coded/partial-derivative group, the ranking is not what the growth curves of Section 8.3.3 alone would predict: `pderiv_std` is marginally faster than `pderiv_bc` on this corpus (0.598 ms against 0.613 ms mean), even though `pderiv_bc`'s bit-coded state is asymptotically smaller. Section 8.4.3 returns to this point together with the pathological-scale data, where the same two parsers are compared at far larger input lengths.

Timing by outcome. `BENCH_SPLIT_OUTCOME`'s second table (Table 8.10) shows the same run split by the entry's verified membership outcome.

Parser	Matches	Match time (s)	Non-matches	Non-match time (s)
`deriv_std_rec`	5 379	67.913	5 442	9.620
`deriv_std_loop`	5 380	62.660	5 442	9.773
`deriv_bc`	5 444	13.162	5 444	5.014
`pderiv_bc`	5 444	5.208	5 444	1.525
`pderiv_std`	5 444	5.078	5 444	1.496

Table 8.10: Total wall time by verified outcome, same run as Table 8.9. Every parser spends most of its time on matching entries, not non-matching ones.

Every parser's total is dominated by the matching side, by a factor of roughly 3× to 7× depending on the parser — expected, since a non-match can often be rejected early in the derivation, while a match forces the parser all the way to constructing a tree. The split is also where the two standard variants' near-identical means in Table 8.9 resolve into a real difference: `deriv_std_loop`'s match-side total (62.660 s) is noticeably lower than `deriv_std_rec`'s (67.913 s), even though their non-match totals are almost the same (9.773 s against 9.620 s). The two variants implement the same algorithm; the recursive variant's overhead is specifically on the matching path, where the parse tree is actually built.

Encoding, deriving, decoding. The bit-coded parser (`deriv_bc`) reports its mean as three components rather than one: `mean_encode` (building the initial bit-annotated `Regex`), `mean_derive` (the derivative loop itself), and `mean_decode` (walking the final bit-code into a parse tree). On the full corpus these are 0.117 ms, 1.504 ms, and 0.005 ms respectively — the derivative loop is the cost, encoding is a small fixed overhead paid once, and decoding is close to free because it is a single linear walk over an already-small bit-code rather than a second derivation pass. The two standard variants and the two partial-derivative variants do not split this way: `deriv_std`'s backward-pass reconstruction is not separable from its forward pass the way `deriv_bc`'s decode step is, and the partial-derivative variants build their tree incrementally as they go rather than in a separate final step.

Translation cost

Section 8.3.1 separated pattern-structure cost into two phases paid at different times: translation and the parser's own derivative loop. Everything above measures the second phase only — `bench_corpus_time` and `bench_pathological_time` both time an already-translated `Regex`. `bench_corpus_translate` and `bench_pathological_translate` mea-

sure the first phase on its own, deduplicated by distinct pattern rather than by entry, since translation cost depends only on the pattern, not on which input is run against it afterwards.

```
cargo bench --bench bench_corpus_translate
cargo bench --bench bench_pathological_translate
```

Table 8.11 reports the S2ELab side and Table 8.12 the pathological side, both from a real run.

	Patterns	Mean (ms)	Mean nodes	Max nodes
S2ELab (full-match)	463	0.015	502.5	18 707

Table 8.11: Translation cost per distinct pattern, full default scope.

Translation on S2ELab is already cheap at 0.015 ms mean, and even its single largest pattern (18 707 translated nodes) translates in well under a millisecond: no S2ELab pattern comes close to `core_regex_tagged`'s node cap of 20 000 (Section 7.4.2), so this axis is essentially slack for the real corpus.

Cat.	Name	Rows swept	Rejected	Mean (ms)	Max nodes
A	Nested quantifiers	16	0	0.001	84
B	Overlapping alternation	15	0	0.002	389
C	Bounded rep over star	7	0	0.052	9 881
D	Anchored-failure style	20	0	0.005	905
E	Prefix-overlap alternation	8	0	0.001	40
F	Alternation-star + literal	7	0	0.003	756
H	Many adjacent stars/groups	4	0	0.049	3 889
I	Trivial pattern, large input	10	0	0.017	4 001
J	Large pattern, trivial input	17	**9**	0.265	13 999
K	Large pattern, large input	4	**2**	0.009	404
L	Trivial pattern, trivial input	33	0	0.003	388

Table 8.12: Translation cost over the pathological set's own default sweep (pre-recalibration `suggested_p` for J/K; see discussion below). *Rejected*: rows that hit `core_regex_tagged`'s 20 000-node cap at the swept pattern size and translated to nothing.

The prediction from Section 8.3.1 is confirmed and sharpened: **C** is the single most translation-expensive category by far (mean 0.052 ms, 9,881 max nodes, both an order of magnitude past most other categories), consistent with it being built around `core_regex`'s own bound-unrolling. **J** and **K**, the categories swept to a large pattern size, show something the time/memory benches could only show indirectly as uniform "crashed" entries (Section 8.3.2's discussion): **9 of J's 17** swept rows and **2 of K's 4** rejected outright at translation, both at their old `suggested_p` top end (10 000/100 000 for J, 1 000 for K). A rejected row produces no entry for any of the five parsers to be compared on, which is why the uniform per-parser crash counts in Section 8.3.2 are a translation artefact, not a parser-timing result — the real bottleneck for roughly half of J's and K's routine entries was never the derivative loop at all. Both categories' `suggested_p` have since been lowered (J: [100, 500, 1000, 2000], K: [300], down from [100, 1000, 10000, 100000] and [1000]) to keep this category measuring what it is meant to — parser throughput on large-but-translatable patterns — rather than mostly re-discovering the translation cap; a fresh run against the new values is pending.

8.3.3 Memory

Allocation counters

`bench_corpus_memory` replaces the system allocator with a wrapper that counts allocated bytes, and reports three figures per parser:

- **total bytes**: every allocation during the whole corpus pass, including freed and re-allocated bytes. Measures churn.
- **average bytes per entry**: a macro-average, the same correction as `mean` in Table 8.9 (Section 8.1.8) — each pattern's own average is taken first, then those per-pattern averages are averaged unweighted.
- **max peak**: the highest live-byte high-water mark seen on any single entry. Measures worst-case memory. Unlike the other two figures, this one is a flat, pooled-over-every-entry maximum, not a macro-average.

Like `bench_corpus_time`, it is a plain reporting binary and reads the same two environment variables, `BENCH_PATTERN_LIMIT` and `BENCH_MATCH_INTENT`. Allocation counting sees heap traffic only: each worker's call-thread stack is invisible to it, so a pattern whose cost is dominated by recursion depth rather than allocation would under-report here — the figures below are heap numbers, not a general memory bound.

```
cargo bench --bench bench_corpus_memory
BENCH_PATTERN_LIMIT=5   cargo bench --bench bench_corpus_memory
BENCH_MATCH_INTENT=both cargo bench --bench bench_corpus_memory
```

Table 8.13 reports a full run over the same default 463-pattern, 10 888-entry scope as Table 8.9.

Parser	Total bytes	Avg bytes/entry	Max peak
`deriv_std_rec`	26.77 GB	2.551 MB	582.4 MB
`deriv_std_loop`	26.91 GB	2.563 MB	582.4 MB
`deriv_bc`	14.80 GB	1.345 MB	5.1 MB
`pderiv_bc`	4.44 GB	0.404 MB	3.8 MB
`pderiv_std`	4.34 GB	0.395 MB	2.5 MB

Table 8.13: Heap allocation over the full default scope (463 patterns, 10 888 entries), one pass per parser. `Total bytes` is summed over the whole pass; `avg bytes/entry` is the macro-average; `max peak` is the single worst entry's high-water mark.

The gap between the two constructions is confirmed at full corpus scale and is starkest in peak allocation: both standard variants peak at 582.4 MB on their single worst entry, against 5.1 MB for `deriv_bc` — a factor of about 114× — because `deriv_std` retains its full derivation sequence for the backward pass while `deriv_bc` carries only a flat bit vector. The same pattern holds in total and average bytes, though less sharply (roughly 1.8× and 1.9×).

The partial-derivative pair tells a different story from the one the growth curves of Section 8.3.3 alone would suggest: `pderiv_std` is *lower* than `pderiv_bc` on every column here — average bytes (0.395 MB against 0.404 MB) and peak (2.5 MB against 3.8 MB) alike — the reverse of the timing gap on this same corpus (Section 8.4.3 returns to this). At the

short input lengths S2ELab's given examples actually reach, the partial-derivative frontier is already small regardless of construction, so `pderiv_bc`'s extra bit-tracking machinery shows up here as pure overhead rather than a saving; nothing in this table contradicts the asymptotic bound-state argument of Section 8.3.3, since that argument is about how the two scale as input length grows, not about which is smaller at the input lengths this particular corpus happens to contain.

Growth: two canonical expressions

To show *why* memory behaves as measured, three growth demos drive the parsers on two canonical expressions and plot the size of the live state per input character.

The first expression is `a*`, the smallest non-trivial star expression. The second is `(a+(b+ab))*`, which forces every parser to hold multiple residuals simultaneously.

```
cargo run --release --example demo_growth_deriv
cargo run --release --example demo_growth_pderiv
cargo run --release --example demo_growth_eval
cargo run --release --example demo_simp_branches
```

Table 8.14 reports the live-state size at three input lengths per expression. For `deriv_std` and `deriv_bc` the figure is the node count of the current residual. For `pderiv_std` and `pderiv_bc` it is the total node count summed across the whole frontier.

	a^* on a^n			$(a+(b+ab))^*$ on $(ab)^n$		
Parser	n=1	n=8	n=16	n=1	n=8	n=16
`deriv_std`	4	39	79	16	539	9179
`deriv_bc`	2	2	2	12	143	2303
`pderiv_std`	4	4	4	22	160	2560
`pderiv_bc`	2	2	2	18	128	2048

Table 8.14: Live-state size per parser at three input lengths.

Three patterns are visible.

First, on `a*` the `deriv_std` column grows linearly in n, while the other three are bounded. This is the cost of the two-pass structure: the standard parser keeps every intermediate expression alive until the backward pass reconstructs the tree, so its state size is a function of input length. The bit-coded and partial-derivative variants do not carry a history, so their state does not grow.

Second, on `(a+(b+ab))*`, all four grow, but at very different rates. At $n = 16$, `deriv_std` has grown to 9179 nodes, `deriv_bc` to 2303, and the two partial-derivative variants to 2560 and 2048. The gap between the two derivative-based variants is the cost of carrying reconstruction information through each step, which the bit-coded variant does not.

Third, the two partial-derivative variants stay close to each other across both expressions. Their live state is the frontier of residuals, not a single expression; the accumulation is linear in the number of distinct residuals reachable, which is bounded by the pattern's own size.

`demo_simp_branches` shows why simplification never fires its $r+r = r$ rule on `(a+(b+ab))*`. Every step has *distinct* branches by bit-annotation even when the structural shape is iden-

tical. This is a property of the bit-coded representation, not a deficiency of the simplifier: the branches genuinely differ, and merging them would lose the reconstruction information that makes single-pass decoding possible.

```
cargo run --release --example demo_simp_branches
```

A figure showing the branch-count series for `(a+(b+ab))*`, using the pgfplots coordinates the demo already prints, can be added here.

Peak memory on the corpus

The max-peak column from Table 8.13 answers whether any S2ELab pattern causes a peak that would be infeasible on a production machine. At full corpus scale the answer is no: the highest single-entry peak is 582.4 MB, for the two standard derivative variants, well within the memory of a typical workstation; the other three parsers peak in single-digit megabytes. S2ELab's given examples are short enough that none of its 10 888 entries approaches a memory-infeasible regime for any of the five parsers.

That is a property of this corpus's own input lengths, not a general bound on the standard construction: Table 8.7's pathological-scale figures, on inputs several orders of magnitude longer, show the same two-pass retention pushing both standard variants to crash or OOM already at one million characters, while the bit-coded and partial-derivative variants keep growing smoothly into the low gigabytes out to ten million — large, but a controlled, roughly linear growth rather than the standard variants' outright failure.

8.4 Discussion of Results

8.4.1 Cost of the standard construction

The two standard derivative parsers (`deriv_std_rec` and `deriv_std_loop`) are consistently the slowest and memory-heaviest of the four. This is a direct consequence of the two-pass structure described in Chapter 5: every intermediate expression is stored, so the whole derivation sequence stays alive until the backward pass completes. On `a*` input of length n, the standard parser's state size grows linearly in n, while the bit-coded and partial-derivative parsers stay constant.

On the full S2ELab corpus the two standard variants are close in mean time (Table 8.9: 7.379 ms against 6.902 ms) but not identical, and the outcome split (Table 8.10) shows where the difference actually lives: on matching entries — where a parse tree is built — `deriv_std_loop`'s total (62.660 s) is measurably below `deriv_std_rec`'s (67.913 s), while their non-match totals are almost the same. The recursion-versus-iteration difference between the two is specifically a cost of the matching path, not a flat per-call overhead.

8.4.2 Cost of bit-coding

For the standard-derivative family, bit-coding is an unqualified win on the full S2ELab corpus: `deriv_bc`'s mean time (1.653 ms) is roughly 4–4.5× below either standard variant's, and its peak allocation (5.1 MB) is about 114× below theirs (582.4 MB, Table 8.13). The trade-off it makes is that the forward pass cannot simplify away redundant branches without losing reconstruction information. The `demo_simp_branches` output shows this

clearly: the branch counts grow as expected but the distinctness count stays at 1 after bit stripping, meaning every branch is retained because it carries a different bit annotation. The retention is not wasted: it is exactly what makes single-pass reconstruction possible, and `deriv_bc`'s own encode/derive/decode split (Section 8.3.2) shows that decode — the step that cashes in that retained information — is close to free (0.005 ms mean), so the retention is not even an expensive one.

8.4.3 Cost of partial derivatives

The two partial-derivative parsers are the fastest and most memory-light of the five on the full S2ELab corpus, both clearly ahead of either derivative-based variant. They split the residual into a set of states rather than combining into one expression, which keeps each state small and avoids the alternation growth that the standard derivative parser suffers.

Within the pair, bit-coding's advantage is *not* unconditional the way it is for the standard-derivative family above, and which variant wins on time depends on more than just scale. On the full S2ELab corpus, `pderiv_std` is marginally ahead of `pderiv_bc` on every figure measured: mean time (0.598 ms against 0.613 ms), average bytes (0.395 MB against 0.404 MB), and peak bytes (2.5 MB against 3.8 MB) — at S2ELab's short input lengths, `pderiv_bc`'s bit-tracking is pure overhead on both axes. At Figure 7 scale (Table 8.7), the real run splits by *row*, not by scale: on P60 (`(a|b|ab)*` over 'a'-repeated input), `pderiv_std` stays faster than `pderiv_bc` at every point tested — 2.72 s against 3.26 s at 1 M, 13.68 s against 16.88 s at 5 M, 27.62 s against 33.86 s at 10 M, a stable 20–23% gap that does not narrow with input length. On P61 (the same pattern over 'ab'-repeated input) the ranking flips entirely: `pderiv_bc` is faster than `pderiv_std` at every point tested instead — 2.35 s against 2.50 s at 500 K, 11.96 s against 12.71 s at 2.5 M, 23.73 s against 24.61 s at 5 M, a smaller but equally consistent 4–6% gap in the opposite direction. Peak memory at this scale has not been measured in a dedicated run (Table 8.7's caption), so no memory-crossover claim is made here; only the time comparison above is from real data. A plausible reading, not yet verified against the implementation, is that 'a'-repeated input keeps more alternative decompositions simultaneously live at every step than 'ab'-repeated input does against the same pattern, and `pderiv_bc`'s per-branch bit-tracking cost scales with exactly that — but the honest finding is simply that *which* input shape is used changes which variant wins, not only how large the gap is, and generalising from one of P60/P61 to "bit-coding eventually wins" would not have survived checking the other row. Bit-coding's cost for the partial-derivative construction is a real, fixed per-step overhead; whether it pays for itself depends on the pattern and input together, not on scale alone — unlike the standard-derivative family above, where bit-coding wins unconditionally because it removes an $O(n)$ liability outright rather than trading one small constant for another.

The price of partial derivatives, independent of which variant, is that they produce a non-POSIX parse tree rather than a POSIX one. This is a specification difference, not an implementation deficiency, and it is the reason the two families are compared separately in Section 8.2: within each family, the two variants must agree, but between families they are allowed to differ.

8.4.4 When each parser is appropriate

The four parsers are not interchangeable. Table 8.15 summarises the decision.

Requirement	Parser	Rationale
POSIX + long patterns	`deriv_bc`	Only POSIX variant with bounded state; wins unconditionally, not just in some regime (Section 8.4.2).
POSIX + short/shallow	`deriv_std_loop`	Lowest code complexity; acceptable when input is small.
non-POSIX, default choice	`pderiv_std`	Matched or beat `pderiv_bc` on every real measurement taken so far — the full S2ELab corpus and both Figure 7 rows up to 10 M characters (Section 8.4.3); simpler code, no bit-tracking to maintain.
non-POSIX, `pderiv_bc` worth trying	`pderiv_bc`	Beat `pderiv_std` on one of the two Figure 7 rows (P61, 'ab'-repeated) at every scale tested, never on the other (P60, 'a'-repeated) — which one depends on the input's own structure, not on scale, so this is a profile-and-compare recommendation, not a default.
Formal verification/teaching	`deriv_std_rec`	Closest to the paper's definition.

Table 8.15: Parser recommendation by requirement.

The frontend, not the parser, is the limiting factor for the small "other" share of Chapter 7's rejection count (Table **??**): patterns that exceed the bound cap of 1000 repetitions, or that mix a bound with a nested alternation, are outside the current translation. Raising the cap is a per-pattern trade-off, not a parser choice.

8.4.5 Threats to validity

Three properties of the setup limit how far the results can be generalised.

Corpus scope. The one real corpus is S2ELab: human-authored regular expressions from one dataset's own domain and convention of writing a pattern, each already paired with its own author-chosen match/non-match examples. A corpus drawn from a different domain, or one assembled rather than author-paired with examples, might exhibit different acceptance rates and different timing profiles. The default scope within S2ELab is also a choice, not the whole dataset: every table in this chapter measures only the 463 patterns a real author wrapped in `^...$` (Section 8.1.3), on the grounds that an unwrapped pattern's intended matching scope is genuinely ambiguous; a different, defensible choice there (`BENCH_MATCH_INTENT=both`) would fold in the remaining 115 patterns and could shift the absolute numbers, though not the ordering among parsers, which does not depend on which patterns are included.

The pathological set. Each of its 113 rows is hand-curated to isolate one specific complexity mechanism at a controlled scale (Section 8.1.9); it does not claim to represent the distribution of pattern shapes any real author would write, only to show how each parser's cost grows once that one mechanism is pushed past where a real corpus would reliably exercise it. Most categories sweep a single parameter in isolation, so a pattern that combines two stress mechanisms at once — deep nesting *and* input length *and* ambiguity together, say — is not directly represented, even though each mechanism on its own is.

Measurement model. `bench_corpus_memory` and `bench_pathological_memory` measure allocation counts, not wall-clock memory pressure. On systems with a garbage collector or a page cache, the figures would interact with those subsystems in ways this evaluation does not model. The evaluator runs on a single machine, so the wall-clock figures in Table 8.9 and Table 8.7 are CPU- and cache-specific. The S2ELab timing and memory tables are drawn from the full corpus, and so now is the pathological set's own per-category table (Table 8.6) and its Figure 7 representatives (Table 8.7, P60/P61); peak memory at Figure 7 scale specifically has not been measured in a dedicated run, so no memory claim is made there, only time. The remaining four individually-swept representatives (Table 8.8: `FIG7C`/P20, `FIG_A`/P01, `FIG_B`/P09, `FIG_F`/P25+P27+P28) are not yet run at the scale they are configured for, and the numbers reported for those rows should be read as pending rather than as a claim about their own behaviour until that run lands.

8.4.6 Summary

On both timing and memory, the partial-derivative parsers dominate the two standard-derivative parsers by a wide and consistent margin, on both the real corpus and the pathological set. Within the standard-derivative family, bit-coding dominates unconditionally: it removes an $O(n)$ liability, so `deriv_bc` is faster and lighter than both `deriv_std_rec` and `deriv_std_loop` at every scale measured, and both standard variants fail outright — by two different, independently confirmed mechanisms, stack overflow for the recursive variant and heap exhaustion for the iterative one — on every Figure 7-scale input tested, not only on especially large ones. Within the partial-derivative family, no single direction holds: `pderiv_std` is marginally faster and lighter than `pderiv_bc` on the real corpus's short inputs and on P60 at every Figure 7 scale tested up to 10 M characters, while `pderiv_bc` is consistently faster on P61 at every scale tested instead (Section 8.4.3) — the same pattern over different input shapes, not a larger-input-wins story. The choice of construction (standard vs. partial derivative) matters more than the choice of variant (standard vs. bit-coded) in every case measured; the choice of variant matters, but which way it cuts for the partial-derivative family is not yet predictable from scale alone, and neither variant should be assumed to dominate the other without measuring the specific pattern and input in question.

All five parsers correctly classify every case in both the full S2ELab corpus and the full pathological set, confirmed by a real run of the agreement check (Table 8.4): 100% membership agreement on both datasets, with zero within-family shape disagreements printed by either run. That confirmation came with one real correction along the way: the first pathological run flagged a genuine dataset-labeling bug in two category-H rows (`P29`, `P31`), traced to a non-match construction that cannot work against an unbounded-length pattern, fixed, and re-verified at 100% agreement (Section 8.2's discussion has the full account). The POSIX-vs-non-POSIX *Diff* where it does appear is a specification

difference rather than an implementation bug.

9 Related Work and Conclusion

9.1 Related Work

9.2 Conclusion

Bibliography

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